

Adaptively Secure IBE from Lattices with Asymptotically Better Efficiency[†]

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Abstract. Current adaptively secure identity-based encryption (IBE) constructions from lattices are unable to achieve a good balance among the master public key size, secret key size, modulus and reduction loss. All existing lattice-based IBE schemes share a common restriction: the modulus is quadratic in the trapdoor norm.

In this work, we remove this restriction and present a new adaptively secure IBE scheme from lattices in the standard model, which improves the state-of-the-art construction proposed by Ables et al. (TCC 2021) and achieves asymptotically better efficiency. More precisely, we achieve the asymptotically minimal number of public vectors among all the existing schemes, along with a significantly smaller modulus compared to the scheme by Ables et al. (TCC 2021). Furthermore, our scheme enjoys the smallest Gaussian width of the secret key among all existing schemes and has the same tightness as Ables et al.'s scheme.

We propose a novel cross-multiplication design for our IBE scheme, along with several novel tools and techniques, including: (a) a homomorphic computation algorithm that outputs BGG+-style encoding with two distinct-norm trapdoors; (b) a sampling algorithm with hybrid Gaussian outputs; and (c) a partial rerandomization algorithm. These new tools and techniques are general and could find rich applications in lattice-based cryptography.

Keywords: Lattice-based cryptography · Identity-based encryption · GSW-style encryption · BGG+-style encoding · Sampling algorithm.

1 Introduction

Identity-based encryption (IBE), proposed by Shamir [34] as a way to simplify public key and certificate management, is a generalization of public key encryption, where the public key can be an arbitrary string, such as a name, a telephone number, or an email address. Since its first realization proposed by Boneh and Franklin [8], various IBEs based on bilinear maps [6, 7, 20, 36, 37], quadratic

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residues modulo composite [10, 15], and lattices [1, 2, 14, 17, 21, 23, 25, 38, 39] have been proposed.

Two major security notions, selective security and adaptive security, have been studied in the literature. The former requires the adversary to choose the challenge identity before seeing the master public key (**mpk**), while the latter does not have this restriction. Adaptive security offers stronger protection and is more desirable in practical settings. However, realizing this notion is quite challenging, especially when aiming for efficiency comparable to that of selectively secure designs in the plain model.

Prior constructions from bilinear groups have achieved this goal using the powerful dual-system framework [37]. However, it remains uncertain whether the dual-system framework can be instantiated from other assumptions, particularly from post-quantum candidates such as lattices. In the post-quantum context, although there are adaptively secure lattice-based IBEs, the current instantiations are unable to achieve a good balance between **mpk** size, secret key size, modulus, and reduction loss, i.e., either with large **mpk** and reduction loss, small modulus and secret key, or with small **mpk** and (almost) tight reduction, yet large modulus and secret key. (See Tab. 1 for details.) Achieving a good balance among these aspects is a crucial step toward realizing a practical post-quantum IBE. In this work, we focus on constructing adaptively secure lattice-based IBE with compact **mpk**, small modulus, small secret key size, and tight reduction simultaneously in the standard model. Below we discuss the challenges faced by current approaches and introduce our new ideas.

Challenges in Current Techniques. In Eurocrypt 2010, Agrawal et al. [2] constructed an efficient adaptively secure IBE in the standard model. However, this construction is not compact in the sense that the public parameter contains $O(\lambda)$ number of basic matrices, where λ is the security parameter. To deal with this issue, Yamada [38] constructed IBE schemes from lattices based on the partitioning technique and reduced the number of the public matrices to $O(\lambda^{1/\tau})$, where $\tau \in \mathbb{N}$ is an arbitrary constant. However, these schemes require a super-polynomial LWE modulus. Subsequently, Katsumata and Yamada [25] proposed a more efficient IBE scheme from the ring LWE (RLWE) assumption with asymptotically the same number of public matrices, but only with polynomial modulus. Later, Yamada [39] proposed two new constructions and reduced the number of the public parameters to $\omega(\log \lambda)$ in his second construction. However, his second construction relies on the Barrington’s Theorem [4] to compute an NC1 boolean circuit, which can be done in polynomial time in theory yet would not be expected to be efficient in practice.

Besides, there exists a bootstrapping technique by [13, 16] which can transform any selectively secure IBE into an adaptively secure one without blowing up the **mpk** at all. However, the resulting scheme is not considered (even close to) practical as each ciphertext consists of ℓ garbled circuits (ℓ is the bit length of ID). In a separate line of work, Boyen and Li [12], along with the subsequent work by Lai et al. [27], presented adaptively secure IBE schemes with (almost) tight security from lattices. Their constructions follow the Katz-Wang frame-

work [26] but fail to achieve a compact **mpk**. Specifically, their constructions require homomorphic computation of a PRF, which leads to the need for encoding the PRF’s seed into the **mpk**. Both constructions use a bit-by-bit encoding that requires $O(\lambda)$ additional random public matrices in the **mpk**.

Recently, Abla et al. [1] proposed a more compact IBE scheme (ALWW-IBE) with only $\omega(1)$ ring vectors in the public parameters. Moreover, they further improved previous works in two aspects: (1) every component in their construction is explicit, i.e., without relying on Barrington’s Theorem [4] and (2) they achieved a tighter security reduction. Despite these theoretical advancements, the construction in [1] still has a distance to practicality. The main drawback of their construction is that it requires a large modulus (about at least $\tilde{O}(n^{12})$, where n is the dimension of the underlying ring). This drawback also affects the total size of the **mpk**¹, the total size of the ciphertext, the running time of the scheme, and the concrete hardness of the underlying RLWE problem. Specifically, their construction is based on the partitioning technique with a subtle design of partition function and homomorphic computation in cyclotomic rings. However, this paradigm involves heavy homomorphic evaluation of the partition function. What’s worse, the modulus is at least the quadratic of the norm of the trapdoor after homomorphic evaluation due to the noise re-randomization as described in [25]. It seems that the existing approach, particularly the noise re-randomization, inherently induces the modulus to be quadratic of the trapdoor norm, and thus harshly increases the modulus. This raises a natural question:

Can we remove the quadratic restriction of modulus on the trapdoor norm, and thus design an adaptively secure lattice-based IBE that inherits the compactness and tightness of ALWW-IBE, but with small modulus?

1.1 Our Contributions

In this work, we provide an affirmative answer to this question. Particularly, we remove the quadratic restriction of modulus, and thus obtain an adaptively secure IBE from lattices in the standard model, with the same compactness and tightness of ALWW-IBE, but with significantly smaller modulus. The key innovation lies in a novel cross-multiplication design, supported by several novel tools and techniques. We believe that these tools can find broad applications in other lattice-based primitives and thus are of general interests. Below we summarize our two major contributions, and present our new techniques in Sect. 1.2.

- Our IBE scheme, like [1], achieves the asymptotically minimal number of basic vectors in **mpk** among all the existing schemes, and as tight as the ALWW-IBE, but with significantly smaller modulus compared to [1]. Furthermore, our scheme enjoys the smallest Gaussian width of the secret key among all existing schemes (including [1]). For a detailed overview and comparison, please refer to Tab. 1.

¹ The number of matrices in **mpk** is unchanged. But due to a large modulus, the size of each matrix is large and leads to a large **mpk**.

Table 1. Comparison with previous lattice-based IBE constructions with adaptive security in the standard model.

Scheme	$ \text{mpk} $ # of $\mathcal{R}_q^k$ vec. [#]	RLWE param $\frac{1}{\alpha} = \frac{q}{\sigma_{\text{RLWE}}}$	Gaussian width of the sk_{id}	$ \text{sk}_{\text{id}} , \text{ct} $ # of $\mathcal{R}^k, \mathcal{R}_q^k$ vec.	Reduction cost
[2]+ [11]	$O(\lambda)$	$\tilde{O}(n^{5.5})$	$\tilde{O}(n^2)$	$O(1)$	$O(\epsilon^2/qQ)$
[12]	$O(\lambda)$	superpoly(n)	$\tilde{O}(n^3)$	$O(1)$	$O(\epsilon/\lambda)$
[27]	$O(\lambda)$	$O(n^8)$	$O(n^{3.5})$	$O(1)$	$O(\epsilon/\lambda)$
[38]	$O(\lambda^{1/\tau})^*$	superpoly(n)	superpoly(n)	$O(1)$	$O(\epsilon^{\tau+1}/\ell Q^\tau)$
[25]	$O(\lambda^{1/\tau})$	$O(n^{2.5+2\tau})$	$O(n^\tau)$	$O(1)$	$O(\lambda^{\tau-1}\epsilon^\tau/Q^\tau)^{\tau+1}$
[39] I+ [24]	$\omega(\log^2(\lambda))$	$\tilde{O}(n^{5.5})$	$\tilde{O}(n^{2.5})$	$O(1)$	$O(\epsilon^{v+1}/Q^v)^\Xi$
[39] II	$\omega(\log(\lambda))$	poly(n) [†]	poly(n)	$O(1)$	$O(\epsilon^2/\ell^2 Q)$
[1]	$\omega(1)$	$\tilde{O}(n^{9.5+\frac{4}{\kappa}})^{\ddagger, \S}$	$\tilde{O}(n^{4.5+\frac{2}{\kappa}})$	$O(1)$	$O(\epsilon^2/Q)^*$
Ours	$\omega(1)$	$\tilde{O}(n^{7+\frac{2}{\kappa}})^{\ddagger}$	$\tilde{O}(n^{1.5})$	$O(1)$	$O(\epsilon^2/Q)^*$

Notations: $|\text{mpk}|$, $|\text{ct}|$, and $|\text{sk}_{\text{id}}|$ denote the size of the master public key, ciphertext, and secret key of the IBE. λ , n , q , σ_{RLWE} denote the security parameter, ring dimension, modulus, and Gaussian parameter of RLWE. Q and ϵ denote the number of key extraction queries and the advantage in attacking the IBE scheme. All the schemes set the ring dimension $n = \Theta(\lambda)$. To measure the reduction cost, we show the advantage of the RLWE algorithm constructed from the adversary against the corresponding IBE scheme, just like [39]. To be fair, we calculate the reduction cost by employing the technique of Bellare and Ristenpart [5] for all schemes.

[#] $\mathcal{R}_q = \mathbb{Z}_q[x]/(x^n + 1)$ is a polynomial ring, and $k = \lceil \log_b q \rceil$ where b is a small constant (e.g., $b = 2$) or a polynomial in n (e.g., n^v for any real v).

^{*} $\tau \in \mathbb{N}$ is a constant that can be chosen arbitrarily. Since the reduction cost is exponential in τ , this value is typically set very small (e.g., $\tau = 2$ or 3).

^{Ξ} $v > 1$ is a constant that can be set small, depending on the underlying error correcting code.

[†] poly(n) denotes some fixed but large polynomial. It is hard to determine an explicit bound for comparison due to the implicit construction of the work.

[‡] $\kappa \geq 1$ can be any constant that satisfies $n^{\frac{1}{\kappa}} > 3 + \kappa$, e.g., 2 or 4, depending on the parameters of the underlying error correcting code.

^{$\S$} Note that [1, Section 3] does their analysis in both the plain model and the CRS model. A smaller RLWE parameter ($\tilde{O}(n^{7.5+\frac{4}{\kappa}})$) can be achieved in the CRS model. Here, we only compare the parameters (in [1] and ours) in the plain model. Our analysis in the plain model can be easily adapted to the CRS model, which results in a smaller RLWE parameter. We refer to Appendix E for the detailed parameter analysis in both models.

^{*} In the ALWW-IBE paper [1], the authors obtain $T' = T + \min\{\tilde{O}(\lambda^{1/\kappa}Q/\epsilon), O(\lambda^{1+3/\kappa})Q^{\kappa+3}\}$ and $\epsilon' = O(\epsilon/\lambda^{1/\kappa}Q)$ using the bit-security framework [31]. Here we analyse their reduction cost with the most common technique [5] without relying on the bit-security framework. We note that our IBE could have the same reduction cost as the ALWW-IBE in the bit-security model, as it contains the same partition function as in the ALWW-IBE. The analysis in the bit-security model of our IBE is similar to [1, Lemma 5.4] and we omit it in our paper.

- Technically, we propose three novel tools and techniques to obtain our IBE scheme. They are (a) homomorphic computation outputting BGG+-style encoding with two distinct-norm trapdoors; (b) sampling algorithm with hybrid Gaussian outputs; and (c) partial re-randomization. These tools and techniques are general, not only restricted to ring settings, but also can be applied to LWE-based IBE as shown in Appendix F. We believe it could find rich application scenarios in lattice-based cryptography.

1.2 Techniques Overview

Our IBE follows the framework of the ALWW-IBE and we remove the quadratic restriction of modulus by using a novel cross-multiplication design with the help of several novel tools/techniques including a) homomorphic computation outputting BGG+-style encoding with two distinct-norm trapdoors; b) sampling algorithm with (D_r, D_σ) -hybrid outputs; c) partial re-randomization. In this subsection, we recap the ALWW-IBE framework and provide a high level overview of the ideas of our construction.

Recap of ALWW-IBE [1]. For a polynomial ring $\mathcal{R} = \mathbb{Z}[x]/(x^n + 1)$, the ALWW-IBE construction follows the general framework [2] of constructing lattice-based IBE which associates each identity id a vector

$$\mathbf{pk}_{\text{id}}^\top = [\mathbf{b}^\top | F(\text{id})^\top] \in \mathcal{R}_q^{2k}, \quad (1)$$

where $\mathbf{b} \in \mathcal{R}_q^k$ is a vector chosen uniformly at random. One of the main technical contributions of [1] is that they propose a succinct hash function and reduce the size of the public vectors to $\omega(1)$. More concretely, they use the equality test function $\text{Equal}_\beta(x^\alpha)$ that outputs 1 if $\alpha = \beta$ and 0 otherwise, and compute the function $F(\text{id})$ as

$$F(\text{id}) = -\mathbf{c}_\gamma + \sum_{i \in [t]} \sum_{j \in [L+1]} H\text{-Equal}_j(\mathbf{c}_i) \cdot x^{f(i,j)}, \quad (2)$$

where $\mathbf{c}_\gamma, \{\mathbf{c}_i\}_{i \in [t]} \in \mathcal{R}_q^k$ are public vectors, $f : [t] \times [L+1] \rightarrow [n]$ is a function related to an error correcting code with position index L , and $H\text{-Equal}_j(\mathbf{c}_i)$ is a homomorphic computation of $\text{Equal}_j(\cdot)$ from the public vectors $\{\mathbf{c}_i\}_{i \in [t]}$. The number of the public vectors is $\omega(1)$, i.e., $t = \omega(1)$. During the security proof, the reduction algorithm first prepares some random monomials $x^\gamma, \{x^i\}_{i \in [t]} \in \mathcal{R}$ and random matrices $\mathbf{R}_\gamma, \{\mathbf{R}_i\}_{i \in [t]} \in \mathcal{R}^{k \times k}$ with a small spectral norm, and sets the public vectors by “BGG+-style encoding” form [9] as

$$\mathbf{c}_\gamma^\top = \mathbf{b}^\top \mathbf{R}_\gamma + x^\gamma \mathbf{g}^\top \in \mathcal{R}_q^k, \quad \mathbf{c}_i^\top = \mathbf{b}^\top \mathbf{R}_i + x^i \mathbf{g}^\top \in \mathcal{R}_q^k, \quad (3)$$

where $\mathbf{g} = [1|b|b^2|\dots|b^{k-1}] \in \mathcal{R}_q^k$ is the gadget vector with well-known trapdoor $\mathbf{T}_\mathbf{g}$ [29]. Then, the equality test function can be homomorphically computed as

$$H\text{-Equal}_j(\mathbf{c}_i) = \mathbf{b}^\top \mathbf{R}_{i,j} + \text{Equal}_j(x^i) \mathbf{g}^\top.$$

Further, the function $F(\text{id})$ in Eq. (2) can be homomorphically computed as

$$\begin{aligned} F(\text{id})^\top &= -(\mathbf{b}^\top \mathbf{R}_\gamma + x^\gamma \mathbf{g}^\top) + \sum_{i \in [t]} \sum_{j \in [L+1]} (\mathbf{b}^\top \mathbf{R}_{i,j} + \text{Equal}_j(x^i) \mathbf{g}^\top) \cdot x^{f(i,j)} \\ &= \mathbf{b}^\top \left(\underbrace{\sum_{i,j} \mathbf{R}_{i,j} \cdot x^{f(i,j)} - \mathbf{R}_\gamma}_{:= \mathbf{R}_{\text{id}}} \right) + \left(\underbrace{\sum_{i,j} \text{Equal}_j(x^i) \cdot x^{f(i,j)} - x^\gamma}_{:= H(\text{id}), \text{invertible}} \right) \mathbf{g}^\top \\ &= \mathbf{b}^\top \mathbf{R}_{\text{id}} + H(\text{id}) \mathbf{g}^\top. \end{aligned} \quad (4)$$

The sampling vector for identity id in Eq. (1) is now converted into

$$\mathbf{pk}_{\text{id}}^\top = [\mathbf{b}^\top | \mathbf{b}^\top \mathbf{R}_{\text{id}} + H(\text{id})\mathbf{g}^\top] \in \mathcal{R}_q^{2k}. \quad (5)$$

For any uniformly random $u \in \mathcal{R}_q$, the reduction algorithm can sample a short vector $\mathbf{x}$ satisfying $\mathbf{pk}_{\text{id}}^\top \cdot \mathbf{x} = u$ using the public trapdoor $\mathbf{T}_{\mathbf{g}}$ if and only if $H(\text{id}) \neq 0$. In more detail, we first sample a perturbation² $\mathbf{p} \in \mathcal{R}^{2k}$ following the idea of Gaussian convolution by Peikert [32]. Then, by the Gaussian sampling algorithm [29] and public trapdoor $\mathbf{T}_{\mathbf{g}}$, we can obtain a short vector $\tilde{\mathbf{x}}$ such that $\mathbf{g}^\top \cdot \tilde{\mathbf{x}} = H(\text{id})^{-1} \cdot (u - \mathbf{pk}_{\text{id}}^\top \cdot \mathbf{p})$ under the condition that $H(\text{id}) \neq 0$ and $H(\text{id})$ is invertible. Finally, the secret key for identity id is a short vector $\mathbf{x}$ defined as

$$\underbrace{[\mathbf{b}^\top | \mathbf{b}^\top \mathbf{R}_{\text{id}} + H(\text{id})\mathbf{g}^\top]}_{\mathbf{pk}_{\text{id}}^\top} \cdot \underbrace{\left(\mathbf{p} + \begin{bmatrix} -\mathbf{R}_{\text{id}} \\ \mathbf{I} \end{bmatrix} \cdot \tilde{\mathbf{x}} \right)}_{\mathbf{sk}_{\text{id}} (:= \mathbf{x})} = u.$$

By the definition of $\mathbf{x}$, the size of $\mathbf{x}$ is *linear* in the norm of the matrix $\mathbf{R}_{\text{id}}$ ³. We also note that $\mathbf{x}$ is close to a spherical Gaussian distribution.

The IBE ciphertext of a message m is akin to a dual Regev ciphertext [21, 33] which consists of two parts:

$$c_0 = u \cdot v + e_0 + \left\lceil \frac{q}{2} \right\rceil \cdot m \in \mathcal{R}_q, \quad \mathbf{c}_1 = \mathbf{pk}_{\text{id}} \cdot v + \mathbf{e}_1 \in \mathcal{R}_q^{2k},$$

where $v \in \mathcal{R}_q$ is a secret and $e_0 \leftarrow D_{\mathcal{R}, \sigma_0}$, $\mathbf{e}_1 \leftarrow D_{\mathcal{R}^{2k}, \sigma_1}$ are some errors. Note that in the security reduction, the challenge ciphertext can be seen as $(\mathbf{c}_1^*)^\top = [\mathbf{b}^\top | \mathbf{b}^\top \mathbf{R}_{\text{id}^*}] \cdot v + \mathbf{e}_1$ because $H(\text{id}^*) = 0$, then the challenge ciphertext $\mathbf{c}_1^*$ can be simulated by a re-randomization⁴ algorithm as follows:

$$(\mathbf{c}_1^*)^\top = \text{ReRand} \left([\mathbf{I}_k | \mathbf{R}_{\text{id}^*}], \mathbf{b} \cdot v + \mathbf{e}'_0, \sigma_0, \frac{\sigma_1}{2\sigma_0} \right) = \mathbf{b}^\top [\mathbf{I}_k | \mathbf{R}_{\text{id}^*}] \cdot v + (\mathbf{e}'_1)^\top,$$

where $\mathbf{e}'_0 \leftarrow D_{\mathcal{R}^k, \sigma_0}$ and the distribution of $\mathbf{e}'_1$ is statistically close to the discrete Gaussian distribution $D_{\mathcal{R}^{2k}, \sigma_1}$. For the re-randomization algorithm, it requires $\frac{\sigma_1}{2\sigma_0} \geq \|\mathbf{I}_k | \mathbf{R}_{\text{id}^*}\|$, thus the size of the error $\mathbf{e}_1 \leftarrow D_{\mathcal{R}^{2k}, \sigma_1}$ is *linear* in the norm of the matrix $\mathbf{R}_{\text{id}^*}$ ⁵. During the decryption step, the user (who owns the secret

² We need to take a perturbation, since the output secret key $\mathbf{x}$ should follow a spherical Gaussian distribution to ensure that no information about the trapdoor matrix $\mathbf{R}_{\text{id}}$ is revealed.

³ The size of the short vector $\tilde{\mathbf{x}}$ depends on the public trapdoor $\mathbf{T}_{\mathbf{g}}$ and has a small norm. The perturbation $\mathbf{p}$ acts as a mask for $\mathbf{R}_{\text{id}}$ and does not have a noticeable impact on the overall size of $\mathbf{x}$. So we don't need to pay attention to them here.

⁴ On input a vector $\mathbf{b} + \mathbf{x} \in \mathcal{R}^k$, a matrix $\mathbf{V} \in \mathcal{R}^{k \times l}$, two reals σ_0 and σ_1 such that $\sigma_1 \geq 2\sigma_0 \cdot \|\mathbf{V}\|$ and $\mathbf{x} \leftarrow D_{\mathcal{R}, \sigma_0}$, the re-randomization algorithm $\text{ReRand}(\mathbf{V}, \mathbf{b} + \mathbf{x}, \sigma_0, \frac{\sigma_1}{2\sigma_0})$ outputs $\mathbf{b}^\top \mathbf{V} + (\mathbf{x}')^\top$, where the distribution of $\mathbf{x}' \in \mathcal{R}^l$ is statistically close to the discrete Gaussian with width σ_1 .

⁵ The Gaussian parameter σ_0 is only related to the hardness of the RLWE assumption, not related to $\mathbf{R}_{\text{id}^*}$, so we don't consider it here.

key $\mathbf{sk}_{\text{id}} := \mathbf{x}$) can compute

$$c_0 - \mathbf{c}_1^\top \cdot \mathbf{x} = \left\lceil \frac{q}{2} \right\rceil \cdot m + \underbrace{e_0 - \mathbf{e}_1^\top \cdot \mathbf{x}}_{\text{error term}}. \quad (6)$$

To ensure decryption correctness, the modulus q should be larger than the error term.

As mentioned above, the sizes of the error $\mathbf{e}_1$ and the secret key $\mathbf{x}$ are both *linear* in the norm of the matrix $\mathbf{R}_{\text{id}}$, so the size of error term would be at least *quadratic* of the norm of the matrix $\mathbf{R}_{\text{id}}$. Recall that the matrix $\mathbf{R}_{\text{id}}$ is generated by homomorphic computation of the partition function⁶ $H(\cdot)$ as in Eq. (4), which itself has a relatively large norm due to the complex circuit of the partition function. Therefore, the restriction, i.e., the modulus q should be at least the *quadratic* of the trapdoor matrix $\mathbf{R}_{\text{id}}$'s norm, leads to large modulus.

Remove the quadratic restriction of modulus. Our goal is to remove the quadratic restriction of the modulus q on the trapdoor matrix $\mathbf{R}_{\text{id}}$'s norm. At a high level, our idea can be represented by Fig. 1. Specifically, the error term⁷

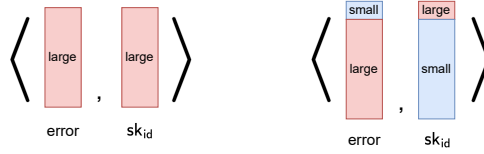


Fig. 1. The error term of ALWW-IBE (left) and ours (right), where $\langle \cdot, \cdot \rangle$ represents vector inner product.

in ALWW-IBE (Eq. (6)) is the inner product of a large error $\mathbf{e}_1$ (i.e., *linear* in $\mathbf{R}_{\text{id}}$'s norm) and a large secret key $\mathbf{x}$ (i.e., *linear* in $\mathbf{R}_{\text{id}}$'s norm). Our idea is to design the error term to be the inner product of a $(D_{\sigma_0}, D_{\sigma_1})$ -hybrid error with $\sigma_1 \gg \sigma_0$ and a (D_r, D_σ) -hybrid secret key with $r \gg \sigma$ ⁸, where D_θ represents a Gaussian distribution with width θ and $(D_{\theta_0}, D_{\theta_1})$ -hybrid represents that the first part is sampled according to D_{θ_0} and the second part is sampled according to D_{θ_1} . In our design, the error term is a cross-multiplication, i.e., “small $\times$ large + large $\times$ small”, thus removing the quadratic restriction.

We approach our idea in two parts: (1) obtaining a (D_r, D_σ) -hybrid secret key; (2) obtaining a $(D_{\sigma_0}, D_{\sigma_1})$ -hybrid error. Below, we provide some details of our techniques.

⁶ We call $H(\cdot)$ as “partition function” since $H(\text{id}) = 0$ if and only if $\text{id} = \text{id}^*$.

⁷ We ignore the term e_0 because it only depends on the hardness of the RLWE problem and it has a small norm.

⁸ To further shrink the size of the secret key, we only take the D_σ part (small) as the user's secret key in our IBE construction. It does not affect the decryption, since anyone can compute the D_r part (large) after it gets the D_σ part.

(1) *Obtaining a (D_r, D_σ) -hybrid secret key.* To achieve this, we first design a series of homomorphic computation algorithms to output a BGG+-style encoding that encodes the partition function $H(\text{id})$ and has two trapdoors with distinct norms. Using such a BGG+-style encoding as our new sampling vector, we design a new sampling algorithm with (D_r, D_σ) -hybrid outputs.

(1.1) *Homomorphic computation outputting BGG+-style encoding with two distinct-norm trapdoors.*

Homomorphic computation of partition function in GSW-style encryption form. First, instead of setting the public parameters as in Eq. (3), we use the “GSW-style encryption” form [22] to construct as follows

$$\mathbf{C}_\gamma = \mathbf{A}\mathbf{R}_\gamma + x^\gamma \mathbf{G} \in \mathcal{R}_q^{2 \times 2k}, \quad \mathbf{C}_i^\top = \mathbf{A}\mathbf{R}_i + x^i \mathbf{G} \in \mathcal{R}_q^{2 \times 2k}, \quad (7)$$

where $\mathbf{R}_\gamma, \mathbf{R}_i \in \mathcal{R}^{2k \times 2k}$ is some random matrices with small spectral norm, $\mathbf{G} = \begin{bmatrix} \mathbf{g}^\top & \mathbf{0}^\top \\ \mathbf{0}^\top & \mathbf{g}^\top \end{bmatrix} \in \mathcal{R}_q^{2 \times 2k}$ is the public gadget matrix, and $\mathbf{A} = \begin{bmatrix} \mathbf{a}^\top \\ \mathbf{a}^\top \mathbf{s} + \mathbf{e}^\top \end{bmatrix} \in \mathcal{R}_q^{2 \times 2k}$ is the public key of the GSW-style encryption scheme, consisting of RLWE samples: a uniformly random vector $\mathbf{a} \in \mathcal{R}_q^{2k}$, a secret $s \in \mathcal{R}_q$ and an error vector $\mathbf{e} \in \mathcal{R}^{2k}$. Similar to Eq. (4), the function $F(\text{id})$ can be homomorphically computed by the public parameters in Eq. (7) as

$$F(\text{id}) = \mathbf{A}\mathbf{R}_{\text{id}} + H(\text{id})\mathbf{G} \in \mathcal{R}_q^{2 \times 2k}. \quad (8)$$

Note that the function $F(\text{id})$ in Eq. (4) is a BGG+-style *encoding* of the partition function $H(\text{id})$ and cannot be decrypted. In contrast, the function $F(\text{id})$ in Eq. (8) is a GSW-style *encryption* of the partition function $H(\text{id})$ and is decryptable.

Homomorphic transformation from GSW-style encryption to BGG+-style encoding. To generate the sampling vector similar to Eq. (5), we first sample a uniform vector $\mathbf{b} \in \mathcal{R}_q^k$, and prepare an additional public vector in the “BGG+-style encoding” form [9] (similar to Eq. (3)) as

$$\mathbf{c}_s^\top = \mathbf{b}^\top \mathbf{R}_s + s \cdot \mathbf{g}^\top,$$

where $\mathbf{R}_s \in \mathcal{R}^{k \times k}$ is a random matrix with small spectral norm and $s \in \mathcal{R}_q$ is the secret key of the GSW-style homomorphic encryption scheme. Then, we use the decryptable property of the GSW-style ciphertext and do a homomorphic “incomplete decryption” of the GSW-style ciphertext $F(\text{id})$ in Eq. (8) in two steps: (1) write the last k columns of the matrix $\mathbf{R}_{\text{id}}$ as $\hat{\mathbf{R}}_{\text{id}} \in \mathcal{R}^{2k \times k}$, and take the last k columns of the ciphertext $F(\text{id})$ as $\begin{bmatrix} \hat{\mathbf{c}}_0^\top \\ \hat{\mathbf{c}}_1^\top \end{bmatrix} = \begin{bmatrix} \mathbf{a}^\top \hat{\mathbf{R}}_{\text{id}} \\ (\mathbf{a}^\top \mathbf{s} + \mathbf{e}^\top) \hat{\mathbf{R}}_{\text{id}} + H(\text{id})\mathbf{g}^\top \end{bmatrix}$; (2) compute the following equation

$$\begin{aligned} \widehat{F(\text{id})}^\top &= \hat{\mathbf{c}}_1^\top - \mathbf{c}_s^\top \cdot \mathbf{g}^{-1}(\hat{\mathbf{c}}_0^\top) = \hat{\mathbf{c}}_1^\top - (\mathbf{b}^\top \mathbf{R}_s + s \cdot \mathbf{g}^\top) \cdot \mathbf{g}^{-1}(\hat{\mathbf{c}}_0^\top) \\ &= \mathbf{b}^\top \cdot -\mathbf{R}_s \cdot \mathbf{g}^{-1}(\hat{\mathbf{c}}_0^\top) + (\hat{\mathbf{c}}_1^\top - s \cdot \hat{\mathbf{c}}_0^\top) \\ &= \mathbf{b}^\top \cdot \underbrace{-\mathbf{R}_s \cdot \mathbf{g}^{-1}(\hat{\mathbf{c}}_0^\top)}_{\mathbf{R}} + H(\text{id})\mathbf{g}^\top + \underbrace{\mathbf{e}^\top \hat{\mathbf{R}}_{\text{id}}}_{\mathbf{e}_{\text{id}}^\top} \quad (9) \\ &= \mathbf{b}^\top \mathbf{R} + H(\text{id})\mathbf{g}^\top + \mathbf{e}_{\text{id}}^\top. \end{aligned}$$

Note that the real GSW-style decryption first computes $\hat{\mathbf{c}}_1 - s \cdot \hat{\mathbf{c}}_0$ and then do a rounding for the final element to recover the message. Eq. (9) actually does a homomorphic computation of the first step (This is where our “incomplete decryption” comes from). It homomorphically transforms a GSW-style ciphertext (Eq. (8)) into a BGG+-style encoding (Eq. (9)) of the same message $H(\text{id})$.

Now we get the sampling vector $[\mathbf{b}^\top | \widehat{F(\text{id})}^\top]$ for the identity id as follows

$$\mathbf{pk}_{\text{id}}^\top = [\mathbf{b}^\top | \mathbf{b}^\top \mathbf{R} + H(\text{id})\mathbf{g}^\top + \mathbf{e}_{\text{id}}^\top] \in \mathcal{R}_q^{2k}. \quad (10)$$

Note that $\mathbf{e}_{\text{id}}$ has a large norm (i.e., *linear* in $\mathbf{R}_{\text{id}}$'s norm), while $\mathbf{R}$ is a fresh matrix and thus has a small norm. Compared to Eq. (5), our new sampling vector in Eq. (10) has two trapdoors with distinct norms (i.e., small $\mathbf{R}$ and large $\mathbf{e}_{\text{id}}$), which allows us to sample a (D_r, D_σ) -hybrid secret key in the next step.

(1.2) *Sampling algorithm with (D_r, D_σ) -hybrid outputs.* Now we can use our new vector $\mathbf{pk}_{\text{id}}$ in Eq. (10) to sample the secret key. Generally speaking, for a given uniform $u \in \mathcal{R}_q$, we design a new sampling algorithm, to sample a small-norm (i.e., D_σ) vector $\mathbf{x} \in \mathcal{R}^{2k}$ and a large-norm (i.e., D_r , where $r \gg \sigma$) error $w \in \mathcal{R}$ such that $\mathbf{pk}_{\text{id}}^\top \cdot \mathbf{x} = u + w$.

In more detail, we first sample a perturbation $\mathbf{p} \in \mathcal{R}^{2k}$ that masks the trapdoor matrix $\mathbf{R}$ by Gaussian convolution. Using the trapdoor $\mathbf{T}_{\mathbf{g}}$, we obtain a short vector $\tilde{\mathbf{x}}$ such that $\mathbf{g}^\top \cdot \tilde{\mathbf{x}} = H(\text{id})^{-1} \cdot (u - \mathbf{pk}_{\text{id}}^\top \cdot \mathbf{p})$ under the condition that $H(\text{id}) \neq 0$ and $H(\text{id})$ is invertible. Then, we get a short vector $\mathbf{x}$ satisfying

$$\underbrace{[\mathbf{b}^\top | \mathbf{b}^\top \mathbf{R} + H(\text{id})\mathbf{g}^\top + \mathbf{e}_{\text{id}}^\top]}_{\mathbf{pk}_{\text{id}}^\top} \cdot \underbrace{\left(\mathbf{p} + \begin{bmatrix} -\mathbf{R} \\ \mathbf{I} \end{bmatrix} \cdot \tilde{\mathbf{x}} \right)}_{:=\mathbf{x}} = u + \underbrace{\mathbf{e}_{\text{id}}^\top \cdot \tilde{\mathbf{x}}}_{\text{error}}.$$

Obviously, the error would leak some information about the trapdoor $\mathbf{e}_{\text{id}}$. To solve this problem, a trivial idea is to add a perturbation that masks the trapdoor $\mathbf{e}_{\text{id}}$, just like the perturbation $\mathbf{p}$ masking the trapdoor $\mathbf{R}$. Concretely, we first sample a perturbation $h \in \mathcal{R}$ and compute $u' = u + h$, then follow the same steps as above except for replacing u with u' , we get

$$\underbrace{[\mathbf{b}^\top | \mathbf{b}^\top \mathbf{R} + H(\text{id})\mathbf{g}^\top + \mathbf{e}_{\text{id}}^\top]}_{\mathbf{pk}_{\text{id}}^\top} \cdot \underbrace{\left(\mathbf{p} + \begin{bmatrix} -\mathbf{R} \\ \mathbf{I} \end{bmatrix} \cdot \tilde{\mathbf{x}} \right)}_{\mathbf{x}} = u + \underbrace{(h + \mathbf{e}_{\text{id}}^\top \cdot \tilde{\mathbf{x}})}_{\text{error}}.$$

It is true that the error would not leak any information about $\mathbf{e}_{\text{id}}$, just as $\mathbf{x}$ does not reveal $\mathbf{R}$. However, both $\mathbf{x}$ and the error use the same source of randomness, i.e., $\tilde{\mathbf{x}}$, which results in that $\mathbf{x}$ and the error both have the same Gaussian width. This violates our design goal that $\mathbf{x}$ and the error have different widths. Therefore, we introduce a new source of randomness to deal with this issue. Specifically, we additionally sample a perturbation $\bar{\mathbf{p}} \in \mathcal{R}^k$ and compute $u^* =$

$u' + \mathbf{e}_{\text{id}}^\top \cdot \bar{\mathbf{p}}$, then follow the previous steps except for replacing u' with u^* , we get

$$\underbrace{[\mathbf{b}^\top | \mathbf{b}^\top \mathbf{R} + H(\text{id})\mathbf{g}^\top + \mathbf{e}_{\text{id}}^\top]}_{\mathbf{pk}_{\text{id}}^\top} \cdot \underbrace{\left(\mathbf{p} + \begin{bmatrix} -\mathbf{R} \\ \mathbf{I} \end{bmatrix} \cdot \tilde{\mathbf{x}} \right)}_{\mathbf{x}, \text{ small}} = u + \underbrace{(h + \mathbf{e}_{\text{id}}^\top \cdot (\bar{\mathbf{p}} + \tilde{\mathbf{x}}))}_{:=w, \text{ large error}}.$$

Then $\begin{bmatrix} \tilde{\mathbf{x}} \\ \mathbf{x} \end{bmatrix} = \begin{bmatrix} \bar{\mathbf{p}} \\ \mathbf{p} \end{bmatrix} + \begin{bmatrix} \mathbf{I} \\ \mathbf{T} \end{bmatrix} \tilde{\mathbf{x}}$ where $\mathbf{T} = \begin{bmatrix} -\mathbf{R} \\ \mathbf{I} \end{bmatrix}$. Now both $\mathbf{x}$ and $\bar{\mathbf{x}}$ use the same source of randomness $\tilde{\mathbf{x}}$, allowing us to simultaneously guarantee they have the same width and are independent. Consequently, the source of randomness $\bar{\mathbf{x}}$ in the error w is unrelated to $\mathbf{x}$, and we conclude that the two outputs ($w \in \mathcal{R}, \mathbf{x} \in \mathcal{R}^{2k}$) are independent and have different Gaussian widths.

We combine the large w and the small $\mathbf{x}$ into a (D_r, D_σ) -hybrid secret key.

(2) *Obtaining a $(D_{\sigma_0}, D_{\sigma_1})$ -hybrid error.* To achieve this, we pick an IBE ciphertext with $(D_{\sigma_0}, D_{\sigma_1})$ -hybrid errors. To simulate such a ciphertext in the security reduction, we run the re-randomization algorithm to generate partial ciphertext (i.e., the D_{σ_1} part) and incorporate some tricks.

(2.1) *IBE ciphertext with $(D_{\sigma_0}, D_{\sigma_1})$ -hybrid errors.* Our IBE scheme outputs ciphertext $(c_0, \mathbf{c}_1)$ where $\mathbf{c}_1$ has $(D_{\sigma_0}, D_{\sigma_1})$ -hybrid errors. We have that $\sigma_1 \gg \sigma_0$ which corresponds to the (D_r, D_σ) -hybrid secret key and follows our idea of “cross-multiplication”.

$$c_0 = u \cdot v + e_0 + \left\lceil \frac{q}{2} \right\rceil \cdot m \in \mathcal{R}_q, \quad \mathbf{c}_1 = d \cdot \begin{bmatrix} 1 \\ \mathbf{pk}_{\text{id}} \end{bmatrix} \cdot v + \begin{bmatrix} e_1 \\ \mathbf{e}_2 \end{bmatrix} \in \mathcal{R}_q^{2k+1},$$

where $e_1 \leftarrow D_{\mathcal{R}, \sigma_0}$ is a small error and $\mathbf{e}_2 \leftarrow D_{\mathcal{R}^{2k}, \sigma_1}$ is a large error. Note that we add an invertible element $d \in \mathcal{R}_q$ to align with the corresponding term in the security reduction in the next step. Similarly, we modify the input of the secret key sampling algorithm (in (1.2)) from u to $d^{-1}u$, thus we have $\mathbf{pk}_{\text{id}}^\top \cdot \mathbf{x} = d^{-1}u + w$.

(2.2) *Partial re-randomization.* In the security reduction, we use the fact that $H(\text{id}^*) = 0$ and $\mathbf{pk}_{\text{id}^*}^\top = [\mathbf{b}^\top | \mathbf{b}^\top \mathbf{R}^* + \mathbf{e}_{\text{id}^*}^\top]$ (defined in Eq. (10)). To simulate the challenge ciphertext $(\mathbf{c}_1^*)^\top = d \cdot [1 | \mathbf{b}^\top | \mathbf{b}^\top \mathbf{R}^* + \mathbf{e}_{\text{id}^*}^\top] \cdot v + [e_1 | \mathbf{e}_2^\top]$, a direct approach is to run the re-randomization algorithm as follows:

$$\begin{aligned} (\mathbf{c}_1^*) &= \text{ReRand} \left(\begin{bmatrix} \mathbf{I}_{k+1} & \begin{bmatrix} \mathbf{R}_{\text{id}^*}^* \\ \mathbf{e}_{\text{id}^*}^\top \end{bmatrix} \end{bmatrix}, d \cdot \begin{bmatrix} 1 \\ \mathbf{b} \end{bmatrix} \cdot v + \mathbf{e}'_0, \sigma_0, \frac{\sigma_1}{2\sigma_0} \right) \\ &= d \cdot [1 | \mathbf{b}^\top] \cdot \begin{bmatrix} \mathbf{I}_{k+1} & \begin{bmatrix} \mathbf{e}_{\text{id}^*}^\top \\ \mathbf{R}^* \end{bmatrix} \end{bmatrix} \cdot v + \mathbf{e}' \\ &= d \cdot [1 | \mathbf{b}^\top | \mathbf{b}^\top \mathbf{R}^* + \mathbf{e}_{\text{id}^*}^\top] \cdot v + \mathbf{e}' \end{aligned}$$

where the distribution $\mathbf{e}'$ is statistically close to the discrete Gaussian $D_{\mathcal{R}^{2k+1}, \sigma_1}$. However, this does not match our desired ciphertext distribution since each component of the $2k+1$ vector $\mathbf{e}'$ follows the same Gaussian distribution.

To solve this problem, we first use the re-randomization algorithm to generate a partial ciphertext:

$$\begin{aligned} (\tilde{\mathbf{c}}_1^*)^\top &= \text{ReRand} \left(\begin{bmatrix} \mathbf{I}_k & \mathbf{R}^* \\ \mathbf{0}^\top & \mathbf{e}_{\text{id}^*}^\top \end{bmatrix}, d \cdot \begin{bmatrix} \mathbf{b} \\ 1 \end{bmatrix} \cdot v + \mathbf{e}'_0, \sigma_0, \frac{\sigma_1}{2\sigma_0} \right) \\ &= d \cdot [\mathbf{b}^\top | 1] \cdot \begin{bmatrix} \mathbf{I}_k & \mathbf{R}^* \\ \mathbf{0}^\top & \mathbf{e}_{\text{id}^*}^\top \end{bmatrix} \cdot v + (\mathbf{e}'_0)^\top \\ &= d \cdot [\mathbf{b}^\top | \mathbf{b}^\top \mathbf{R}^* + \mathbf{e}_{\text{id}^*}^\top] \cdot v + (\mathbf{e}'_0)^\top. \end{aligned}$$

Then, we concatenate the last element of $d \cdot \begin{bmatrix} \mathbf{b} \\ 1 \end{bmatrix} \cdot v + \mathbf{e}'_0$ ⁹ and the partial ciphertext $\tilde{\mathbf{c}}_1^*$, to generate the ciphertext $\mathbf{c}_1^*$ as follows:

$$(\mathbf{c}_1^*)^\top = d \cdot [1 | \mathbf{b}^\top | \mathbf{b}^\top \mathbf{R}^* + \mathbf{e}_{\text{id}^*}^\top] \cdot v + [e'_1 | \mathbf{e}'_2],$$

where $e'_1 \leftarrow D_{\mathcal{R}, \sigma_0}$ is the last element of the small error $\mathbf{e}'_0$ and the distribution of $\mathbf{e}'_2$ is statistically close to the discrete Gaussian distribution $D_{\mathcal{R}^{2k}, \sigma_1}$. For the re-randomization algorithm, it requires $\frac{\sigma_1}{2\sigma_0} \geq \left\| \begin{bmatrix} \mathbf{I}_k & \mathbf{R}^* \\ \mathbf{0}^\top & \mathbf{e}_{\text{id}^*}^\top \end{bmatrix} \right\|$, thus the error $\mathbf{e}_2 \sim D_{\mathcal{R}^{2k}, \sigma_1}$ is related to the vector $\mathbf{e}_{\text{id}^*}$ and has a large norm.

We obtain a $(D_{\sigma_0}, D_{\sigma_1})$ -hybrid error, consisting of small e_1 and large $\mathbf{e}_2$.

Realizing our idea. During the decryption step, the user can compute

$$c_0 - \mathbf{c}_1^\top \cdot \begin{bmatrix} -w \\ \mathbf{x} \end{bmatrix} = \left\lfloor \frac{q}{2} \right\rfloor \cdot m + (e_0 + \underbrace{e_1 \cdot w}_{\text{small} \times \text{large}} - \underbrace{\mathbf{e}_2^\top \cdot \mathbf{x}}_{\text{large} \times \text{small}}). \quad (11)$$

Compared to the error term in Eq. (6), our error term in Eq. (11) has a cross-multiplication characteristic, i.e., “small $\times$ large + large $\times$ small”, thus removing the quadratic restriction of the modulus q on the trapdoor norm.

2 Preliminaries

Notations. We denote $\mathbb{Z}, \mathbb{N}$ and $\mathbb{R}$ as the set of integers, the set of natural numbers and the set of real numbers, respectively. We use bold uppercase letters (e.g., $\mathbf{A}$) to denote matrices, and bold lowercase letters (e.g., $\mathbf{a}$) for column vectors. We use $\|\mathbf{a}\|$ to denote the Euclidean norm of vector $\mathbf{a}$ and define $\|\mathbf{A}\| := \sup_{\|\mathbf{x}\|=1} \|\mathbf{A}\mathbf{x}\|$. We denote the horizontal concatenation of two vectors $\mathbf{a}, \mathbf{b}$ by $[\mathbf{a}^\top | \mathbf{b}^\top]$. We use $\tilde{\mathbf{A}}$ to denote the Gram-Schmidt orthogonalization of $\mathbf{A}$. For a (quotient) polynomial ring $\mathcal{R}$ over $\mathbb{Z}$ and a set $S \subset \mathbb{Z}$, we denote $S_{\mathcal{R}} \subseteq \mathcal{R}$ as the set of elements in $\mathcal{R}$ with all coefficients in S . For a positive integer k , let $[k]$ be the set of integers $\{0, 1, \dots, k-1\}$ and $[-k, k] := \{-k, \dots, -1, 0, 1, \dots, k\}$. We say a function $f : \mathbb{N} \rightarrow \mathbb{R}$ is negligible if $\forall c > 0, \exists \lambda_0 \in \mathbb{N}, \forall \lambda > \lambda_0, f(\lambda) <$

⁹ Consider $d \cdot \begin{bmatrix} \mathbf{b} \\ 1 \end{bmatrix} \cdot v + \mathbf{e}'_0$ as RLWE samples. We add $d \in \mathcal{R}_q$ to mask the secret v in the extra “1” term. Thus, we can view the last element as $d \cdot v + e'_1$, which constitutes a single RLWE sample.

$1/\lambda^c$ and we use $\text{negl}(\lambda)$ to denote any negligible function. We use “PPT” to denote probabilistic polynomial time. For a distribution D , we use $d \leftarrow D$ to denote sampling d according to D and use $d \sim D$ to denote that d follows the distribution D . For a set S , we use $U(S)$ to denote the uniform distribution over S and use $s \xleftarrow{\$} S$ to denote $s \leftarrow U(S)$. For any probabilistic algorithm $\mathcal{A}$, we use $y \xleftarrow{\$} \mathcal{A}(x)$ as running $\mathcal{A}$ with fresh randomness on input x and assigning the output to y . For any two random variables X and Y with support Ω , denote their statistical distance as $\Delta(X, Y) = \frac{1}{2} \sum_{s \in \Omega} |\Pr[X = s] - \Pr[Y = s]|$.

2.1 Identity-Based Encryption (IBE)

Syntax. We recall the standard syntax of IBE [8, 34] in the following. An identity-based encryption scheme Π with identity space $\mathcal{ID}$ consists of four PPT algorithms (Setup , KeyGen , Enc , Dec) as follows.

- $\text{Setup}(1^\lambda)$: Given the security parameter λ , it outputs the master public key mpk and the master secret key msk .
- $\text{KeyGen}(\text{mpk}, \text{msk}, \text{id})$: Given (mpk, msk) and an identity $\text{id} \in \mathcal{ID}$, it outputs the secret key sk_{id} .
- $\text{Enc}(\text{mpk}, \text{id}, m)$: Given the master public key mpk , an identity $\text{id} \in \mathcal{ID}$, and a message m , it outputs a ciphertext ct .
- $\text{Dec}(\text{mpk}, \text{sk}_{\text{id}}, \text{ct})$: Given the master public key mpk , the secret key sk_{id} , and a ciphertext ct , it outputs a message m' or $\perp$.

Correctness. We say an IBE scheme Π is correct, if for all $\text{id} \in \mathcal{ID}$ and all m in the specified message space, it holds that $\Pr[\text{Dec}(\text{mpk}, \text{sk}_{\text{id}}, \text{ct}) \neq m] = \text{negl}(\lambda)$, where the probability is taken over the randomness used in $(\text{mpk}, \text{msk}) \xleftarrow{\$} \text{Setup}(1^\lambda)$, $\text{sk}_{\text{id}} \xleftarrow{\$} \text{KeyGen}(\text{mpk}, \text{msk}, \text{id})$ and $\text{ct} \xleftarrow{\$} \text{Enc}(\text{mpk}, \text{id}, m)$.

Adaptive anonymous security. We consider the adaptive anonymous security notion for IBE as in [39], which implies the adaptive (non-anonymous) security considered in [1]. This security is defined by the following game between a challenger and an adversary $\mathcal{A}$.

Setup: At the beginning of the game, the challenger runs $\text{Setup}(1^\lambda)$ to get (mpk, msk) and sends mpk to $\mathcal{A}$.

Phase 1: $\mathcal{A}$ may adaptively make key extraction queries. When $\mathcal{A}$ submits id , the challenger returns $\text{sk}_{\text{id}} \xleftarrow{\$} \text{KeyGen}(\text{mpk}, \text{msk}, \text{id})$.

Challenge: At some point, $\mathcal{A}$ outputs a message μ and an identity id^* , on which it wishes to be challenged. Then, the challenger picks a random bit $\text{coin} \xleftarrow{\$} \{0, 1\}$ and a random ciphertext $\text{ct}_1^* \leftarrow \mathcal{C}$ from the ciphertext space. If $\text{coin} = 0$, it runs $\text{ct}_0^* \xleftarrow{\$} \text{Enc}(\text{mpk}, \text{id}^*, \mu)$ and gives the challenge ciphertext ct_0^* to $\mathcal{A}$. If $\text{coin} = 1$, it gives ct_1^* to $\mathcal{A}$.

Phase 2: $\mathcal{A}$ continues to make key queries with a restriction that $\text{id} \neq \text{id}^*$.

Guess: Finally, $\mathcal{A}$ outputs a bit $\widehat{\text{coin}}$ for coin.

The advantage of $\mathcal{A}$ is defined as $|\Pr[\widehat{\text{coin}} = \text{coin}] - \frac{1}{2}|$. We say that the scheme satisfies adaptively-anonymous security if the advantage of any PPT $\mathcal{A}$ is negligible.

2.2 Lattices and Gaussians

Lattices. An n -dimensional (full rank) lattice $\Lambda \subseteq \mathbb{R}^n$ is the set of all integer linear combinations of some set of n linearly independent basis vectors $\mathbf{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\} \subseteq \mathbb{R}^n$, $\Lambda = \{\mathbf{B}\mathbf{x} \mid \mathbf{x} \in \mathbb{Z}^n\}$. We denote the dual lattice of Λ by $\Lambda^* := \{\mathbf{y} \in \text{span}(\Lambda) \mid \langle \mathbf{y}, \mathbf{x} \rangle \in \mathbb{Z}, \forall \mathbf{x} \in \Lambda\}$. For a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, define $\Lambda_q^\perp(\mathbf{A}) := \{\mathbf{e} \in \mathbb{Z}^m : \mathbf{A}\mathbf{e} = \mathbf{0} \bmod q\}$, which is a full-rank m -dimensional integer lattice. We omit q when it is clear from the context.

Gaussian distributions. The Gaussian function $\rho : \mathbb{R}^m \rightarrow (0, 1]$ is defined as $\rho(\mathbf{x}) = \exp(-\pi \cdot \langle \mathbf{x}, \mathbf{x} \rangle)$. Applying a linear transformation given by an invertible matrix $\mathbf{B}$ yields $\rho_{\mathbf{B}}(\mathbf{x}) = \rho(\mathbf{B}^{-1}\mathbf{x}) = \exp(-\pi \cdot \mathbf{x}^\top \Sigma^{-1} \mathbf{x})$, where $\Sigma = \mathbf{B}\mathbf{B}^\top$. Since $\rho_{\mathbf{B}}$ is exactly determined by Σ , we also write it as $\rho_{\sqrt{\Sigma}}$. For a lattice Λ and $\mathbf{c} \in \text{span}(\Lambda)$, the discrete Gaussian distribution $D_{\Lambda+\mathbf{c}, \sqrt{\Sigma}}$ is defined as: for any $\mathbf{x} \in \Lambda + \mathbf{c}$,

$$D_{\Lambda+\mathbf{c}, \sqrt{\Sigma}}(\mathbf{x}) = \frac{\rho_{\sqrt{\Sigma}}(\mathbf{x})}{\rho_{\sqrt{\Sigma}}(\Lambda + \mathbf{c})}.$$

When $\Sigma = \sigma^2 \mathbf{I}$ for some real $\sigma > 0$, we write $\rho_{\sqrt{\Sigma}}$, $D_{\Lambda+\mathbf{c}, \sqrt{\Sigma}}$ as ρ_σ , $D_{\Lambda+\mathbf{c}, \sigma}$.

We recall the definition of smoothing parameter of lattices as follows.

Definition 1 ([30], smoothing parameter). For any $\epsilon > 0$, any n -dimensional lattice Λ , the smoothing parameter $\eta_\epsilon(\Lambda)$ is the smallest real $s > 0$ such that $\rho_{1/s}(\Lambda^* \setminus \{\mathbf{0}\}) \leq \epsilon$.

For the discrete Gaussian over lattices, we have the following tail bounds.

Lemma 1 ([25]). For $\sigma > \eta_\epsilon(\mathbb{Z}^n)$, $t \geq 0$, $\mathbf{x} \in \mathbb{Z}^n$, $\Pr_{\mathbf{e} \leftarrow D_{\mathbb{Z}^n, \sigma}}[|\mathbf{e}^\top \mathbf{x}| \geq t] \leq 2e^{-\pi \cdot \frac{t^2}{\|\mathbf{x}\|^2 \sigma^2}}$.

Lemma 2 ([30]). For n -dimensional lattice Λ , $\epsilon \in (0, \frac{1}{2})$, $s \geq \eta_\epsilon(\Lambda)$, we have $\Pr_{\mathbf{x} \leftarrow D_{\Lambda, s}}[\|\mathbf{x}\| > s\sqrt{n}] \leq 2^{-n}$.

The following lemma can be used to re-establish the discrete Gaussian distribution after multiplying with a matrix.

Lemma 3 ([25], noise re-randomization). Let q, l, m be positive integers and σ_1 be a positive real satisfying $\sigma_1 > \max\{\eta_\epsilon(\mathbb{Z}^m), \eta_\epsilon(\mathbb{Z}^l)\}$. Let $\mathbf{b} \in \mathbb{Z}_q^m$ be arbitrary and $\mathbf{x}$ chosen from $D_{\mathbb{Z}^m, \sigma_1}$. Then for any $\mathbf{V} \in \mathbb{Z}^{m \times l}$ and positive real $\sigma_2 > \|\mathbf{V}\|$, there exists a PPT algorithm $\text{ReRand}(\mathbf{V}, \mathbf{b} + \mathbf{x}, \sigma_1, \sigma_2)$ that outputs $\mathbf{b}'^\top = \mathbf{b}^\top \mathbf{V} + \mathbf{x}'^\top \in \mathbb{Z}_q^l$ where the statistical distance of the discrete Gaussian $D_{\mathbb{Z}^l, 2\sigma_1\sigma_2}$ and the distribution of $\mathbf{x}'$ is within 8ϵ .

2.3 Rings and Ideal Lattices

Rings. In this paper, we consider the polynomial ring $\mathcal{R} := \mathbb{Z}[x]/(x^n + 1)$ where n (the degree of $\mathcal{R}$) is a power of 2. Any element $a \in \mathcal{R}$ can be denoted as $a = \sum_{i=0}^{n-1} a_i x^i$, where $a_i \in \mathbb{Z}$. For any prime integer q , we denote $\mathcal{R}_q$ as $\mathcal{R}/q\mathcal{R} = \mathbb{Z}_q[x]/(x^n + 1)$ and $\mathcal{R}_q^\times$ as the set of invertible elements of $\mathcal{R}_q$.

Coefficient Embedding. We define a coefficient map $\phi : \mathcal{R} \rightarrow \mathbb{Z}^n$ that sends a ring element $a = \sum_{i=0}^{n-1} a_i x^i \in \mathcal{R}$ to its coefficient vector $\mathbf{a} = [a_0 | \cdots | a_{n-1}]^\top \in \mathbb{Z}^n$. Furthermore, we can define another map $\text{Rot} : \mathcal{R} \rightarrow \mathbb{Z}^{n \times n}$ that sends $a \in \mathcal{R}$ to a matrix in $\mathbb{Z}^{n \times n}$ such that the i -th row is $\phi(a \cdot x^{i-1} \bmod (x^n + 1))^\top \in \mathbb{Z}^n$. We can extend the maps ϕ and Rot to ring vectors and matrices. The norms of ring matrices (or vectors) are defined by their corresponding coefficient embedding matrices (or vectors), i.e., for any $\mathbf{A} \in \mathcal{R}^{s \times t}$, $\|\mathbf{A}\| := \|\text{Rot}(\mathbf{A})\|$. The Gram-Schmidt orthogonalization of ring matrices is defined analogously, i.e., for any $\mathbf{A} \in \mathcal{R}^{s \times t}$, $\tilde{\mathbf{A}} := \widetilde{\text{Rot}(\mathbf{A})}$.

Discrete Gaussian over rings. For $\Sigma = \mathbf{B}\mathbf{B}^\top$ where $\mathbf{B}$ is an invertible matrix over $\mathbb{R}$, we denote $D_{\mathcal{R}^k, \sqrt{\Sigma}}$ as the distribution of sampling $\mathbf{x} \leftarrow D_{\mathbb{Z}^{nk}, \sqrt{\Sigma}}$ and output $\phi^{-1}(\mathbf{x}) \in \mathcal{R}^k$. When Σ is a matrix over $\mathbb{R}[x]/(x^n + 1)$, we use the notation $D_{\mathcal{R}^k, \sqrt{\Sigma}}$ to denote $D_{\mathcal{R}^k, \text{Rot}(\sqrt{\Sigma})}$, where Rot can be naturally extended to $\mathbb{R}[x]/(x^n + 1)$.

We extend Lemma 1 to the ring setting via the following lemma.

Corollary 1 (Corollary of Lemma 1). *For positive integer k , $\sigma > \eta_\epsilon(\mathbb{Z}^{nk})$, real number $t \geq 0$, $\mathbf{x} \in \mathcal{R}^k$, we have $\Pr_{\mathbf{e} \leftarrow D_{\mathcal{R}^k, \sigma}}[|\phi(\mathbf{e}^\top \mathbf{x})_j| \geq t] \leq 2e^{-\pi \cdot \frac{t^2}{\|\mathbf{x}\|^2 \sigma^2}}$.*

The proof of Corollary 1 is trivial and we omit it due to space limit.

Lemma 4 ([18], Fact 6). *For a positive integer ρ and a matrix $\mathbf{R}$ uniformly chosen from $[-\rho, \rho]_{\mathcal{R}}^{s \times t}$, there exists a constant C ($1 < C < 2$) such that for any positive number ω , we have $\Pr[\|\mathbf{R}\| \geq C\rho\sqrt{n}(\sqrt{s} + \sqrt{t} + \omega)] \leq 2e^{-\pi\omega^2}$.*

Lemma 5 ([25], Regularity lemma). *Let n be a power of two, $q \equiv 3 \bmod 8$ be a prime and l, k, ρ be positive integers that $\rho < \frac{1}{2}\sqrt{q/n}$. For $\mathbf{A} \xleftarrow{\$} \mathcal{R}_q^{k' \times k}$ and $\mathbf{R} \xleftarrow{\$} [-\rho, \rho]_{\mathcal{R}}^{k \times l}$, we have $\Delta((\mathbf{A}, \mathbf{A}\mathbf{R}), (\mathbf{A}, U(\mathcal{R}_q^{k' \times l}))) \leq \frac{l}{2} \cdot \left(\frac{q^{k'}}{(2\rho+1)^k}\right)^{n/2}$.*

Ring learning with errors. The Learning With Errors (LWE) problem was introduced by Regev [33] and we will use the ring version of it, namely RLWE [28].

Definition 2 ([28], RLWE). *For positive integers $n = n(\lambda)$, $k = k(n)$, a prime integer $q = q(n) > 2$, an error distribution $\chi = \chi(n)$ over $\mathcal{R}$, and an algorithm $\mathcal{B}$, consider the experiment where a secret bit $\text{coin} \xleftarrow{\$} \{0, 1\}$ is chosen and then $\mathcal{B}(\{(u_i, y_i)\}_{i=1}^k)$ is called when $\text{coin} = 0$, $\mathcal{B}(\{(u_i, u_i \cdot v + e_i)\}_{i=1}^k)$ is called*

when $\text{coin} = 1$ where $u_1, \dots, u_k, y_1, \dots, y_k, v \xleftarrow{\$} \mathcal{R}_q$ and $e_1, \dots, e_k \xleftarrow{\$} \chi$. Finally, $\mathcal{B}$ outputs a bit coin' and $\mathcal{B}$ wins the experiment if $\text{coin}' = \text{coin}$. The advantage of $\mathcal{B}$ is defined as $|\Pr[\text{coin}' = \text{coin}] - \frac{1}{2}|$. We say the $\text{RLWE}_{n,k,q,\chi}$ assumption holds if the advantage of any PPT $\mathcal{B}$ is negligible.

We note that the RLWE problem is at least as hard as some worst-case lattice problem. We refer to Lemma 23 for details.

2.4 Lattice Trapdoor Over Rings and Gaussian Sampling

We recall the public gadget and its trapdoor defined in [29] and related lemmas.

Definition 3 ([29]). For positive integers b and $k \geq k' = \lceil \log_b q \rceil$, the public gadget vector is defined as $\mathbf{g} := [1|b|b^2|\dots|b^{k'-1}|\mathbf{0}^\top]^\top \in \mathcal{R}_q^k$, and there is a publicly known matrix $\mathbf{T}_{\mathbf{g}}$ such that $\text{Rot}(\mathbf{T}_{\mathbf{g}}) \in \mathbb{Z}^{nk \times nk}$ is a basis of the lattice $\Lambda^\perp(\text{Rot}(\mathbf{g}^\top))$ and $\|\widetilde{\mathbf{T}_{\mathbf{g}}}\| \leq \sqrt{b^2 + 1}$.

Lemma 6 ([1]). For integers k, q, b satisfying the requirements of Def. 3, on input a vector $\mathbf{u} \in \mathcal{R}_q^k$, there exists a deterministic polynomial-time algorithm $\mathbf{g}^{-1}(\cdot)$ which outputs the matrix $\mathbf{R} = \mathbf{g}^{-1}(\mathbf{u}^\top)$ such that $\mathbf{R} \in [-b, b]_{\mathcal{R}}^{k \times k}$, $\mathbf{g}^\top \cdot \mathbf{R} = \mathbf{u}^\top$ and $\|\mathbf{R}\| \leq nkb$.

For the $\mathbf{G}$ matrix defined in [29], we have the following corollary of Lemma 6. We note that Lemma 6 and Corollary 2 can also hold in the $\mathbb{Z}_q$ setting, we refer to Appendix A for details.

Corollary 2. For the gadget matrix $\mathbf{G} = \begin{bmatrix} \mathbf{g}^\top & \mathbf{0}^\top \\ \mathbf{0}^\top & \mathbf{g}^\top \end{bmatrix} \in \mathcal{R}_q^{2 \times 2k}$, on input a matrix $\mathbf{U} \in \mathcal{R}_q^{2 \times 2k}$, there exists a deterministic polynomial-time algorithm $\mathbf{G}^{-1}(\cdot)$ which outputs the matrix $\mathbf{R} = \mathbf{G}^{-1}(\mathbf{U})$ such that $\mathbf{R} \in [-b, b]_{\mathcal{R}}^{2k \times 2k}$, $\mathbf{G} \cdot \mathbf{R} = \mathbf{U}$ and $\|\mathbf{R}\| \leq 2nkb$.

Next, we recall a general trapdoor generation method in the ring setting as defined in [1] together with some useful lemmas for the trapdoor pair.

Lemma 7 ([1]). For positive integers ρ, q such that $\rho < \frac{1}{2}\sqrt{q/n}$, $k \geq 2\log_\rho q$, there exists a polynomial time algorithm $\text{TrapGen}(n, k, \rho, b, q)$ that outputs a vector $\mathbf{b} \in \mathcal{R}_q^k$ and a matrix $\mathbf{T}_{\mathbf{b}} \in \mathcal{R}^{k \times k}$, where $\text{Rot}(\mathbf{b}^\top) \in \mathbb{Z}^{n \times nk}$ is a full-rank matrix and $\text{Rot}(\mathbf{T}_{\mathbf{b}}) \in \mathbb{Z}^{nk \times nk}$ is a basis of $\Lambda^\perp(\text{Rot}(\mathbf{b}^\top))$, such that $\mathbf{b}$ is statistically close to $U(\mathcal{R}_q^k)$ and $\text{Rot}(\mathbf{b}^\top) \in \mathbb{Z}_q^{n \times nk}$ has full (column) rank, and $\|\widetilde{\mathbf{T}_{\mathbf{b}}}\| \leq O(b\rho\sqrt{n\log_\rho q})$.

Lemma 8 ([21], adapted to the ring setting). For a trapdoor pair $(\mathbf{b}, \mathbf{T}_{\mathbf{b}}) \in \mathcal{R}_q^k \times \mathcal{R}^{k \times k}$ satisfying $\mathbf{b}^\top \cdot \mathbf{T}_{\mathbf{b}} = \mathbf{0}^\top \bmod q$ and $\text{Rot}(\mathbf{b}^\top) \in \mathbb{Z}^{n \times nk}$ is a full-rank matrix, $\sigma \geq \eta_\epsilon(\mathbb{Z}^{nk}) \cdot \|\widetilde{\mathbf{T}_{\mathbf{b}}}\|$ and a given target $u \xleftarrow{\$} \mathcal{R}_q$, there exists a preimage sampling algorithm $\text{SamPre}(\mathbf{b}, \mathbf{T}_{\mathbf{b}}, u, \sigma)$ that outputs a short vector $\mathbf{x}$ such that $\mathbf{b}^\top \cdot \mathbf{x} = u \bmod q$ and the distribution of $\mathbf{x}$ is statistically close to $D_{\mathcal{R}^k, \sigma}$.

Lemma 9 ([21], adapted to the ring setting). *For a trapdoor pair $(\mathbf{b}, \mathbf{T}_\mathbf{b}) \in \mathcal{R}_q^k \times \mathcal{R}^{k \times k}$ satisfying $\mathbf{b}^\top \cdot \mathbf{T}_\mathbf{b} = \mathbf{0}^\top \bmod q$ and $\text{Rot}(\mathbf{b}^\top) \in \mathbb{Z}^{n \times nk}$ is a full-rank matrix, and let $\epsilon \in (0, \frac{1}{2})$ and $\sigma \geq \eta_\epsilon(\mathbb{Z}^{nk}) \cdot \|\widetilde{\mathbf{T}_\mathbf{b}}\|$. Then, for $\mathbf{x} \leftarrow D_{\mathcal{R}^k, \sigma}$, the distribution of the syndrome $u = \mathbf{b}^\top \cdot \mathbf{x} \bmod q$ is statistically close to uniform distribution over $\mathcal{R}_q$.*

Finally, we recall the following lemma about perturbation of non-spherical Gaussians, which originates from [32].

Lemma 10 ([19, 29, 40], adapted to the ring setting). *For $\mathbf{T} \in \mathcal{R}^{m \times k}$, $\sigma^2 \geq (s^2 + \eta_\epsilon^2(\mathbb{Z}^{nk})) \cdot (\|\mathbf{T}\|^2 + 1)$, $\Sigma_p = \sigma^2 \mathbf{I}_m - s^2 \mathbf{T} \mathbf{T}^\top$, the following two distributions are statistically close:*

- sample $\mathbf{z} \leftarrow D_{\mathcal{R}^k, s}$, $\mathbf{p} \leftarrow D_{\mathcal{R}^m, \sqrt{\Sigma_p}}$, output $\mathbf{p} + \mathbf{T}\mathbf{z}$;
- output $\mathbf{x} \leftarrow D_{\mathcal{R}^m, \sigma}$.

2.5 Homomorphic Encryption and Homomorphic Computation

Here, we describe the GSW-style [22] homomorphic encryption scheme over ring elements and δ -expanding homomorphic computation.

The Ring-GSW homomorphic encryption scheme. We first recall the Ring-GSW homomorphic encryption scheme in [22], which consists of six algorithms (Setup, KeyGen, Enc, Dec, Add, Mul).

- Setup(1^λ): on input the security parameter λ , do:
 - set the ring dimension $n := n(\lambda)$ which is a power of 2, the modulus $q := q(n)$, the base $b := b(n)$, the positive integer $\rho := \rho(n)$, $k := \lceil \log_b q \rceil$, and the noise distribution χ over $\mathcal{R}$.
 - output the public parameter $pp := (n, q, b, \rho, k, \chi)$. We assume that pp is an input to all the following algorithms.
- KeyGen(pp): on inputs the public parameter pp , do:
 - pick $\mathbf{a} \xleftarrow{\$} \mathcal{R}_q^{2k}$, $s \xleftarrow{\$} \mathcal{R}_q$, $\mathbf{e} \leftarrow \chi^{2k}$.
 - set the encryption key $\mathbf{ek} := \mathbf{A} = \begin{bmatrix} \mathbf{g}^\top & \mathbf{0}^\top \\ \mathbf{a}^\top & s + \mathbf{e}^\top \end{bmatrix}$, the decryption key $\mathbf{dk} := s$.
- Enc($\mathbf{ek}, m$): on input encryption key $\mathbf{ek} := \mathbf{A}$, a message $m \in \{0, 1\}_{\mathcal{R}}$, do:
 - sample a random matrix $\mathbf{R} \xleftarrow{\$} [-\rho, \rho]_{\mathcal{R}}^{2k \times 2k}$.
 - set the public gadget matrix $\mathbf{G} = \begin{bmatrix} \mathbf{g}^\top & \mathbf{0}^\top \\ \mathbf{0}^\top & \mathbf{g}^\top \end{bmatrix} \in \mathcal{R}_q^{2 \times 2k}$.
 - output the ciphertext $\mathbf{C} = \mathbf{A}\mathbf{R} + m\mathbf{G} \in \mathcal{R}_q^{2 \times 2k}$.
- Dec($\mathbf{dk}, \mathbf{C}$): on input the decryption key $\mathbf{dk}$ and a ciphertext $\mathbf{C}$, takes the last column of ciphertext and split it into two elements (c_0, c_1) , do:
 - compute $\lfloor \frac{1}{b^{k-1}} \cdot (c_1 - c_0 \cdot s) \rfloor$.

– $\text{Add}(\mathbf{C}_1, \mathbf{C}_2)$: on input two ciphertexts $\mathbf{C}_1, \mathbf{C}_2$, output:

$$\mathbf{C}_{\text{Add}} = \mathbf{C}_1 + \mathbf{C}_2 = \mathbf{A}(\mathbf{R}_1 + \mathbf{R}_2) + (m_1 + m_2)\mathbf{G}.$$

– $\text{Mul}(\mathbf{C}_1, \mathbf{C}_2)$: on input two ciphertexts $\mathbf{C}_1, \mathbf{C}_2$, output:

$$\begin{aligned} \mathbf{C}_{\text{Mul}} &= \mathbf{C}_1 \cdot \mathbf{G}^{-1}(\mathbf{C}_2) = (\mathbf{A}\mathbf{R}_1 + m_1\mathbf{G}) \cdot \mathbf{G}^{-1}(\mathbf{A}\mathbf{R}_2 + m_2\mathbf{G}) \\ &= \mathbf{A}(\mathbf{R}_1 \cdot \mathbf{G}^{-1}(\mathbf{A}\mathbf{R}_2 + m_2\mathbf{G}) + \mathbf{R}_2 m_1) + (m_1 m_2)\mathbf{G}. \end{aligned}$$

The following definition expresses the quality of the homomorphic evaluation by the factor δ , which is useful in the noise analysis of the IBE scheme.

Definition 4 ([39], δ -expanding evaluation). *Let k be a natural number. We say that the deterministic algorithms $(\text{PubEval}, \text{TrapEval})$ as in are δ -expanding for a function family $\mathcal{F} = \{f : \mathcal{X}^t \rightarrow \mathcal{Y}\}$ where $\mathcal{X}, \mathcal{Y} \subset \mathcal{R}$ if they are efficient and satisfy the following properties:*

- $\text{PubEval}(\{\mathbf{C}_i \in \mathcal{R}_q^{2 \times 2k}\}_{i \in [t]}, f \in \mathcal{F}) = \mathbf{C}_f \in \mathcal{R}_q^{2 \times 2k}$
- $\text{TrapEval}(\mathbf{A} \in \mathcal{R}_q^{2 \times 2k}, \{\mathbf{R}_i \in \mathcal{R}^{2k \times 2k}\}_{i \in [t]}, \{z_i\}_{i \in [t]}, f \in \mathcal{F}) = \mathbf{R}_f \in \mathcal{R}^{2k \times 2k}$

for $\mathbf{z} = [z_1 | \dots | z_t]^\top \in \mathcal{X}^t$. We require the following holds:

$$\text{PubEval}(\{\mathbf{A}\mathbf{R}_i + z_i\mathbf{G}\}_{i \in [t]}, f) = \mathbf{A}\mathbf{R}_f + f(\mathbf{z})\mathbf{G},$$

and we have $\|\mathbf{R}_f\| \leq \delta \cdot \max_{i \in [t]} \{\|\mathbf{R}_i\|\}$.

3 Homomorphic Computation Outputting BGG+-style Encoding with Two Distinct-norm Trapdoors

In TCC 2021, Abla et al. [1] proposed a specific equality test function and a succinct partition function (using the equality test function as a sub-function), which can be homomorphically computed using $\omega(1)$ public vectors. In this section, we first homomorphically compute the equality test function and the partition function in the GSW-style encryption form (defined in Sect. 2.5). By the decryptable property of the GSW-style ciphertext, we design an incomplete decryption function to homomorphically transform the GSW-style encryption to a BGG+-style encoding (defined in Def. 9), which has two trapdoors with distinct norms.

3.1 Homomorphic Computation of Equality Test function in GSW-style Encryption Form

As the introduction mentions, this work focuses on the cyclotomic rings of power-of-two, which have simpler mathematical structures. Let $\mathcal{R} = \mathbb{Z}[x]/\Phi_m(x)$ be the m -th cyclotomic ring, modulus q be co-prime to m , and $\mathcal{R}_q = \mathcal{R}/q\mathcal{R}$. For this setting, we have $\Phi_m(x) = x^n + 1$ where $n = \varphi(m) = m/2$.

Definition 5 (Equality test function). Define function $\text{Equal}_\beta(\cdot)$ parameterized by $\beta \in [m]$ as follows: on input $x^\alpha \in \mathcal{R}$, the function $\text{Equal}_\beta(x^\alpha)$ outputs 1 if $\alpha \equiv \beta \pmod m$ and 0 otherwise.

In [1], the authors provide algorithms to homomorphically compute the equality test functions and analyze the expansion factor of their algorithms. We adapt their algorithms to our Ring-GSW homomorphic encryption setting and provide new algorithms ($\text{PubEval}_E, \text{TrapEval}_E$) in Construction 2. Due to space limit, we put our algorithms in Appendix B. We bound the expansion factor of our algorithms with the following theorem.

Theorem 1. The algorithms ($\text{PubEval}_E, \text{TrapEval}_E$) in Construction 2 are $2mn(kb)^2$ -expanding with respect to the function family $\{\text{Equal}_\beta\}_{\beta \in [m]}$.

The proof of this theorem is an adaptation of the proof in [1]. Due to space limit, we put the proof in Sect. B.1.

3.2 Homomorphic Computation of Partition function in GSW-style Encryption Form

In this subsection, we first recall the definition of the partition function and the partition function family¹⁰ in [1] that will be used in our IBE construction.

Definition 6 ([1], partition function). Let p, t, L, η, m, n be integers such that $tp \leq n$, $m = 2n$, $L + 1 \leq m^\eta$ ¹¹; $\text{ECC} : \mathcal{D} \rightarrow \mathbb{Z}_p^L$ be a function¹², whose image is indexed by $\{1, \dots, L\}$ and $\text{ECC}(z)[0] = 0$ for every $z \in \mathcal{D}$. For any $(\alpha, \beta) \in [L + 1]^t \times \mathbb{Z}_p^t$, the partition function $H_{\alpha, \beta}^{\mathcal{R}, t} : \mathcal{D} \rightarrow \mathcal{R}$ as $H_{\alpha, \beta}^{\mathcal{R}, t}(z) := \sum_{i \in [t]} (x^{ip + \text{ECC}(z)[\alpha_i]} - x^{ip + \beta_i})$.

Abla et al. [1] gave another form of the partition function for homomorphic computation as follows.

Definition 7 ([1]). Let p, t, L, η, m, n be integers such that $tp \leq n$, $m = 2n$, $L + 1 \leq m^\eta$ where numbers in $[L + 1]$ can be represented in m -ary; $\text{ECC} : \mathcal{D} \rightarrow \mathbb{Z}_p^L$ be a function whose image is indexed by $\{1, \dots, L\}$ and $\text{ECC}(z)[0] = 0$ for every $z \in \mathcal{D}$. For each $j \in [L + 1]$, denote j 's m -ary representation as $(j[0], \dots, j[\eta - 1])$. Let $\mathcal{R} := \mathbb{Z}[x]/(x^n + 1)$. For any $z \in \mathcal{D}$, on input $(\{\alpha_{i, i'}\}_{i \in [t], i' \in [\eta]} \in [m]^{t \times \eta}, \beta \in \mathcal{R}_q)$, the partition function G_z is defined as

$$G_z(\{\alpha_{i, i'}\}_{i \in [t], i' \in [\eta]}, \tilde{\beta}) := -\tilde{\beta} + \sum_{i \in [t], j \in [L + 1]} \left(\prod_{i' \in [\eta]} \text{Equal}_{j[i']} (x^{\alpha_{i, i'}}) \right) \cdot x^{ip + \text{ECC}(z)[j]}.$$

¹⁰ We provide some supplementary notes about the “partition” property in Sect. B.2.

¹¹ Note that m, η and the relationship $L + 1 \leq m^\eta$ are actually not used in this definition. We keep them in this definition because they will be used in another form of the partition function as in Def. 7.

¹² In [1], ECC is an error correcting code with relative distance γ . Here we consider a more general definition for simplicity.

Note that the function defined in Def. 6 and Def. 7 can be equal for certain parameters. More precisely, when $\{\alpha_{i,i'}\}_{i' \in [\eta]}$ is the m -ary representation of α_i and $\tilde{\beta} = \sum_{i \in [t]} x^{ip+\beta_i}$, we have that $H_{\alpha,\beta}^{\mathcal{R},t}(z) = G_z(\{\alpha_{i,i'}\}_{i \in [t], i' \in [\eta]}, \tilde{\beta})$. We put the full derivation process of this equation into Sect. B.3.

Definition 8 ([1], partition function family). For any $t' \in [t]$, the class $\mathcal{H}^{\mathcal{R},t,t'}$ is defined as

$$\mathcal{H}^{\mathcal{R},t,t'} = \left\{ H_{\alpha,\beta}^{\mathcal{R},t} : \alpha' \in ([L+1] \setminus \{0\})^{t'}, \beta' \in \mathbb{Z}_p^{t'}, \alpha^\top = (\alpha'^\top, \mathbf{0}^\top), \beta^\top = (\beta'^\top, \mathbf{0}^\top) \right\},$$

where $\mathbf{0}^\top = (0, \dots, 0) \in \mathbb{Z}_p^{t-t'}$, i.e., padding 0's to match the dimension t . Furthermore, define $\mathcal{H}^{\mathcal{R},t} = \cup_{t' \in [t]} \mathcal{H}^{\mathcal{R},t,t'}$.

In [1], the authors provide algorithms to homomorphically compute this partition function and analyze the expansion factor of their algorithms. We adapt their algorithms to our Ring-GSW homomorphic encryption setting and provide new algorithms ($\text{PubEval}_G, \text{TrapEval}_G$) in Construction 3. Due to space limit, we put our algorithms in Appendix B. We bound the expansion factor of our algorithms with the following theorem.

Theorem 2. For parameters as stated in Def. 6, and assuming the underlying algorithms ($\text{PubEval}_E, \text{TrapEval}_E$) are δ -expanding for $\{\text{Equal}_j\}_{j \in [m]}$, the algorithms ($\text{PubEval}_G, \text{TrapEval}_G$) in Construction 3 are $(L+1)t \cdot \eta \cdot 2nkb \cdot \delta$ -expanding with respect to the function family $\{G_z\}_{z \in \mathcal{D}}$.

The proof of this theorem is an adaptation of the proof in [1]. Due to space limit, we put the proof in Sect. B.4.

3.3 Homomorphic transformation from GSW-style encryption to BGG+-style encoding

In this subsection, we first recall the BGG+-style encoding in [9]. Then, we design an incomplete decryption function for a GSW-style ciphertext and its homomorphic computation algorithm. Specifically, this homomorphic computation algorithm takes in a GSW-style ciphertext for some message m and transforms it into a BGG+-style encoding of m with some additional errors (with the help of a BGG+-style encoding for the decryption key s).

First, we recall the BGG+-style encoding over ring elements.

Definition 9 ([9], BGG+-style encoding). Given a public vector $\mathbf{b} \in \mathcal{R}_q^k$ and a positive integer ρ , to encode $\alpha \in \mathcal{R}$, we choose a random small-norm matrix $\mathbf{R}_\alpha \xleftarrow{\$} [-\rho, \rho]_{\mathcal{R}}^{k \times k}$, and define the encoding as $\mathbf{c}^\top := \mathbf{b}^\top \cdot \mathbf{R}_\alpha + \alpha \cdot \mathbf{g}^\top$.

Then, we give our new definition of the incomplete decryption function.

Definition 10 (Incomplete decryption function). On input a GSW-style ciphertext (as in Sect. 2.5) $\mathbf{C} \in \mathcal{R}_q^{2 \times 2k}$ with $\begin{bmatrix} \mathbf{c}_0^\top \\ \mathbf{c}_1^\top \end{bmatrix}$ as the last k -columns and the corresponding decryption key $s \in \mathcal{R}_q$, the incomplete decryption function is defined as $\widetilde{\text{Dec}}(\mathbf{C}, s) := \mathbf{c}_1 - s \cdot \mathbf{c}_0$.

Next, we provide homomorphic computation algorithms for $\widetilde{\text{Dec}}$ and analyze the expansion bound.

Construction 1 On input a GSW-style ciphertext (as in Sect. 2.5) $\mathbf{C}_m = \begin{bmatrix} \mathbf{a}^\top \\ \mathbf{a}^\top s + \mathbf{e}^\top \end{bmatrix} \cdot \mathbf{R}_m + m \cdot \mathbf{G}$ for $m \in \mathcal{R}$ and a BGG+-style encoding (as in Def. 9) $\mathbf{c}_s^\top = \mathbf{b}^\top \cdot \mathbf{R}_s + s \cdot \mathbf{g}^\top$ for $s \in \mathcal{R}$, we construct $(\text{Eval}^{\text{Pub}}, \text{Eval}^{\text{Trap}}, \text{Eval}^{\text{Error}})$ for the incomplete decryption function $\widetilde{\text{Dec}}$ as follows.

$\text{Eval}^{\text{Pub}}(\mathbf{C}_m, \mathbf{c}_s)$: output $\mathbf{c}_1^\top - \mathbf{c}_s^\top \cdot \mathbf{g}^{-1}(\mathbf{c}_0^\top)$, where $\begin{bmatrix} \mathbf{c}_0^\top \\ \mathbf{c}_1^\top \end{bmatrix}$ is $\mathbf{C}_m$'s last k -columns.

$\text{Eval}^{\text{Trap}}(\mathbf{R}_s, \mathbf{C}_m)$: output $-\mathbf{R}_s \cdot \mathbf{g}^{-1}(\mathbf{c}_0^\top)$, where $\mathbf{c}_0$ is the vector defined in Eval^{Pub} .

$\text{Eval}^{\text{Error}}(\mathbf{e}, \mathbf{R}_m)$: output $\mathbf{e}^\top \cdot \hat{\mathbf{R}}_m$, where $\hat{\mathbf{R}}_m$ is the last k -columns of $\mathbf{R}_m$.

Theorem 3. In the algorithms $(\text{Eval}^{\text{Pub}}, \text{Eval}^{\text{Trap}}, \text{Eval}^{\text{Error}})$ for the incomplete decryption function $\widetilde{\text{Dec}}$, the 2-norm of the output of the $\text{Eval}^{\text{Trap}}$ is bounded by $nkb \cdot \|\mathbf{R}_s\|$, the 2-norm of the output of the $\text{Eval}^{\text{Error}}$ is bounded by $\|\mathbf{e}\| \cdot \|\mathbf{R}_m\|$.

Proof. It is easy to see that

$$\begin{aligned} \mathbf{c}_1^\top - \mathbf{c}_s^\top \cdot \mathbf{g}^{-1}(\mathbf{c}_0^\top) &= \mathbf{c}_1^\top - (\mathbf{b}^\top \cdot \mathbf{R}_s + s \cdot \mathbf{g}^\top) \cdot \mathbf{g}^{-1}(\mathbf{c}_0^\top) \\ &= \mathbf{b}^\top \cdot (-\mathbf{R}_s) \cdot \mathbf{g}^{-1}(\mathbf{c}_0^\top) + (\mathbf{c}_1^\top - s \cdot \mathbf{c}_0^\top) \\ &= \mathbf{b}^\top \cdot \underbrace{(-\mathbf{R}_s) \cdot \mathbf{g}^{-1}(\mathbf{c}_0^\top)}_{\text{Eval}^{\text{Trap}}} + m \cdot \mathbf{g}^\top + \underbrace{\mathbf{e}^\top \cdot \hat{\mathbf{R}}_m}_{\text{Eval}^{\text{Error}}}. \end{aligned}$$

By Lemma 6, we have $\|-\mathbf{R}_s \cdot \mathbf{g}^{-1}(\mathbf{c}_0^\top)\| \leq nkb \cdot \|\mathbf{R}_s\|$. By the definition of $\hat{\mathbf{R}}_m$ as in $\text{Eval}^{\text{Error}}$ algorithm, we have $\|\mathbf{e}^\top \cdot \hat{\mathbf{R}}_m\| \leq \|\mathbf{e}\| \cdot \|\hat{\mathbf{R}}_m\| \leq \|\mathbf{e}\| \cdot \|\mathbf{R}_m\|$.

4 Sampling Algorithms with (D_r, D_σ) -Hybrid Outputs

In this section, we follow the idea of the sampling algorithms in [2] and design new sampling algorithms with (D_r, D_σ) -hybrid outputs. Due to space limitation, we recall the sampling algorithms from [2] in Sect. C.1.

Our sampling algorithm would like to obtain $(w, \mathbf{x})$ which is a (D_r, D_σ) -hybrid pair where $r \gg \sigma$, satisfying

$$\mathbf{f}^\top \mathbf{x} = u + w \bmod q, \quad (12)$$

for any given uniformly random $u \in \mathcal{R}_q$. The vector $\mathbf{f}$ is

$$\mathbf{f}^\top = [\mathbf{b}^\top | \mathbf{b}^\top \mathbf{R} + y \mathbf{g}^\top + \mathbf{e}^\top] \in \mathcal{R}_q^{2k} \quad (13)$$

where $\mathbf{b} \in \mathcal{R}_q^k$, $\mathbf{R} \in \mathcal{R}^{k \times k}$ is a matrix with small entries, $\mathbf{e} \in \mathcal{R}^k$ is a vector with (relatively) large entries, $y \in \mathcal{R}_q^\times$ is an invertible element, and $\mathbf{g} = [1|b|\dots|b^{k-1}]^\top \in \mathcal{R}_q^k$ is the specific gadget vector as defined in Def. 3, whose trapdoor $\mathbf{T}_g$ is publicly known.

4.1 Our New Sampling Algorithms

Our sampling algorithms with (D_r, D_σ) -hybrid outputs consist of two parts: $\text{SampleLeft}_{\text{hybrid}}$ and $\text{SampleRight}_{\text{hybrid}}$.

Let $(\mathbf{b}, \mathbf{T}_\mathbf{b}) \in \mathcal{R}_q^k \times \mathcal{R}^{k \times k}$ be a trapdoor pair satisfying the requirements of Lemma 7 and SamPre be the algorithm in Lemma 8. Consider the following two algorithms.

$\text{SampleLeft}_{\text{hybrid}}(\mathbf{b}, \mathbf{c} \in \mathcal{R}_q^k, \mathbf{T}_\mathbf{b} \in \mathcal{R}^{k \times k}, u \in \mathcal{R}_q, \sigma, r \in \mathbb{R})$

- sample $w \leftarrow D_{\mathcal{R}, r}$, $\mathbf{x}_2 \leftarrow D_{\mathcal{R}^k, \sigma}$, set $\tilde{u} = u + w$.
- run $\mathbf{x}_1 \leftarrow \text{SamPre}(\mathbf{b}, \mathbf{T}_\mathbf{b}, u', \sigma)$ where $u' = \tilde{u} - \mathbf{c}^\top \mathbf{x}_2$.
- output $(w, \mathbf{x} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix})$.

$\text{SampleRight}_{\text{hybrid}}(\mathbf{b}, \mathbf{g} \in \mathcal{R}_q^k, \mathbf{e} \in \mathcal{R}^k, \mathbf{T}_\mathbf{g}, \mathbf{R} \in \mathcal{R}^{k \times k}, y \in \mathcal{R}_q^\times, u \in \mathcal{R}_q, \sigma, s, r \in \mathbb{R})$

- sample $h \leftarrow D_{\mathcal{R}, \sqrt{\Sigma_h}}$ where $\Sigma_h := r^2 - \sigma^2 \mathbf{e}^\top \mathbf{e}$.
- sample $\mathbf{p}' = \begin{bmatrix} \bar{\mathbf{p}} \\ \mathbf{p} \end{bmatrix} \leftarrow D_{\mathcal{R}^{3k}, \sqrt{\Sigma_p}}$ where $\bar{\mathbf{p}} \in \mathcal{R}^k$, $\mathbf{p} \in \mathcal{R}^{2k}$, and $\Sigma_p := \sigma^2 \mathbf{I}_{3k} - s^2 \mathbf{T}' \mathbf{T}'^\top$ for $\mathbf{T} = \begin{bmatrix} -\mathbf{R} \\ \mathbf{I}_k \end{bmatrix} \in \mathcal{R}^{2k \times k}$ and $\mathbf{T}' = \begin{bmatrix} \mathbf{I}_k \\ \mathbf{T} \end{bmatrix} \in \mathcal{R}^{3k \times k}$.
- set $u^* = u + h + \mathbf{e}^\top \bar{\mathbf{p}}$.
- compute $v = y^{-1}(u^* - \mathbf{f}^\top \mathbf{p})$ where $\mathbf{f}$ is defined in Eq. (13).
- run $\mathbf{z} \leftarrow \text{SamPre}(\mathbf{g}, \mathbf{T}_\mathbf{g}, v, s)$.
- compute $w = h + \mathbf{e}^\top (\bar{\mathbf{p}} + \mathbf{z})$ and $\mathbf{x} = \mathbf{p} + \mathbf{T}\mathbf{z}$.
- output $(w, \mathbf{x})$.

4.2 Analysis of Our New Sampling Algorithms

In this subsection, we analyze our new sampling algorithms and prove the output distributions of these two algorithms are statistically close.

Theorem 4. *For any trapdoor pair $(\mathbf{b}, \mathbf{T}_\mathbf{b}) \in \mathcal{R}_q^k \times \mathcal{R}^{k \times k}$ satisfying $\mathbf{b}^\top \cdot \mathbf{T}_\mathbf{b} = \mathbf{0}^\top \bmod q$ and $\text{Rot}(\mathbf{b}^\top) \in \mathbb{Z}^{n \times nk}$ is a full-rank matrix, any $\mathbf{c} \in \mathcal{R}_q^k$, $\sigma \geq \eta_\epsilon(\mathbb{Z}^{nk}) \cdot \|\widetilde{\mathbf{T}_\mathbf{b}}\|$ and any real $r > 0$, define $\mathbf{f}^\top = [\mathbf{b}^\top | \mathbf{c}^\top]$, then the following two distributions are statistically close*

- $\{(\mathbf{f}, w, \mathbf{x}, u) : u \xleftarrow{\$} \mathcal{R}_q, (w, \mathbf{x}) \leftarrow \text{SampleLeft}_{\text{hybrid}}(\mathbf{b}, \mathbf{c}, \mathbf{T}_\mathbf{b}, u, \sigma, r)\}$.
- $\{(\mathbf{f}, w, \mathbf{x}, u) : w \leftarrow D_{\mathcal{R}, r}, \mathbf{x} \leftarrow D_{\mathcal{R}^{2k}, \sigma}, u = \mathbf{f}^\top \mathbf{x} - w\}$.

Proof. Consider the first distribution $(\mathbf{f}, w, \mathbf{x}, u)$. We omit $\mathbf{f}$ since it is fixed in this distribution and we only consider the $(w, \mathbf{x} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix}, u)$ part where $u \xleftarrow{\$} \mathcal{R}_q$, $(w, \mathbf{x})$ are sampled using the algorithm $\text{SampleLeft}_{\text{hybrid}}(\mathbf{b}, \mathbf{c}, \mathbf{T}_\mathbf{b}, u, \sigma, r)$. Note that the $\text{SampleLeft}_{\text{hybrid}}$ algorithm samples $w \leftarrow D_{\mathcal{R}, r}$, $\mathbf{x}_2 \leftarrow D_{\mathcal{R}^k, \sigma}$ and $\mathbf{x}_1 \leftarrow \text{SamPre}(\mathbf{b}, \mathbf{T}_\mathbf{b}, u + w - \mathbf{c}^\top \mathbf{x}_2, \sigma)$.

Hybrid 1. Now we consider a hybrid distribution $(w, \mathbf{x} = [\mathbf{x}_1], u)$ where $w, \mathbf{x}_2$ are sampled as above, $\mathbf{x}_1 \leftarrow D_{\mathcal{R}^k, \sigma}, u' = \mathbf{b}^\top \mathbf{x}_1$ and finally compute $u = u' - w + \mathbf{c}^\top \mathbf{x}_2$. We note that Hybrid 1 and the first distribution are statistically close due to the following claim.

Claim. For any trapdoor pair $(\mathbf{b}, \mathbf{T}_\mathbf{b}) \in \mathcal{R}_q^k \times \mathcal{R}^{k \times k}$ satisfying $\mathbf{b}^\top \cdot \mathbf{T}_\mathbf{b} = \mathbf{0}^\top \bmod q$ and $\text{Rot}(\mathbf{b}^\top) \in \mathbb{Z}^{n \times nk}$ is a full-rank matrix, $\sigma \geq \eta_\epsilon(\mathbb{Z}^{nk}) \cdot \|\widetilde{\mathbf{T}_\mathbf{b}}\|$, the following two distributions are statistically close:

- sample $u' \xleftarrow{\$} \mathcal{R}_q$, then sample $\mathbf{x}_1 \leftarrow \text{SamPre}(\mathbf{b}, \mathbf{T}_\mathbf{b}, u', \sigma)$, output $(\mathbf{x}_1, u')$;
- sample $\mathbf{x}_1 \leftarrow D_{\mathcal{R}^k, \sigma}$, then compute $u' = \mathbf{b}^\top \mathbf{x}_1$, output $(\mathbf{x}_1, u')$.

It is obvious that this claim follows from Lemma 8 and Lemma 9.

Note that Hybrid 1 is exactly the second distribution in Theorem 4. This completes the proof. $\square$

Theorem 5. For a vector $\mathbf{b} \in \mathcal{R}_q^k$, the public gadget trapdoor pair $(\mathbf{g}, \mathbf{T}_\mathbf{g}) \in \mathcal{R}_q^k \times \mathcal{R}^{k \times k}$ satisfying $\mathbf{g}^\top \cdot \mathbf{T}_\mathbf{g} = \mathbf{0}^\top \bmod q$ and $\text{Rot}(\mathbf{g}^\top) \in \mathbb{Z}^{n \times nk}$ is a full-rank matrix, any matrix $\mathbf{R} \in \mathcal{R}^{k \times k}$ and vector $\mathbf{e} \in \mathcal{R}^k$, any invertible element $y \in \mathcal{R}_q^\times$, $s \geq \eta_\epsilon(\mathbb{Z}^{nk}) \cdot \|\widetilde{\mathbf{T}_\mathbf{g}}\|$, $\sigma^2 \geq (s^2 + \eta_\epsilon^2(\mathbb{Z}^{nk})) \cdot (\|\mathbf{T}'\|^2 + 1)$ where $\mathbf{T}' = [\mathbf{I}_k]$ and $\mathbf{T} = [\mathbf{R}]$, $r^2 \geq (\sigma^2 + \eta_\epsilon^2(\mathbb{Z}^{nk})) \cdot (\|\mathbf{e}\|^2 + 1)$, define $\mathbf{f}^\top = [\mathbf{b}^\top | \mathbf{b}^\top \mathbf{R} + y\mathbf{g}^\top + \mathbf{e}^\top]$, then the following two distributions are statistically close

- $\{(\mathbf{f}, w, \mathbf{x}, u) : u \xleftarrow{\$} \mathcal{R}_q, (w, \mathbf{x}) \leftarrow \text{SampleRight}_{\text{hybrid}}(\mathbf{b}, \mathbf{g}, \mathbf{e}, \mathbf{T}_\mathbf{g}, \mathbf{R}, y, u, \sigma, s, r)\}$;
- $\{(\mathbf{f}, w, \mathbf{x}, u) : w \leftarrow D_{\mathcal{R}, r}, \mathbf{x} \leftarrow D_{\mathcal{R}^{2k}, \sigma}, u = \mathbf{f}^\top \mathbf{x} - w\}$.

Proof. Consider the first distribution $(\mathbf{f}, w, \mathbf{x}, u)$. We omit $\mathbf{f}$ since it is fixed in this distribution and we only consider the $(w, \mathbf{x}, u)$ part where $u \xleftarrow{\$} \mathcal{R}_q$, $(w, \mathbf{x})$ are sampled using the algorithm $\text{SampleRight}_{\text{hybrid}}(\mathbf{b}, \mathbf{g}, \mathbf{e}, \mathbf{T}_\mathbf{g}, \mathbf{R}, y, u, \sigma, s, r)$.

Hybrid 1. Now we consider a hybrid distribution $(w, \mathbf{x}, u)$ where $w, \mathbf{x}$ are sampled as above. We swap the sampling order of u and u^* where u^* is used in $\text{SampleRight}_{\text{hybrid}}$, i.e., we first sample $u^* \xleftarrow{\$} \mathcal{R}_q$, $h \leftarrow D_{\mathcal{R}, \sqrt{\Sigma_h}}$, $\mathbf{p}' = [\mathbf{p}] \leftarrow D_{\mathcal{R}^{3k}, \sqrt{\Sigma_p}}$ and then compute $u = u^* - h - \mathbf{e}^\top \mathbf{p}$. Clearly, the first distribution and Hybrid 1 are identically distributed.

Hybrid 2. Now we consider the second hybrid distribution $(w, \mathbf{x}, u)$ where $w, \mathbf{x}$ are sampled as above. We swap the order of sampling u^* and v where v is used in $\text{SampleRight}_{\text{hybrid}}$, i.e., we first sample $v \xleftarrow{\$} \mathcal{R}_q$, set $u^* = yv + \mathbf{f}^\top \mathbf{p}$ and set u as above. We claim that Hybrid 1 and Hybrid 2 are identically distributed. Since $y \in \mathcal{R}_q^\times$ is invertible and $v \xleftarrow{\$} \mathcal{R}_q$, the term yv follows a uniform distribution over $\mathcal{R}_q$, then $u^* = yv + \mathbf{f}^\top \mathbf{p}$ as well.

Hybrid 3. Now we consider the third hybrid distribution $(w, \mathbf{x}, u)$ where $w, \mathbf{x}$ are sampled as above. We swap the order of sampling v and $\mathbf{z}$, i.e., we first

sample $\mathbf{z} \leftarrow D_{\mathcal{R}^k, s}$, then compute $v = \mathbf{g}^\top \mathbf{z}$ and set u as above. We note that Hybrid 3 and Hybrid 2 are statistically close due to the following claim.

Claim. For the gadget trapdoor pair $(\mathbf{g}, \mathbf{T}_\mathbf{g}) \in \mathcal{R}_q^k \times \mathcal{R}^{k \times k}$ satisfying $\mathbf{g}^\top \cdot \mathbf{T}_\mathbf{g} = \mathbf{0}^\top \bmod q$ and $\text{Rot}(\mathbf{g}^\top) \in \mathbb{Z}^{n \times nk}$ is a full-rank matrix, $s \geq \eta_\epsilon(\mathbb{Z}^{nk}) \cdot \|\widetilde{\mathbf{T}_\mathbf{g}}\|$, the following two distributions are statistically close:

- sample $v \xleftarrow{\$} \mathcal{R}_q$, then sample $\mathbf{z} \leftarrow \text{SamPre}(\mathbf{g}, \mathbf{T}_\mathbf{g}, v, s)$, output $(\mathbf{z}, v)$;
- sample $\mathbf{z} \leftarrow D_{\mathcal{R}^k, s}$, then compute $v = \mathbf{g}^\top \mathbf{z}$, output $(\mathbf{z}, v)$.

It is obvious that this claim follows from Lemma 8 and Lemma 9.

Note that in Hybrid 3, w is computed by $h + \mathbf{e}^\top(\bar{\mathbf{p}} + \mathbf{z})$ and $\mathbf{x}$ is computed by $\mathbf{x} = \mathbf{p} + \mathbf{T}\mathbf{z}$ where $\mathbf{p}' = \begin{bmatrix} \bar{\mathbf{p}} \\ \mathbf{p} \end{bmatrix} \leftarrow D_{\mathcal{R}^{3k}, \sqrt{\Sigma_p}}$, $\mathbf{z} \leftarrow D_{\mathcal{R}^k, s}$, $h \leftarrow D_{\mathcal{R}, \sqrt{\Sigma_h}}$. We define $\bar{\mathbf{x}} := \bar{\mathbf{p}} + \mathbf{I}_k \cdot \mathbf{z}$, then $w = h + \mathbf{e}^\top \bar{\mathbf{x}}$. We further define $\mathbf{x}' := \begin{bmatrix} \bar{\mathbf{x}} \\ \mathbf{x} \end{bmatrix}$, then $\mathbf{x}' = \begin{bmatrix} \bar{\mathbf{p}} + \mathbf{I}_k \mathbf{z} \\ \mathbf{p} + \mathbf{T}\mathbf{z} \end{bmatrix} = \mathbf{p}' + \mathbf{T}'\mathbf{z}$ in Hybrid 3.

By routine computation, we have that

$$u = u^* - h - \mathbf{e}^\top \bar{\mathbf{p}} = \mathbf{f}^\top \mathbf{x} - w. \quad (14)$$

We put the full derivation process of Eq. (14) in Sect. C.2. Then we compute $u = \mathbf{f}^\top \mathbf{x} - w$ in the following hybrids.

Hybrid 4. Now we consider the fourth hybrid distribution $(w, \mathbf{x}, u)$ where u is sampled as above, $\mathbf{x}' = \begin{bmatrix} \bar{\mathbf{x}} \\ \mathbf{x} \end{bmatrix} \leftarrow D_{\mathcal{R}^{3k}, \sigma}$ and w is computed as above. Note that Hybrid 4 is statistically close to Hybrid 3 due to the following claim.

Claim. For a matrix $\mathbf{T}' \in \mathcal{R}^{3k \times k}$, $\sigma^2 \geq (s^2 + \eta_\epsilon^2(\mathbb{Z}^{nk})) \cdot (\|\mathbf{T}'\|^2 + 1)$, $\Sigma_p = \sigma^2 \mathbf{I}_{3k} - s^2 \mathbf{T}' \mathbf{T}'^\top$, the following two distributions are statistically close:

- sample $\mathbf{z} \leftarrow D_{\mathcal{R}^k, s}$, $\mathbf{p}' \leftarrow D_{\mathcal{R}^{3k}, \sqrt{\Sigma_p}}$, output $\mathbf{p}' + \mathbf{T}'\mathbf{z}$;
- output $\mathbf{x}' \leftarrow D_{\mathcal{R}^{3k}, \sigma}$.

Directly, it comes from Lemma 10.

Hybrid 5. Now we consider the last hybrid distribution $(w, \mathbf{x}, u)$ where $\mathbf{x}, u$ are sampled as above and $w \leftarrow D_{\mathcal{R}, r}$. Note that Hybrid 5 is statistically close to Hybrid 4 due to the following claim.

Claim. For a matrix $\mathbf{e} \in \mathcal{R}^{1 \times k}$, $r^2 \geq (\sigma^2 + \eta_\epsilon^2(\mathbb{Z}^{nk})) \cdot (\|\mathbf{e}\|^2 + 1)$, $\Sigma_h = r^2 - \sigma^2 \mathbf{e}^\top \mathbf{e}$, the following two distributions are statistically close:

- sample $\bar{\mathbf{x}} \leftarrow D_{\mathcal{R}^k, \sigma}$, $h \leftarrow D_{\mathcal{R}, \sqrt{\Sigma_h}}$, output $h + \mathbf{e}^\top \bar{\mathbf{x}}$;
- output $w \leftarrow D_{\mathcal{R}, r}$.

Directly, it comes from Lemma 10.

Note that in Hybrid 5, $\mathbf{x}' = \begin{bmatrix} \bar{\mathbf{x}} \\ \mathbf{x} \end{bmatrix} \leftarrow D_{\mathcal{R}^{3k}, \sigma}$ which means that $\mathbf{x}$ follows the distribution of $D_{\mathcal{R}^{2k}, \sigma}$, $u = \mathbf{f}^\top \mathbf{x} - w$ from Hybrid 3 and w follows the distribution of $D_{\mathcal{R}, r}$. This means that the distribution in Hybrid 5 is exactly the same as the second distribution in Theorem 5. This completes the proof. $\square$

Theorem 6. For any trapdoor pair $(\mathbf{b}, \mathbf{T}_\mathbf{b}) \in \mathcal{R}_q^k \times \mathcal{R}^{k \times k}$ satisfying $\mathbf{b}^\top \cdot \mathbf{T}_\mathbf{b} = \mathbf{0}^\top \bmod q$ and $\text{Rot}(\mathbf{b}^\top) \in \mathbb{Z}^{n \times nk}$ is a full-rank matrix, the gadget trapdoor pair $(\mathbf{g}, \mathbf{T}_\mathbf{g}) \in \mathcal{R}_q^k \times \mathcal{R}^{k \times k}$ satisfying $\mathbf{g}^\top \cdot \mathbf{T}_\mathbf{g} = \mathbf{0}^\top \bmod q$ and $\text{Rot}(\mathbf{g}^\top) \in \mathbb{Z}^{n \times nk}$ is a full-rank matrix, any matrix $\mathbf{R} \in \mathcal{R}^{k \times k}$ and vector $\mathbf{e} \in \mathcal{R}^k$, any invertible element $y \in \mathcal{R}_q^\times$, $s \geq \eta_\epsilon(\mathbb{Z}^{nk}) \cdot \|\mathbf{T}_\mathbf{g}\|$, $\sigma^2 \geq (s^2 + \eta_\epsilon^2(\mathbb{Z}^{nk})) \cdot (\|\mathbf{T}'\|^2 + 1)$ for $\mathbf{T} = \begin{bmatrix} -\mathbf{R} \\ \mathbf{I}_k \end{bmatrix} \in \mathcal{R}^{2k \times k}$ and $\mathbf{T}' = \begin{bmatrix} \mathbf{I}_k \\ \mathbf{T} \end{bmatrix} \in \mathcal{R}^{3k \times k}$, $r^2 \geq (\sigma^2 + \eta_\epsilon^2(\mathbb{Z}^{nk})) \cdot (\|\mathbf{e}\|^2 + 1)$, $\sigma \geq \eta_\epsilon(\mathbb{Z}^{nk}) \cdot \|\widetilde{\mathbf{T}_\mathbf{b}}\|$, define $\mathbf{f}^\top = [\mathbf{b}^\top | \mathbf{c}^\top] = [\mathbf{b}^\top | \mathbf{b}^\top \mathbf{R} + y\mathbf{g}^\top + \mathbf{e}^\top]$, then the distributions of outputs from $\text{SampleLeft}_{\text{hybrid}}$ and $\text{SampleRight}_{\text{hybrid}}$ are statistically close. More precisely, the following two distributions are statistically close

- $\{(\mathbf{f}, w, \mathbf{x}, u) : u \xleftarrow{\$} \mathcal{R}_q, (w, \mathbf{x}) \leftarrow \text{SampleLeft}_{\text{hybrid}}(\mathbf{b}, \mathbf{c}, \mathbf{T}_\mathbf{b}, u, \sigma, r)\};$
- $\{(\mathbf{f}, w, \mathbf{x}, u) : u \xleftarrow{\$} \mathcal{R}_q, (w, \mathbf{x}) \leftarrow \text{SampleRight}_{\text{hybrid}}(\mathbf{b}, \mathbf{g}, \mathbf{e}, \mathbf{T}_\mathbf{g}, \mathbf{R}, y, u, \sigma, s, r)\}.$

Proof. Directly from Theorem 4 and Theorem 5. $\square$

5 IBE Scheme

In this section, combining the homomorphic computation of the partition function in the GSW-style encryption form and the homomorphic transformation from GSW-style encryption to BGG+-style encoding (Sect. 3), our new sampling algorithms with (D_r, D_σ) -hybrid outputs (Sect. 4), we provide an adaptively secure lattice-based IBE scheme with smaller modulus, smaller secret key size than ALWW-IBE [1]. Particularly, we provide the construction in Sect. 5.1, the security proof in Sect. 5.2 and parameter analysis in Sect. 5.3.

5.1 Construction

Let $n := n(\lambda)$, $q := q(n)$, $b := b(n)$, $\rho := \rho(n)$, $k := k(n)$, $s := s(n)$, $\sigma_0 := \sigma_0(n)$, $\sigma'_0 := \sigma'_0(n)$, $\sigma_1 := \sigma_1(n)$, $\sigma := \sigma(n)$, $r := r(n)$ be parameters that are specified later. Let $\mathcal{R} = \mathbb{Z}[x]/(x^n + 1)$, $\mathcal{R}_q = \mathcal{R}/q\mathcal{R}$ and $\chi := D_{\mathcal{R}, \sigma'_0}$.

Let the identity space of the scheme be $\mathcal{ID} = \{0, 1\}^l$ for some $l \in \mathbb{N}$ and the message space be $\mathcal{M} = \{0, 1\}^n \subset \mathcal{R}$. Let GSW be the Ring-GSW homomorphic encryption scheme in Sect. 2.5. We define the IBE scheme (Setup, KeyGen, Enc, Dec) in the following.

Setup(1^λ). On input a security parameter λ , do:

1. $pp := (n, q, b, \rho, k, \chi) \leftarrow \text{GSW.Setup}(1^\lambda)$.
2. $(\text{ek} := \mathbf{A} = \begin{bmatrix} \mathbf{a}^\top & \mathbf{a}^\top \\ \mathbf{a}^\top & \mathbf{s} + \mathbf{e}^\top \end{bmatrix}, \text{dk} := s) \leftarrow \text{GSW.KeyGen}(pp)$.
3. $(\mathbf{b} \in \mathcal{R}_q^k, \mathbf{T}_\mathbf{b} \in \mathcal{R}^{k \times k}) \leftarrow \text{TrapGen}(pp)$.
4. $\left\{ \{ \mathbf{C}_{\alpha_{i,i'}} \}_{i \in [t], i' \in [\eta]}, \mathbf{C}_{\tilde{\beta}} \right\} \leftarrow \text{GSW.Enc}(\text{ek}, 0)$.
5. $\hat{\mathbf{c}} \xleftarrow{\$} \mathcal{R}_q^k$, $\mathbf{c}_s \xleftarrow{\$} \mathcal{R}_q^k$, $d \xleftarrow{\$} \mathcal{R}_q^\times$, $u \xleftarrow{\$} \mathcal{R}_q$.
6. output $\text{mpk} := (\mathbf{b}, \{ \mathbf{C}_{\alpha_{i,i'}} \}_{i \in [t], i' \in [\eta]}, \mathbf{C}_{\tilde{\beta}}, \hat{\mathbf{c}}, \mathbf{c}_s, \mathbf{A}, d, u)$, $\text{msk} := (\mathbf{T}_\mathbf{b}, s)$.

KeyGen($\text{mpk}, \text{msk}, \text{id}$). On inputs the master public key mpk , the master secret key msk , and an identity $\text{id} \in \mathcal{ID}$, do:

1. define G_{id} as the function as in Def. 7 with index id .
2. compute $\mathbf{C}_G = \text{PubEval}_G(\{\{\mathbf{C}_{\alpha_{i,i'}}\}_{i \in [t], i' \in [\eta]}, \mathbf{C}_{\tilde{\beta}}\}, G_{\text{id}}) \in \mathcal{R}_q^{2 \times 2k}$.
3. compute $F(\text{id}) = \text{Eval}^{\text{Pub}}(\mathbf{C}_G, \mathbf{c}_s) \in \mathcal{R}_q^k$.
4. sample $(w, \mathbf{x}) \leftarrow \text{SampleLeft}_{\text{hybrid}}(\mathbf{b}, \hat{\mathbf{c}} + F(\text{id}), \mathbf{T}_{\mathbf{b}}, d^{-1}u, \sigma, r)$, satisfying

$$[\mathbf{b}^\top | \hat{\mathbf{c}}^\top + F(\text{id})^\top] \cdot \mathbf{x} = d^{-1}u + w.$$

5. output $\text{sk}_{\text{id}} := \mathbf{x} \in \mathcal{R}^{2k}$.

Enc(mpk, id, m). On inputs the master public key mpk , an identity id , and a message m , do:

1. set $\mu = m_0 + m_1x + \dots m_{n-1}x^{n-1} \in \mathcal{R}_q$.
2. compute $\mathbf{C}_G = \text{PubEval}_G(\{\{\mathbf{C}_{\alpha_{i,i'}}\}_{i \in [t], i' \in [\eta]}, \mathbf{C}_{\tilde{\beta}}\}, G_{\text{id}}) \in \mathcal{R}_q^{2 \times 2k}$.
3. compute $F(\text{id}) = \text{Eval}^{\text{Pub}}(\mathbf{C}_G, \mathbf{c}_s) \in \mathcal{R}_q^k$.
4. sample $v \xleftarrow{\$} \mathcal{R}_q$, $e_0, e_1 \leftarrow D_{\mathcal{R}, \sigma_0}$, $\mathbf{e}_2 \leftarrow D_{\mathcal{R}^{2k}, \sigma_1}$.
5. compute $c_0 = u \cdot v + e_0 + \lceil \frac{q}{2} \rceil \cdot \mu$, $\mathbf{c}_1 = d \cdot \begin{bmatrix} 1 \\ \mathbf{b} \\ \hat{\mathbf{c}} + F(\text{id}) \end{bmatrix} \cdot v + \begin{bmatrix} e_1 \\ \mathbf{e}_2 \end{bmatrix}$.
6. output the ciphertext $\text{ct} := (c_0, \mathbf{c}_1) \in \mathcal{R}_q \times \mathcal{R}_q^{2k+1}$.

Dec($\text{mpk}, \text{sk}_{\text{id}} = \mathbf{x}, \text{ct} = (c_0, \mathbf{c}_1)$). on inputs the master public key mpk , the user secret key sk_{id} , and the ciphertext ct , do:

1. compute $\mathbf{C}_G = \text{PubEval}_G(\{\{\mathbf{C}_{\alpha_{i,i'}}\}_{i \in [t], i' \in [\eta]}, \mathbf{C}_{\tilde{\beta}}\}, G_{\text{id}}) \in \mathcal{R}_q^{2 \times 2k}$.
2. compute $F(\text{id}) = \text{Eval}^{\text{Pub}}(\mathbf{C}_G, \mathbf{c}_s) \in \mathcal{R}_q^k$.
3. compute $w = [\mathbf{b}^\top | \hat{\mathbf{c}}^\top + F(\text{id})^\top] \cdot \mathbf{x} - d^{-1}u$.
4. output $m = \lfloor \frac{2}{q} \cdot \phi(c_0 - \mathbf{c}_1^\top \cdot \lfloor \frac{-w}{\mathbf{x}} \rfloor) \rfloor \bmod 2$, where the rounding function $\lfloor \cdot \rfloor$ is applied component-wise.

Remark 1. In the **KeyGen** step, we use our new algorithm $\text{SampleLeft}_{\text{hybrid}}$ to sample $(w, \mathbf{x})$ (which are (D_r, D_σ) -hybrid outputs where $r \gg \sigma$). To shrink the size of the secret key, we only take $\mathbf{x}$ (i.e., D_σ part) as the user's secret key. Note that w can be deterministically computed from $\mathbf{x}, \text{mpk}$ and id , i.e., steps 1-3 of the decryption algorithm. This means that during the key query phase of the IBE security game, an adversary is able to compute w itself after obtaining $\mathbf{x}$ from the challenger.

Lemma 11 (Correctness). *For any positive number ω , if $q \geq 5\omega(\sigma_0 + r\sigma_0\sqrt{n} + \sigma_1\sigma\sqrt{2nk})$, then the above IBE scheme has a decryption error at most $6e^{-\pi\omega^2} + 2^{-2nk} + 2^{-n}$.*

Proof. For the Dec algorithm, we show that the error term in decryption would not exceed $q/5$. Specifically, the decryption algorithm calculates

$$\begin{aligned} c_0 - \mathbf{c}_1^\top \cdot \begin{bmatrix} -w \\ \mathbf{x} \end{bmatrix} &= u \cdot v + e_0 + \lceil \frac{q}{2} \rceil \cdot \mu - (d \cdot [1|\mathbf{b}^\top|\widehat{\mathbf{c}}^\top + F(\text{id})^\top] \cdot v + [e_1|\mathbf{e}_2^\top]) \cdot \begin{bmatrix} -w \\ \mathbf{x} \end{bmatrix} \\ &= \lceil \frac{q}{2} \rceil \cdot \mu + \underbrace{(e_0 + w \cdot e_1 - \mathbf{e}_2^\top \cdot \mathbf{x})}_{\text{error term}}. \end{aligned}$$

Then, the following inequalities hold¹³:

- by Lemma 20 and $e_0 \sim D_{\mathcal{R}, \sigma_0}$, we have $\Pr[|\phi(e_0)_j| \geq t_1] \leq 2e^{-\pi \frac{t_1^2}{\sigma_0^2}}$.
- by Theorem 4, we have $\mathbf{x} \sim D_{\mathcal{R}^{2k}, \sigma}$ ¹⁴, $w \sim D_{\mathcal{R}, r}$.
- by Corollary 1 and $w \sim D_{\mathcal{R}, r}$, we have $\Pr[|\phi(w \cdot e_1)_j| \geq t_2] \leq 2e^{-\pi \frac{t_2^2}{\|e_1\|^2 r^2}}$.
- by Lemma 2 and $e_1 \sim D_{\mathcal{R}, \sigma_0}$, $\Pr[\|e_1\| \geq \sigma_0 \sqrt{n}] \leq 2^{-n}$.
- by Corollary 1 and $\mathbf{e}_2 \sim D_{\mathcal{R}^{2k}, \sigma_1}$, we have $\Pr[|\phi(\mathbf{e}_2^\top \cdot \mathbf{x})_j| \geq t_3] \leq 2e^{-\pi \frac{t_3^2}{\|\mathbf{x}\|^2 \sigma_1^2}}$.
- by Lemma 2 and $\mathbf{x} \sim D_{\mathcal{R}^{2k}, \sigma}$, we have $\Pr[\|\mathbf{x}\| \geq \sigma \sqrt{2nk}] \leq 2^{-2nk}$.

Taking $t_1 = \omega \sigma_0$, $t_2 = \omega r \sigma_0 \sqrt{n}$, $t_3 = \omega \sigma_1 \sigma \sqrt{2nk}$, by union bound, we have

$$\Pr[|\phi(e_0 + w \cdot e_1 - \mathbf{e}_2^\top \cdot \mathbf{x})_j| \geq t_1 + t_2 + t_3] \leq 6e^{-\pi \omega^2} + 2^{-2nk} + 2^{-n}.$$

Therefore, if $\omega(\sigma_0 + r \sigma_0 \sqrt{n} + \sigma_1 \sigma \sqrt{2nk}) \leq q/5$, then the decryption error occurs with probability at most $6e^{-\pi \omega^2} + 2^{-2nk} + 2^{-n}$. $\square$

5.2 Security Proof

In this section, we analyze the security of our IBE scheme. Due to space limitations, we put the proofs of all the lemmas involved in Theorem 7 (Lemmas 12 to 15 and 17 to 19) in Appendix D, except for Lemma 16, because the proof of Lemma 16 shows how we run the re-randomization algorithm to simulate a partial ciphertext.

Theorem 7. *The above IBE scheme is adaptively-anonymous security assuming the assumptions $\text{RLWE}_{n, 2k, q, D_{\mathcal{R}, \sigma'_0}}$ and $\text{RLWE}_{n, k+2, q, D_{\mathcal{R}, \sigma_0}}$ hold.*

¹³ In this correctness analysis, we use multiple lemmas and theorems without explicitly specifying the parameters. We note that all the requirements of these lemmas and theorems are met for the parameters in Sect. 5.3.

¹⁴ Actually, Theorem 4 guarantees that $\mathbf{x}$ is statistically close to $D_{\mathcal{R}^{2k}, \sigma}$. Since the statistical distance is negligible, we omit the statistical distance and do the correctness analysis without it.

Proof. Let $\mathcal{A}$ be a PPT adversary that breaks the adaptive security of the IBE scheme. Let $\epsilon = \epsilon(\lambda)$ and $Q = Q(\lambda)$ be its advantage and the number of the key queries. In each game, a value $\text{coin}' \in \{0, 1\}$ is defined. While it is set $\text{coin}' = \widehat{\text{coin}}$ in the first game, these values might be different in the later games. In the following, we define E_i to be the event that $\text{coin}' = \text{coin}$.

Game 0. This is the real security game. Recall that since the ciphertext space is $\mathcal{C} = \mathcal{R}_q \times \mathcal{R}_q^{2k+1}$, in the challenge phase, the challenge ciphertext is set as $\text{ct}^* = (c_0, \mathbf{c}_1^*) \xleftarrow{\$} \mathcal{R}_q \times \mathcal{R}_q^{2k+1}$ if $\text{coin} = 1$. At the end of the game, $\mathcal{A}$ outputs a guess $\widehat{\text{coin}}$ for coin . Finally, the challenger sets $\text{coin}' = \widehat{\text{coin}}$. By definition, we have $|\Pr[E_0] - \frac{1}{2}| = |\Pr[\text{coin}' = \text{coin}] - \frac{1}{2}| = |\Pr[\widehat{\text{coin}} = \text{coin}] - \frac{1}{2}| = \epsilon$.

Game 1. In this game, the challenger performs an additional abort check at the end of the game. First, the challenger chooses a random partitioning function $H \xleftarrow{\$} \mathcal{H}^{\mathcal{R}, t, t'}$ as Def. 8, where $t' = \lceil \log_c \frac{3Q}{\epsilon} \rceil$ ¹⁵. Specifically, the challenger picks random vectors $\alpha' \in [L+1]^{t'}$, $\beta' \in \mathbb{Z}_p^{t'}$, denotes $\alpha = (\alpha', \mathbf{0}') \in [L+1]^t, \beta = (\beta', \mathbf{0}) \in \mathbb{Z}_p^t$, and sets and keeps the partition function:

$$H(\text{id}) := G_{\text{id}}(\alpha, \beta) = H_{\alpha, \beta}^{\mathcal{R}, t}(\text{id}) = \sum_{i \in [t]} \left(x^{ip + \text{ECC}(\text{id})[\alpha_i]} - x^{ip + \beta_i} \right).$$

Then, the challenger checks whether the following condition holds:

$$H(\text{id}^*) = 0 \quad \wedge \quad H(\text{id}^{(1)}) = \dots = H(\text{id}^{(Q)}) = 1,$$

where id^* is the challenge identity, $\text{id}^{(1)}, \dots, \text{id}^{(Q)}$ are identities for which $\mathcal{A}$ has made key extraction queries. If it does not hold, the challenger ignores the output $\widehat{\text{coin}}$ of $\mathcal{A}$, and sets $\text{coin}' \xleftarrow{\$} \{0, 1\}$. If it holds, the challenger sets $\text{coin}' = \widehat{\text{coin}}$.

Lemma 12. $|\Pr[E_1] - \frac{1}{2}| \geq \frac{2}{3} \cdot \frac{1}{p^{t'}} \cdot \epsilon$.

Game 2. In this game, we change the way $\left\{ \{ \mathbf{C}_{\alpha_{i,i'}} \}_{i \in [t], i' \in [\eta]}, \mathbf{C}_{\tilde{\beta}} \right\}$ are chosen.

- in Game 1, $\mathbf{C}_{\alpha_{i,i'}} \leftarrow \text{GSW.Enc}(\text{ek}, 0)$, $\mathbf{C}_{\tilde{\beta}} \leftarrow \text{GSW.Enc}(\text{ek}, 0)$.
- in Game 2, the challenger first picks $\alpha \in [L+1]^t, \beta \in \mathbb{Z}_p^t$ as Game 1, and generates $\mathbf{C}_{\alpha_{i,i'}} \leftarrow \text{GSW.Enc}(\text{ek}, x^{\alpha_{i,i'}})$, $\mathbf{C}_{\tilde{\beta}} \leftarrow \text{GSW.Enc}(\text{ek}, \sum_{i \in [t]} x^{ip + \beta_i})$ with randomness $\mathbf{R}_{\alpha_{i,i'}}, \mathbf{R}_{\tilde{\beta}}$, respectively.

Lemma 13. By the $\text{RLWE}_{n, 2k, q, D_{\mathcal{R}, \sigma_0'}}$ assumption (Def. 2) and the regularity lemma (Lemma 5), we have $|\Pr[E_1] - \Pr[E_2]| = \text{negl}(n)$.

Game 3. In this game, we change the way $\widehat{\mathbf{c}}, \mathbf{c}_s$ are chosen.

- in Game 2, $\widehat{\mathbf{c}}, \mathbf{c}_s \xleftarrow{\$} \mathcal{R}_q^k$.

¹⁵ $c := \frac{1}{1-\gamma}$ where γ is the relative distance of the error correcting code ECC.

- in Game 3, the challenger chooses $\mathbf{R}_0, \mathbf{R}_s \xleftarrow{\$} [-\rho, \rho]_{\mathcal{R}}^{k \times k}$, and sets $\hat{\mathbf{c}} = \mathbf{b}^\top \mathbf{R}_0, \mathbf{c}_s = \mathbf{b}^\top \mathbf{R}_s + s \cdot \mathbf{g}^\top$.

Lemma 14. *By the TrapGen algorithm (Lemma 7), and the regularity lemma (Lemma 5), we have $|\Pr[E_2] - \Pr[E_3]| = \text{negl}(n)$.*

Note. Before entering the next game, we first define the matrix $\mathbf{R}_G \in \mathcal{R}^{2k \times 2k}$ for an identity $\text{id} \in \mathcal{ID}$ as

$$\mathbf{R}_G = \text{TrapEval}_G \left(\mathbf{A}, \left\{ \{\mathbf{R}_{\alpha_{i,i'}}, \mathbf{R}_{\tilde{\beta}}\} \right\}, (\alpha, \beta), G_{\text{id}} \right) \in \mathcal{R}^{2k \times 2k}.$$

By Construction 3 and Theorem 2, we have

$$\mathbf{C}_G = \text{PubEval}_G \left(\left\{ \{\mathbf{C}_{\alpha_{i,i'}}, \mathbf{C}_{\tilde{\beta}}\} \right\}, G_{\text{id}} \right) = \mathbf{A} \mathbf{R}_G + G_{\text{id}}(\alpha, \beta) \cdot \mathbf{G} \in \mathcal{R}_q^{2 \times 2k}.$$

Then, we define the matrix $\mathbf{R}_{\text{id}} \in \mathcal{R}^{k \times k}$ and the vector $\mathbf{e}_{\text{id}} \in \mathcal{R}^k$ as

$$\mathbf{R}_{\text{id}} = \text{Eval}^{\text{Trap}}(\mathbf{R}_s, \mathbf{C}_G) \in \mathcal{R}^{k \times k}, \quad \mathbf{e}_{\text{id}} = \text{Eval}^{\text{Error}}(\mathbf{e}, \mathbf{R}_G) \in \mathcal{R}^k.$$

By Construction 1 and Theorem 3, we have

$$F(\text{id}) = \text{Eval}^{\text{Pub}}(\mathbf{C}_G, \mathbf{c}_s) = \mathbf{b}^\top \mathbf{R}_{\text{id}} + G_{\text{id}}(\alpha, \beta) \cdot \mathbf{g}^\top + \mathbf{e}_{\text{id}}^\top \in \mathcal{R}_q^k.$$

Game 4. In this game, we change the way to generate the public vector $\mathbf{b}$ and to respond to the secret key queries.

- in Game 3, the challenger generates $(\mathbf{b}, \mathbf{T}_{\mathbf{b}}) \leftarrow \text{TrapGen}(pp)$, and answers each of the secret key queries using this trapdoor $\mathbf{T}_{\mathbf{b}}$, i.e., $(w, \mathbf{x}) \leftarrow \text{SampleLeft}_{\text{hybrid}}(\mathbf{b}, \hat{\mathbf{c}} + F(\text{id}), \mathbf{T}_{\mathbf{b}}, u, \sigma, r)$.
- in Game 4, $\mathcal{C}$ samples $\mathbf{b} \xleftarrow{\$} \mathcal{R}_q^k$ uniformly at random instead of running TrapGen algorithm, and answers each of the secret key queries using the newly defined trapdoors $\mathbf{R}_{\text{id}}, \mathbf{e}_{\text{id}}$, i.e., $(w, \mathbf{x}) \leftarrow \text{SampleRight}_{\text{hybrid}}(\mathbf{b}, \mathbf{g}, \mathbf{e}_{\text{id}}, \mathbf{T}_{\mathbf{g}}, \mathbf{R}_0 + \mathbf{R}_{\text{id}}, G_{\text{id}}(\alpha, \beta), u, \sigma, s, r)$.

Lemma 15. *By the TrapGen algorithm (Lemma 7) and the sampling algorithms (Theorem 6), we have $|\Pr[E_3] - \Pr[E_4]| = \text{negl}(n)$.*

Game 5. In this game, we change the way challenge ciphertext $\mathbf{c}_1^*$ is created when $\text{coin} = 0$.

- in Game 4, $\mathbf{c}_1^* = d \cdot \begin{bmatrix} 1 \\ \mathbf{b} \\ F(\text{id}^*) \end{bmatrix} \cdot v + \begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$ for $e_1 \leftarrow D_{\mathcal{R}, \sigma_0}$ and $e_2 \leftarrow D_{\mathcal{R}^{2k}, \sigma_1}$.
- in Game 5, the challenger samples $\mathbf{y} = d \cdot \begin{bmatrix} \mathbf{b} \\ 1 \end{bmatrix} \cdot v + \mathbf{e}'_0$ for $\mathbf{e}'_0 \leftarrow D_{\mathcal{R}^{k+1}, \sigma_0}$, calls the last element of the vector $\mathbf{y}$ as $\bar{y}$, and constructs $\mathbf{c}_1^* = \begin{bmatrix} \bar{y} \\ \mathbf{c}_1^* \end{bmatrix}$, where $\bar{\mathbf{c}}_1^* = \text{ReRand} \left(\begin{bmatrix} \mathbf{I}_k & \mathbf{R}_0 + \mathbf{R}_{\text{id}^*} \\ \mathbf{0}^\top & \mathbf{e}_{\text{id}^*}^\top \end{bmatrix}, \mathbf{y}, \sigma_0, \frac{\sigma_1}{2\sigma_0} \right)$.

Lemma 16. *By Lemma 3, we have $|\Pr[E_4] - \Pr[E_5]| = \text{negl}(n)$.*

Proof. In Game 4, we use the fact that when $\text{id} = \text{id}^*$, $G_{\text{id}}(\alpha, \beta) = 0$, i.e., $F(\text{id}^*)^\top = \mathbf{b}^\top(\mathbf{R}_0 + \mathbf{R}_{\text{id}^*}) + \mathbf{e}_{\text{id}^*}^\top$, then the challenge ciphertext $\mathbf{c}_1^*$ can be re-written as

$$\mathbf{c}_1^* = d \cdot \begin{bmatrix} 1 \\ \mathbf{b} \\ F(\text{id}^*) \end{bmatrix} \cdot v + \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} d \cdot v + e_1 \\ d \cdot \begin{bmatrix} \mathbf{b} \\ F(\text{id}^*) \end{bmatrix} \cdot v + e_2 \end{bmatrix}. \quad (15)$$

In Game 5, by the re-randomization algorithm [ReRand](#), we have $(\bar{\mathbf{c}}_1^*)^\top = d \cdot [\mathbf{b}^\top | 1] \cdot \begin{bmatrix} \mathbf{I}_k & \mathbf{R}_0 + \mathbf{R}_{\text{id}^*} \\ \mathbf{0}^\top & \mathbf{e}_{\text{id}^*}^\top \end{bmatrix} \cdot v + (\mathbf{e}'_2)^\top = d \cdot [\mathbf{b}^\top | \mathbf{b}^\top(\mathbf{R}_0 + \mathbf{R}_{\text{id}^*}) + \mathbf{e}_{\text{id}^*}^\top] \cdot v + (\mathbf{e}'_2)^\top = d \cdot [\mathbf{b}^\top | F(\text{id}^*)^\top] \cdot v + (\mathbf{e}'_2)^\top$ where the distribution of $\mathbf{e}'_2$ is statistically close to the discrete Gaussian distribution $D_{\mathcal{R}^{2k}, \sigma_1}$. By the construction of $\mathbf{c}_1^*$ in Game 5, we have

$$\mathbf{c}_1^* = \begin{bmatrix} \bar{y} \\ \bar{\mathbf{c}}_1^* \end{bmatrix} = \begin{bmatrix} d \cdot v + e'_1 \\ d \cdot \begin{bmatrix} \mathbf{b} \\ F(\text{id}^*) \end{bmatrix} \cdot v + \mathbf{e}'_2 \end{bmatrix}, \quad (16)$$

where e'_1 is the last element of $\mathbf{e}'_0$, i.e., $e'_1 \leftarrow D_{\mathcal{R}, \sigma_0}$. Comparing Eq. (15) and Eq. (16), Game 4 and Game 5 are statistically indistinguishable.

Game 6. In this game, we further change the way challenge ciphertext $(c_0^*, \mathbf{c}_1^*)$ is created when $\text{coin} = 0$.

- in Game 5, the challenger picks $\mathbf{y} = d \cdot [\mathbf{b}] \cdot v + \mathbf{e}'_0$ and sets $c_0^* = u \cdot v + e_0 + \lceil \frac{q}{2} \rceil \cdot \mu$, $\mathbf{c}_1^* = \begin{bmatrix} \bar{y} \\ \bar{\mathbf{c}}_1^* \end{bmatrix}$, where $\bar{\mathbf{c}}_1^* = \text{ReRand} \left(\begin{bmatrix} \mathbf{I}_k & \mathbf{R}_0 + \mathbf{R}_{\text{id}^*} \\ \mathbf{0}^\top & \mathbf{e}_{\text{id}^*}^\top \end{bmatrix}, \mathbf{y}, \sigma_0, \frac{\sigma_1}{2\sigma_0} \right)$.
- in Game 6, the challenger picks $y'_0 \xleftarrow{\$} \mathcal{R}_q$, $\mathbf{y}' \xleftarrow{\$} \mathcal{R}_q^{k+1}$, $c_0^* = y'_0 + e_0 + \lceil \frac{q}{2} \rceil \cdot \mu$, $\mathbf{c}_1^* = \begin{bmatrix} \bar{y} \\ \bar{\mathbf{c}}_1^* \end{bmatrix}$, where $\bar{\mathbf{c}}_1^* = \text{ReRand} \left(\begin{bmatrix} \mathbf{I}_k & \mathbf{R}_0 + \mathbf{R}_{\text{id}^*} \\ \mathbf{0}^\top & \mathbf{e}_{\text{id}^*}^\top \end{bmatrix}, \mathbf{y}' + \mathbf{e}'_0, \sigma_0, \frac{\sigma_1}{2\sigma_0} \right)$.

Lemma 17. *By the RLWE $_{n,k+2,q,D_{\mathcal{R},\sigma_0}}$ assumption (Def. 2), then we have that $|\Pr[E_5] - \Pr[E_6]| = \text{negl}(n)$.*

Game 7. In this game, we further change the way challenge ciphertext $\mathbf{c}_1^*$ is created when $\text{coin} = 0$. Specifically, the challenger picks $\mathbf{y}_1 \xleftarrow{\$} \mathcal{R}_q^k$, $y_2 \xleftarrow{\$} \mathcal{R}_q$, $\mathbf{e}'_2 \leftarrow D_{\mathcal{R}^{2k}, \sigma_1}$, and sets $\bar{\mathbf{c}}_1^* = [\mathbf{y}_1^\top | \mathbf{y}_1^\top(\mathbf{R}_0 + \mathbf{R}_{\text{id}^*}) + y_2 \cdot \mathbf{e}_{\text{id}^*}^\top] + (\mathbf{e}'_2)^\top$.

Lemma 18. *By Lemma 3, we have $|\Pr[E_6] - \Pr[E_7]| = \text{negl}(n)$.*

Game 8. In this game, we change the challenge ciphertext to be a random vector, regardless of whether $\text{coin} = 0$ or 1. It is obvious that $\Pr[E_8] = \frac{1}{2}$.

Lemma 19. *By Lemma 5, we have $|\Pr[E_7] - \Pr[E_8]| = \text{negl}(n)$.*

Analysis. Combining Lemmas 12 to 19, we have

$$\begin{aligned} \left| \Pr[E_8] - \frac{1}{2} \right| &= \left| \Pr[E_1] - \frac{1}{2} + \sum_{i=1}^7 (\Pr[E_{i+1}] - \Pr[E_i]) \right| \\ &\geq \left| \Pr[E_1] - \frac{1}{2} \right| - \sum_{i=1}^7 |\Pr[E_{i+1}] - \Pr[E_i]| \geq \frac{2}{3} \cdot \frac{1}{p^{t'}} \cdot \epsilon - \text{negl}(\lambda). \end{aligned}$$

Since ϵ is noticeable and Q is polynomially bounded, the value $p^{t'}$ is polynomially bounded, and thus $|\Pr[E_8] - \frac{1}{2}|$ is noticeable, which is contradicts to the fact that $\Pr[E_8] = \frac{1}{2}$. This completes the proof. $\square$

5.3 Asymptotic and Concrete Parameters

Due to space limit, we present the constraints on our IBE scheme's parameters in Appendix E. In this section, we present two possible sets of the parameters for our IBE construction in Tab. 2, concretely and asymptotically.

Table 2. Asymptotic and concrete parameters of our IBE scheme^(*).

Definition	Params	(Set I) $b = 2, \rho = 1$		(Set II) $b = \rho = n^{\nu(\natural)}$	
		Asymptotic	Concrete	Asymptotic	Concrete
security parameter	λ		λ		λ
identity length	l	$O(\lambda)$	$\leq 4\lambda$	$O(\lambda)$	$\leq 4\lambda$
bound on the number of key queries	Q	$\text{poly}(\lambda)$	Q	$\text{poly}(\lambda)$	Q
ring dimension	n	$\Theta(\lambda)$	n	$\Theta(\lambda)$	n
length of the gadget vector	k	$O(\log(\lambda))$	$2 \log q$	$O(1)$	$2 \log_b q$
repetition number in partition function	t	$\omega(1)$	$\log_c(3Q)^{(\ddagger)}$	$\omega(1)$	$\log_c(3Q)$
small positive real regarding smoothing parameter	ϵ	$\lambda^{-\omega(1)}$	$2^{-3\lambda}$	$\lambda^{-\omega(1)}$	$2^{-3\lambda}$
ECC parameters ^(*)	L	$O(\lambda^{1+\frac{2}{\kappa}})^{(\dagger)}$	$\frac{(\frac{\sqrt[n]{n}}{\log(\lambda)})^{2l}}{\log(\lambda)}$	$O(\lambda^{1+\frac{2}{\kappa}})$	$\frac{(\frac{\sqrt[n]{n}}{\log(\lambda)})^{2l}}{\log(\lambda)}$
	p	$O(\lambda^{\frac{1}{\kappa}})$	$n^{\frac{1}{\kappa}}$	$O(\lambda^{\frac{1}{\kappa}})$	$n^{\frac{1}{\kappa}}$
	$\mathcal{Y}$	$1 - O(\lambda^{-\frac{1}{\kappa}})$	$1 - \frac{\kappa+3}{\sqrt[n]{n}}$	$1 - O(\lambda^{-\frac{1}{\kappa}})$	$1 - \frac{\kappa+3}{\sqrt[n]{n}}$
Gaussian parameter of e_0, e_1 in IBE ciphertext c_0, c_1	σ_0	$O(\lambda^{0.5})$	σ_0	$O(\lambda^{0.5})$	σ_0
Gaussian parameter of $\mathbf{x}$ in sampling algorithms	σ	$\tilde{O}(\lambda^{1.5})$	$n^{2.5}$	$\tilde{O}(\lambda^{1.5} b^2)$	$n^{2.5} b^2$
Gaussian parameter of w in sampling algorithms	r	$\tilde{O}(\lambda^{5+\frac{2}{\kappa}})$	$n^{6.5+\frac{2}{\kappa}} \lambda$	$\tilde{O}(\lambda^{5+\frac{2}{\kappa}} b^4)$	$n^{6.5+\frac{2}{\kappa}} \lambda b^4$
Gaussian parameter of $\mathbf{e}_2$ in IBE ciphertext c_1	σ_1	$\tilde{O}(\lambda^{5.5+\frac{2}{\kappa}})$	$n^{5+\frac{2}{\kappa}} \cdot \sigma_0$	$\tilde{O}(\lambda^{5.5+\frac{2}{\kappa}} b^4)$	$n^{5+\frac{2}{\kappa}} b^4 \cdot \sigma_0$
system modulus	q	$\tilde{O}(\lambda^{7.5+\frac{2}{\kappa}})$	$n^{9+\frac{2}{\kappa}} \cdot \sigma_0$	$\tilde{O}(\lambda^{7.5+\frac{2}{\kappa}} b^6)$	$n^{9+\frac{2}{\kappa}} b^6 \cdot \sigma_0$

* we note that we present the parameters in the *plain model* here. As said in Appendix E, we can simply calculate the corresponding parameters in the *CRS model*, only the parameters r, σ_1, q would be changed.

* we set the parameter in error correcting code $L, p, \mathcal{Y}$ refer to ALWW-IBE [1].

$\dagger \kappa > 1$ can be any constant that satisfies $n^{\frac{1}{\kappa}} > 3 + \kappa$, depending on how we set parameters of the error correcting code.

$\ddagger c > 1$ is a parameter defined by the relative distance as $c := \frac{1}{1-\overline{r}}$.

$\natural \nu > 0$ is any positive real.

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A Omitted Lemmas and Corollary in Sect. 2

Lemma 20 ([21, 29]). *For $\sigma > \eta_\epsilon(\mathbb{Z})$, $t \geq 0$, $\Pr_{x \leftarrow D_{\mathbb{Z}, \sigma}}[|x| \geq t] \leq 2e^{-\pi \cdot \frac{t^2}{\sigma^2}}$.*

The upper bound of $\eta_\epsilon(\Lambda)$ is given by the well-known lemma below.

Lemma 21 ([21]). *For any n -dimensional lattice Λ with basis $\mathbf{B}$ and any real $\epsilon > 0$, we have $\eta_\epsilon(\Lambda) \leq \|\tilde{\mathbf{B}}\| \cdot \sqrt{\log(2n(1 + 1/\epsilon))/\pi}$. Then for any $\omega(\sqrt{\log n})$ function, there is a negligible function $\epsilon(n)$ for which $\eta_\epsilon(\Lambda) \leq \|\tilde{\mathbf{B}}\| \cdot \omega(\sqrt{\log n})$.*

Lemma 22 ([25]). *Let q be a prime such that $q \equiv 3 \pmod{8}$ and n be a power of 2. Then, all $a \in \mathcal{R}_q$ satisfying $\|\phi(a)\| < \sqrt{q}$ are invertible, i.e., $a \in \mathcal{R}_q^\times$.*

Lemma 23 ([25]). *Let α be the positive real, k be an integer. Let $q \equiv 3 \pmod{8}$ be a prime such that there is another prime $p \equiv 1 \pmod{m}$ satisfying $p \leq q \leq 2p$. Let $\sigma := \alpha q \geq n^{3/2} k^{1/4} \omega(\log^{9/4}(n))$. Then, there is a PPT quantum reduction from $\tilde{O}(n/\alpha)$ -approximate SIVP (or SVP) to $\text{RLWE}_{n,k,q,\chi}$ with $\chi = D_{\mathcal{R},\sigma}$.*

Lemma 24 ([1]). *For integers k, q, b satisfying the requirements of Def. 3, on input a vector $\mathbf{c} \in \mathbb{Z}_q^k \subset \mathcal{R}_q^k$, the algorithm $\mathbf{g}^{-1}(\cdot)$ in Lemma 6 outputs the matrix $\mathbf{R} = \mathbf{g}^{-1}(\mathbf{c}^\top)$ such that $\mathbf{R} \in [-b, b]_{\mathbb{Z}}^{k \times k}$, $\mathbf{g}^\top \cdot \mathbf{R} = \mathbf{c}^\top$ and $\|\mathbf{R}\| \leq kb$.*

Corollary 3. *For the gadget matrix $\mathbf{G} = \begin{bmatrix} \mathbf{g}^\top & \mathbf{0}^\top \\ \mathbf{0}^\top & \mathbf{g}^\top \end{bmatrix} \in \mathcal{R}_q^{2 \times 2k}$, on input a matrix $\mathbf{C} \in \mathbb{Z}_q^{2 \times 2k} \subset \mathcal{R}_q^{2 \times 2k}$, the algorithm $\mathbf{G}^{-1}(\cdot)$ in Corollary 2 outputs the matrix $\mathbf{R} = \mathbf{G}^{-1}(\mathbf{C})$ such that $\mathbf{R} \in [-b, b]_{\mathbb{Z}}^{2k \times 2k}$, $\mathbf{G} \cdot \mathbf{R} = \mathbf{C}$ and $\|\mathbf{R}\| \leq 2kb$.*

B Omitted Details in Sect. 3

B.1 Construction 2 and Proof of Theorem 1

According to [1], the equality test function in Def. 5 can be computed as $\text{Equal}_\beta(x^\alpha) = \frac{1}{m} \cdot \sum_{i=0}^{m-1} (x^{\alpha-\beta})^i$.

Construction 2 *We present algorithms $(\text{PubEval}_E, \text{TrapEval}_E)$ for $\{\text{Equal}_\beta(\cdot)\}_{\beta \in [m]}$ as follows.*

$\text{PubEval}_E(\{\mathbf{C}_\alpha\}, \text{Equal}_\beta)$

- compute $\mathbf{C}' = \mathbf{C}_\alpha \cdot x^{-\beta}$.
- compute $\mathbf{C}_j$ recursively as follows:

$$\mathbf{C}_j = \begin{cases} \mathbf{G}, & j = 0 \\ \mathbf{C}' \cdot \mathbf{G}^{-1}(\mathbf{C}_{j-1}) + \mathbf{G}, & j \geq 1 \end{cases}$$

- output $\mathbf{C}_{m-1} \cdot \mathbf{G}^{-1}(m^{-1} \cdot \mathbf{G})$.

$\text{TrapEval}_E(\mathbf{A}, \{\mathbf{R}_\alpha\}, \{x^\alpha\}, \text{Equal}_\beta)$

- compute $\mathbf{R}' = \mathbf{R}_\alpha \cdot x^{-\beta}$.
- Let $\mathbf{C}_j$'s be matrices as defined in the PubEval_E and $\mathbf{C}_\alpha = \mathbf{A}\mathbf{R}_\alpha + x^\alpha \mathbf{G}$, compute $\mathbf{R}_j$ recursively as follows:

$$\mathbf{R}_j = \begin{cases} \mathbf{0}, & j = 0 \\ \mathbf{R}' \cdot \mathbf{G}^{-1}(\mathbf{C}_{j-1}) + x^{\alpha-\beta} \cdot \mathbf{R}_{j-1}, & j \geq 1 \end{cases}$$

- output $\mathbf{R}_{m-1} \cdot \mathbf{G}^{-1}(m^{-1} \cdot \mathbf{G})$.

Proof of Theorem 1.

Proof. To prove the theorem, we need to show that the 2-norm of the output of the TrapEval_E is bounded by $4mn(kb)^2\|\mathbf{R}_\alpha\|$, and equivalently

$$\|\mathbf{R}_{m-1} \cdot \mathbf{G}^{-1}(m^{-1} \cdot \mathbf{G})\| \leq 4mn(kb)^2 \cdot \|\mathbf{R}_\alpha\|.$$

Since we have the matrix norm $\mathbf{G}^{-1}(m^{-1} \cdot \mathbf{G}) \leq 2kb$ by Corollary 3, and thus proving $\|\mathbf{R}_{m-1}\| \leq 2mnkb \cdot \|\mathbf{R}_\alpha\|$ is enough to complete the proof.

Next, we prove a slightly more general statement by induction: for $j \in [m]$, $\|\mathbf{R}_j\| \leq 2(j+1)nb \cdot \|\mathbf{R}_\alpha\|$, and thus by taking $j = m-1$, we complete the proof.

For the base case $j = 0$, this statement obviously holds as $0 \leq 2nb \cdot \|\mathbf{R}_\alpha\|$ holds trivially as $\mathbf{R}_0 = \mathbf{0}$. We next assume the induction hypothesis that $\|\mathbf{R}_{j^*-1}\| \leq 2j^*nb \cdot \|\mathbf{R}_\alpha\|$ holds for some $j^* - 1$. Our goal is to prove that the statement also holds for j^* , thus implying what we want by induction.

Next, we have

$$\begin{aligned} \|\mathbf{R}_{j^*}\| &= \|\mathbf{R}' \cdot \mathbf{G}^{-1}(\mathbf{C}_{j^*-1}) + x^{\alpha-\beta} \cdot \mathbf{R}_{j^*-1}\| \\ &\leq \|\mathbf{R}' \cdot \mathbf{G}^{-1}(\mathbf{C}_{j^*-1})\| + \|\mathbf{R}_{j^*-1}\| \\ &\leq \|\mathbf{R}'\| \cdot 2nb + 2j^*nb \cdot \|\mathbf{R}_\alpha\| \\ &\leq 2(j^*+1)nb \cdot \|\mathbf{R}_\alpha\|. \end{aligned}$$

The first inequality is from the triangular inequality; the second is by Corollary 2 and the induction hypothesis; the last inequality is by $\|\mathbf{R}'\| = \|\mathbf{R}_\alpha\|$. Thus, we have $\|\mathbf{R}_{m-1}\| \leq 2mnkb \cdot \|\mathbf{R}_\alpha\|$. This complete the proof. $\square$

B.2 Supplementary for the Partition Property of the Partition Function in Sect. 3.2

First, we recall an insight from the work [3], stating that the IBE design with the trapdoor vanishing technique indeed only needs (weak) pairwise independent hash functions plus the random isolation technique of Valiant and Vazirani [35], which can generically replace the prior notions “admissible hash functions” [7, 14] or “abort-resistant hash functions” [36]. To summarize this insight, we state the following lemma.

Lemma 25 ([3]). *Let $Q \subset \{0, 1\}^n$ be an arbitrary subset, A, B be integers such that $B \leq A$, $|Q| \leq \delta B$ for some $\delta \in (0, 1)$, and let $\mathcal{H} : \{0, 1\}^n \rightarrow \mathcal{Y}$ be an almost pairwise independent hash function family which has the following properties:*

1. $\forall x \in \{0, 1\}^n, \Pr_{h \in \mathcal{H}}[h(x) = 0] = \frac{1}{A}$.
2. *For any distinct $x_1 \neq x_2 \in \{0, 1\}^n$, $\Pr_{h \in \mathcal{H}}[h(x_1) = 0 | h(x_2) = 0] < \frac{1}{B}$.*

Then for any element $x \notin Q$, we have

$$\Pr_{h \in \mathcal{H}}[h(x) = 0 \wedge (h(x') \neq 0 \forall x' \in Q)] \in \left(\frac{1-\delta}{A}, \frac{1}{A}\right).$$

Then the following lemma shows that the partition function family defined in Def. 8 is an almost pairwise independent hash function family.

Lemma 26 ([1], Lemma 4.5). *For any code $\text{ECC} : D \rightarrow \mathbb{Z}_p^L$ with relative distance Υ , ring $\mathcal{R}$ with dimension n such that $tp \leq n$. Then for any $t' \leq t$, the hash function family $\mathcal{H}^{\mathcal{R},t,t'}$ as in Def. 8 has following properties:*

1. *For any element $z_1 \in D$, $\Pr_{H \in \mathcal{H}^{\mathcal{R},t,t'}}[H(z_1) = 0] = (1/p)^{t'}$.*
2. *For any distinct elements $z_1 \neq z_2 \in D$, we have*

$$\Pr_{H \in \mathcal{H}^{\mathcal{R},t,t'}}[H(z_1) = 0 | H(z_2) = 0] < (1 - \Upsilon)^{t'}.$$

B.3 Omitted derivation process in Sect. 3.2

$$\begin{aligned} H_{\alpha,\beta}^{\mathcal{R},t}(z) &= - \sum_{i \in [t]} x^{ip+\beta_i} + \sum_{i \in [t]} \sum_{j \in [L+1]} (j \stackrel{?}{=} \alpha_i) \cdot x^{ip+\text{ECC}(z)[j]} \\ &= - \sum_{i \in [t]} x^{ip+\beta_i} + \sum_{i \in [t]} \sum_{j \in [L+1]} \left(\prod_{i' \in [\eta]} j[i'] \stackrel{?}{=} \alpha_{i,i'} \right) \cdot x^{ip+\text{ECC}(z)[j]} \\ &= -\tilde{\beta} + \sum_{i \in [t]} \sum_{j \in [L+1]} \left(\prod_{i' \in [\eta]} \text{Equal}_{j[i']}(\alpha_{i,i'}) \right) \cdot x^{ip+\text{ECC}(z)[j]} \\ &= G_z(\{\alpha_{i,i'}\}_{i \in [t], i' \in [\eta]}, \tilde{\beta}). \end{aligned}$$

B.4 Construction 3 and Proof of Theorem 2

Construction 3 *Given $(\text{PubEval}_E, \text{TrapEval}_E)$ for $\{\text{Equal}_\beta(\cdot)\}_{\beta \in [m]}$ as subroutine, we construct $(\text{PubEval}_G, \text{TrapEval}_G)$ for $\{G_z\}_{z \in \mathcal{D}}$ (defined in Def. 7) as:*

$$\text{PubEval}_G(\{\{\mathbf{C}_{\alpha_{i,i'}}\}_{i \in [t], i' \in [\eta]}, \mathbf{C}_{\tilde{\beta}}\}, G_z)$$

– for $i \in [t]$, $j \in [L+1]$, $i' \in [\eta]$, (homomorphically) compute

$$\mathbf{C}_{i,j,i'} = \begin{cases} \text{PubEval}_E(\mathbf{C}_{\alpha_{i,0}}, \text{Equal}_{j[0]}), & i' = 0 \\ \text{PubEval}_E(\mathbf{C}_{\alpha_{i,i'}}, \text{Equal}_{j[i']}) \cdot \mathbf{G}^{-1}(\mathbf{C}_{i,j,i'-1}), & i' \geq 1 \end{cases}$$

Then, let $\mathbf{C}_{i,j} := \mathbf{C}_{i,j,\eta-1}$.

– output $\mathbf{C}_G := -\mathbf{C}_{\tilde{\beta}} + \sum_{i \in [t], j \in [L+1]} \mathbf{C}_{i,j} \cdot x^{ip+\text{ECC}(z)[j]}$.

$$\text{TrapEval}_G(\mathbf{A}, \{\{\mathbf{R}_{\alpha_{i,i'}}\}_{i \in [t], i' \in [\eta]} \subset \mathcal{R}_q^{2k \times 2k}, \mathbf{R}_{\tilde{\beta}} \in \mathcal{R}_q^{2k \times 2k}\}, \{x^\alpha, \tilde{\beta}\}, G_z)$$

– for $i \in [t]$, $j \in [L+1]$, $i' \in [\eta]$, (homomorphically) compute

$$\mathbf{R}'_{i,j,i'} := \text{TrapEval}_E(\mathbf{A}, \{\mathbf{R}_{\alpha_{i,i'}}\}, \{x^{\alpha_{i,i'}}\}, \text{Equal}_{j[i']}).$$

- for $i' \in [\eta]$, let $\mathbf{C}_{i,j,i'}$ be the matrix in PubEval_G algorithm with $\mathbf{C}_{\alpha_{i,i'}} = \mathbf{A}\mathbf{R}_{\alpha_{i,i'}} + x^{\alpha_{i,i'}}\mathbf{G}$, recursively compute

$$\mathbf{R}_{i,j,i'} = \begin{cases} \mathbf{R}'_{i,j,i'}, & i' = 0 \\ \mathbf{R}'_{i,j,i'} \cdot \mathbf{G}^{-1}(\mathbf{C}_{i,j,i'-1}) + \text{Equal}_{j[i']}(x^{\alpha_{i,i'}}) \cdot \mathbf{R}_{i,j,i'-1}, & i' \geq 1 \end{cases}$$

Then, let $\mathbf{R}_{i,j} := \mathbf{R}_{i,j,\eta-1}$.

- output $\mathbf{R}_G := -\mathbf{R}_{\tilde{\beta}} + \sum_{i \in [t], j \in [L+1]} \mathbf{R}_{i,j}$.

Proof of Theorem 2.

Proof. To prove the theorem, it suffices to prove $\|\mathbf{R}_G\| \leq (L+1)t \cdot \eta \cdot nkb \cdot \delta \cdot \max_{i \in [t], i' \in [\eta]} \{\|\mathbf{R}_{\alpha_{i,i'}}\|, \|\mathbf{R}_{\tilde{\beta}}\|\}$. Since the algorithms (PubEval_E , TrapEval_E) are δ -expanding for function family $\{\text{Equal}_j\}_{j \in [m]}$, we have $\|\mathbf{R}'_{i,j,i'}\| \leq \delta \cdot \max_{i \in [t], i' \in [\eta]} \{\|\mathbf{R}_{\alpha_{i,i'}}\|\}$ for each $i \in [t]$, $j \in [L+1]$, $i' \in [\eta]$. Additionally, we can show $\|\mathbf{R}_{i,j}\| \leq \eta \cdot 2nkb \cdot \delta \cdot \max_{i \in [t], i' \in [\eta]} \{\|\mathbf{R}_{\alpha_{i,i'}}\|\}$ by similar analysis by Theorem 1. Further, it is easy to see that

$$\begin{aligned} \|\mathbf{R}_G\| &\leq \sum_{i \in [t], j \in [L+1]} \|\mathbf{R}_{i,j}\| + \|\mathbf{R}_{\tilde{\beta}}\| \leq (L+1)t \cdot \max_{i \in [t], j \in [L+1]} \{\|\mathbf{R}_{i,j}\|\} + \|\mathbf{R}_{\tilde{\beta}}\| \\ &\leq (L+1)t \cdot \eta \cdot 2nkb \cdot \delta \cdot \max_{i \in [t], i' \in [\eta]} \{\|\mathbf{R}_{\alpha_{i,i'}}\|, \|\mathbf{R}_{\tilde{\beta}}\|\}. \end{aligned}$$

This completes the proof. $\square$

C Omitted details in Sect. 4

C.1 Recall sampling algorithms from [2] for Sect. 4

In [21], the Ajtai trapdoor generation algorithm outputs a pair $(\mathbf{b}, \mathbf{T}_{\mathbf{b}}) \in \mathcal{R}_q^k \times \mathcal{R}^{k \times k}$ satisfying that $\mathbf{b}^\top \cdot \mathbf{T}_{\mathbf{b}} = \mathbf{0}^\top \bmod q$. Later, to improve the efficiency of lattice-based IBE in the standard model, Agrawal, Boneh, and Boyen [2] constructed a family of lattices for which there are two distinct trapdoors for sampling, one is the Ajtai trapdoor, the other one is a “small-norm” randomization matrix $\mathbf{R}$, used in the real world and simulated world, respectively. More concretely, they construct the sampling vector as

$$\mathbf{f}^\top = [\mathbf{b}^\top | \mathbf{b}^\top \mathbf{R} + y\mathbf{g}^\top] \in \mathcal{R}_q^{2k}, \quad (17)$$

where $\mathbf{R} \in \mathcal{R}^{k \times k}$ is a secret trapdoor matrix with small, random entries, $y \in \mathcal{R}_q^\times$ is an invertible element, and $\mathbf{g} = [1|b|\dots|b^{k-1}]^\top \in \mathcal{R}_q^k$ is the specific gadget vector [29], whose trapdoor $\mathbf{T}_{\mathbf{g}}$ is publicly known. For a given uniform $u \in \mathcal{R}_q$, it is easy to sample a short vector $\mathbf{x}$ that satisfying

$$\mathbf{f}^\top \mathbf{x} = u \bmod q.$$

The sampling algorithms consist of two parts: `SampleLeft` and `SampleRight`. They use the `SamPre` algorithm in Lemma 8.

SampleLeft($\mathbf{b}, \mathbf{c} \in \mathcal{R}_q^k, \mathbf{T}_{\mathbf{b}} \in \mathcal{R}^{k \times k}, u \in \mathcal{R}_q, \sigma \in \mathbb{R}$)

- sample $\mathbf{x}_2 \leftarrow D_{\mathcal{R}^k, \sigma}$.
- run $\mathbf{x}_1 \leftarrow \text{SamPre}(\mathbf{b}, \mathbf{T}_{\mathbf{b}}, u', \sigma)$ where $u' = u - \mathbf{c}^\top \mathbf{x}_2$.
- output $\mathbf{x} = [\mathbf{x}_1^\top] \in \mathcal{R}_q^{2k}$.

SampleRight($\mathbf{b}, \mathbf{g} \in \mathcal{R}_q^k, \mathbf{R}, \mathbf{T}_{\mathbf{g}} \in \mathcal{R}^{k \times k}, y \in \mathcal{R}_q^\times, u \in \mathcal{R}_q, \sigma, s \in \mathbb{R}$)

- sample $\mathbf{p} \leftarrow D_{\mathcal{R}^{2k}, \sqrt{\Sigma_p}}$ where $\Sigma_p := \sigma^2 \mathbf{I} - s^2 \mathbf{T} \mathbf{T}^\top$ for $\mathbf{T} = \begin{bmatrix} -\mathbf{R} \\ \mathbf{I} \end{bmatrix}$.
- compute $v = y^{-1}(u - \mathbf{f}^\top \cdot \mathbf{p})$ where $\mathbf{f}$ is computed using Eq. (17).
- run $\mathbf{z} \leftarrow \text{SamPre}(\mathbf{g}, \mathbf{T}_{\mathbf{g}}, v, s)$.
- output $\mathbf{x} = \mathbf{p} + \mathbf{T}\mathbf{z}$.

Lemma 27 ([2]). For $s \geq \eta_\epsilon(\mathbb{Z}^{nk}) \cdot \|\widetilde{\mathbf{T}_{\mathbf{g}}}\|$, $\sigma^2 \geq (s^2 + \eta_\epsilon^2(\mathbb{Z}^{nk})) \cdot (\|\mathbf{T}\|^2 + 1)$, $\sigma \geq \eta_\epsilon(\mathbb{Z}^{nk}) \cdot \|\widetilde{\mathbf{T}_{\mathbf{b}}}\|$, the distributions of outputs from **SampleLeft** and **SampleRight** are statistically close.

C.2 Full derivation process of Eq. (14)

$$\begin{aligned}
u &= u^* - h - \mathbf{e}^\top \bar{\mathbf{p}} \\
&= yv + \mathbf{f}^\top \mathbf{p} - h - \mathbf{e}^\top \bar{\mathbf{p}} \\
&= y\mathbf{g}^\top \mathbf{z} + \mathbf{f}^\top \mathbf{p} - h - \mathbf{e}^\top \bar{\mathbf{p}} \\
&= (y\mathbf{g}^\top + \mathbf{e}^\top) \mathbf{z} - \mathbf{e}^\top \mathbf{z} + \mathbf{f}^\top \mathbf{p} - h - \mathbf{e}^\top \bar{\mathbf{p}} \\
&= (\mathbf{f}^\top \mathbf{T} \mathbf{z} + \mathbf{f}^\top \mathbf{p}) - (h + \mathbf{e}^\top \bar{\mathbf{p}} + \mathbf{e}^\top \mathbf{z}) \\
&= \mathbf{f}^\top \mathbf{x} - w.
\end{aligned}$$

D Omitted Proofs in Theorem 7

Proof of Lemma 12

Proof. Define $\gamma := \Pr[H(\text{id}^*) = 0 \wedge H(\text{id}_i) \neq 0 \forall i \leq Q]$, then we have

$$\begin{aligned}
\gamma &= \Pr[H(\text{id}^*) = 0 \wedge H(\text{id}_1) \neq 0 \wedge \dots \wedge H(\text{id}_Q) \neq 0] \\
&= \Pr[H(\text{id}^*) = 0] - \Pr[H(\text{id}^*) = 0 \wedge (H(\text{id}_1) = 0 \vee \dots \vee H(\text{id}_Q) = 0)] \\
&= \Pr[H(\text{id}^*) = 0] - \Pr[(H(\text{id}^*) = 0 \wedge H(\text{id}_1) = 0) \vee \dots \vee (H(\text{id}^*) = 0 \wedge H(\text{id}_Q) = 0)] \\
&\geq \Pr[H(\text{id}^*) = 0] - \sum_{i=1}^{Q'} \Pr[H(\text{id}^*) = 0 \wedge H(\text{id}_i) = 0] \\
&= \Pr[H(\text{id}^*) = 0] - \sum_{i=1}^{Q'} \Pr[H(\text{id}_i) = 0 | H(\text{id}^*) = 0] \cdot \Pr[H(\text{id}^*) = 0] \\
&= \Pr[H(\text{id}^*) = 0] \left(1 - \sum_{i=1}^Q \Pr[H(\text{id}_i) = H(\text{id}^*)] \right) \\
&= \frac{1}{p^{t'}} (1 - Q \cdot (1 - \gamma)^{t'}).
\end{aligned}$$

Since $|\Pr[E_1] - \frac{1}{2}| \geq \gamma_{\min} \cdot |\Pr[E_0] - \frac{1}{2}| - \frac{1}{2}(\gamma_{\max} - \gamma_{\min})$, $\gamma_{\max} = \Pr[H(\text{id}^*) = 0] = \frac{1}{p^{t'}}$, $\gamma_{\min} = \frac{1}{p^{t'}}(1 - Q \cdot (1 - \gamma)^{t'})$, we have $\gamma_{\max} - \gamma_{\min} = \frac{1}{p^{t'}} \cdot Q \cdot (1 - \gamma)^{t'}$. Then,

$$\left| \Pr[E_1] - \frac{1}{2} \right| \geq \frac{1}{p^{t'}}(1 - Q \cdot (1 - \gamma)^{t'}) \cdot \epsilon - \frac{1}{2} \cdot \frac{1}{p^{t'}} \cdot Q \cdot (1 - \gamma)^{t'}.$$

Since we set $t' = \log_{\frac{1}{1-\gamma}} \frac{3Q}{\epsilon}$, then $(\frac{1}{1-\gamma})^{t'} = \frac{3Q}{\epsilon}$, i.e., $Q \cdot (1 - \gamma)^{t'} = \frac{\epsilon}{3}$,

$$\left| \Pr[E_1] - \frac{1}{2} \right| \geq \frac{1}{p^{t'}} \cdot (1 - \frac{\epsilon}{3}) \cdot \epsilon - \frac{1}{2} \cdot \frac{1}{p^{t'}} \cdot \frac{\epsilon}{3} \geq \frac{5}{6} \cdot \frac{1}{p^{t'}} \cdot \epsilon - \frac{1}{6} \cdot \frac{1}{p^{t'}} \cdot \epsilon = \frac{2}{3} \cdot \frac{1}{p^{t'}} \cdot \epsilon.$$

where $\epsilon < \frac{1}{2}$.

Proof of Lemma 13

Proof. Now we show that the distributions of Game 1 and Game 2 are close by the $\text{RLWE}_{n,2k,q,\chi}$ assumption (Def. 2) and the regularity lemma (Lemma 5). First, we re-write Game 1 and Game 2, and define two intermediate Games, called Game 1+ and Game 2+.

Game 1. In this Game, the challenger first picks $\mathbf{a} \xleftarrow{\$} \mathcal{R}_q^{2k}$, $s \xleftarrow{\$} \mathcal{R}_q$, $\mathbf{e} \leftarrow \chi^{2k}$, and sets $\mathbf{A} = \begin{bmatrix} \mathbf{a}^\top & \mathbf{a}^\top \end{bmatrix} \in \mathcal{R}_q^{2 \times 2k}$. Then, it samples random matrices $\mathbf{R}_{\alpha_{i,i'}}, \mathbf{R}_{\tilde{\beta}} \xleftarrow{\$} [-\rho, \rho]_{\mathcal{R}}^{2k \times 2k}$, and generates the public matrices as follows

$$\mathbf{C}_{\alpha_{i,i'}} := \mathbf{A} \mathbf{R}_{\alpha_{i,i'}} + 0 \cdot \mathbf{G}, \quad \mathbf{C}_{\tilde{\beta}} := \mathbf{A} \mathbf{R}_{\tilde{\beta}} + 0 \cdot \mathbf{G}.$$

Game 1+. In this Game, the challenger first picks random matrix $\mathbf{A} \xleftarrow{\$} \mathcal{R}_q^{2 \times 2k}$, samples random matrices $\mathbf{R}_{\alpha_{i,i'}}, \mathbf{R}_{\tilde{\beta}} \xleftarrow{\$} [-\rho, \rho]_{\mathcal{R}}^{2k \times 2k}$, and generates the public matrices as follows

$$\mathbf{C}_{\alpha_{i,i'}} := \mathbf{A} \mathbf{R}_{\alpha_{i,i'}} + 0 \cdot \mathbf{G}, \quad \mathbf{C}_{\tilde{\beta}} := \mathbf{A} \mathbf{R}_{\tilde{\beta}} + 0 \cdot \mathbf{G}.$$

Game 2+. In this Game, the challenger first picks random matrix $\mathbf{A} \xleftarrow{\$} \mathcal{R}_q^{2 \times 2k}$, samples random matrices $\mathbf{R}_{\alpha_{i,i'}}, \mathbf{R}_{\tilde{\beta}} \xleftarrow{\$} [-\rho, \rho]_{\mathcal{R}}^{2k \times 2k}$, and generates the public matrices as follows

$$\mathbf{C}_{\alpha_{i,i'}} := \mathbf{A} \mathbf{R}_{\alpha_{i,i'}} + x^{\alpha_{i,i'}} \cdot \mathbf{G}, \quad \mathbf{C}_{\tilde{\beta}} := \mathbf{A} \mathbf{R}_{\tilde{\beta}} + \sum_{i \in [t]} x^{ip + \beta_i} \cdot \mathbf{G}.$$

Game 2. In this Game, the challenger first picks $\mathbf{a} \xleftarrow{\$} \mathcal{R}_q^{2k}$, $s \xleftarrow{\$} \mathcal{R}_q$, $\mathbf{e} \leftarrow \chi^{2k}$, and sets $\mathbf{A} = \begin{bmatrix} \mathbf{a}^\top & \mathbf{a}^\top \end{bmatrix} \in \mathcal{R}_q^{2 \times 2k}$. Then, it samples random matrices $\mathbf{R}_{\alpha_{i,i'}}, \mathbf{R}_{\tilde{\beta}} \xleftarrow{\$} [-\rho, \rho]_{\mathcal{R}}^{2k \times 2k}$, and generates the public matrices as follows

$$\mathbf{C}_{\alpha_{i,i'}} := \mathbf{A} \mathbf{R}_{\alpha_{i,i'}} + x^{\alpha_{i,i'}} \cdot \mathbf{G}, \quad \mathbf{C}_{\tilde{\beta}} := \mathbf{A} \mathbf{R}_{\tilde{\beta}} + \sum_{i \in [t]} x^{ip + \beta_i} \cdot \mathbf{G}.$$

By the regularity lemma (Lemma 5), Game 1+ and Game 2+ are statistically indistinguishable. Then we show that the distributions of Game 1 and Game 1+ are computationally indistinguishable by the $\text{RLWE}_{n,2k,q,\chi}$ assumption (Def. 2). We first define an RLWE adversary $\mathcal{B}$ as follows. Upon receiving $2k$ challenge samples $\{(a_i, z_i)\}_{i \in [2k]} \in (\mathcal{R}_q \times \mathcal{R}_q)^{2k}$, the adversary $\mathcal{B}$ simulates the IBE experiment by setting the public matrix $\mathbf{A} = \begin{bmatrix} \mathbf{a}^\top \\ \mathbf{z}^\top \end{bmatrix} \in \mathcal{R}_q^{2 \times 2k}$, sampling random matrices $\mathbf{R}_{\alpha_{i,i'}}, \mathbf{R}_{\tilde{\beta}} \xleftarrow{\$} [-\rho, \rho]_{\mathcal{R}}^{2k \times 2k}$, generating the public matrices

$$\mathbf{C}_{\alpha_{i,i'}} := \mathbf{A} \mathbf{R}_{\alpha_{i,i'}} + 0 \cdot \mathbf{G}, \quad \mathbf{C}_{\tilde{\beta}} := \mathbf{A} \mathbf{R}_{\tilde{\beta}} + 0 \cdot \mathbf{G},$$

and simulating the rest of the experiment according to Game 1. At the end of the game, if $\mathcal{A}$ successfully guesses the correct (simulated) challenge bit from $\mathcal{B}$, then $\mathcal{B}$ guesses that the challenge samples are RLWE samples and outputs 1; otherwise, $\mathcal{B}$ guesses the uniform distribution and outputs 0. It can be seen that $\mathcal{B}$ perfectly simulates the view of $\mathcal{A}$ in Game 1 if $\{(a_i, z_i)\}_{i \in [2k]}$ are valid RLWE samples (i.e., $z_i = a_i s + e_i$) and Game 1+ otherwise (i.e., $z_i \xleftarrow{\$} \mathcal{R}_q$). We therefore conclude that Game 1 and Game 1+ are indistinguishable by the $\text{RLWE}_{n,2k,q,\chi}$ assumption. Similarly, Game 2 and Game 2+ are indistinguishable. Thus, Game 1 and Game 2 are indistinguishable by the $\text{RLWE}_{n,2k,q,\chi}$ assumption and the regularity lemma.

Proof of Lemma 14

Proof. By Lemma 7, the vector $\mathbf{b}$ generated by **TrapGen** algorithm is statistically close to the uniform distribution over $\mathcal{R}_q^k$. Then, for a truly uniform $\mathbf{b}$, the distributions of $(\mathbf{b}, \mathbf{b}^\top \mathbf{R}_0, \mathbf{b}^\top \mathbf{R}_s)$ and $(\mathbf{b}, U(\mathcal{R}_q^k), U(\mathcal{R}_q^k))$ are statistically close by the regularity lemma (Lemma 5). Thus, Game 2 and Game 3 are statistically indistinguishable.

Proof of Lemma 15

Proof. By Lemma 7, the vector $\mathbf{b}$ generated by **TrapGen** algorithm is statistically close to the uniform distribution over $\mathcal{R}_q^k$. By Theorem 6, the distributions of the outputs from **SampleLeft_{hybrid}** and **SampleRight_{hybrid}** algorithms are statistically close. Thus, Game 3 and Game 4 are statistically indistinguishable.

Proof of Lemma 17

Proof. Now we show that the distributions of Game 5 and Game 6 are close by the $\text{RLWE}_{n,k+2,q,D_{\mathcal{R},\sigma_0}}$ assumption (Def. 2). We first define an RLWE adversary $\mathcal{B}'$ as follows.

Instance. $\mathcal{B}'$ is given the RLWE instance $\{(u_i, y_i)\}_{i=0}^{k+1} \in \mathcal{R}_q \times \mathcal{R}_q$. If u_{k+1} is not invertible, then aborts. $\mathcal{B}'$ simulates the IBE experiment.

Setup. To generate the master public key mpk , $\mathcal{B}'$ executes as follows:

- $pp \leftarrow \text{GSW.Setup}(1^\lambda)$.

- $(\mathbf{A}, s) \leftarrow \text{GSW.KeyGen}(pp)$.
- sample $\{\{\mathbf{C}_{\alpha_{i,i'}}\}, \mathbf{C}_{\tilde{\beta}}\}$ as in Game 2.
- set $d := u_{k+1}$, $\mathbf{b} := u_{k+1}^{-1} \cdot [u_1 | \cdots | u_k]$, $u := u_0$.
- sample $\hat{\mathbf{c}}, \mathbf{c}_s$ as in Game 3.
- use the y_0 in the RLWE challenge as the y_0 in the game, then set $\mathbf{y} := [y_1 | \cdots | y_{k+1}]$, $\bar{y} := y_{k+1}$.

Then it outputs $\text{mpk} := (\mathbf{b}, \{\mathbf{C}_{\alpha_{i,i'}}\}, \mathbf{C}_{\tilde{\beta}}, \hat{\mathbf{c}}, \mathbf{c}_s, \mathbf{A}, d, u)$, which simulates the setup phase.

Phase 1 & Phase 2. When $\mathcal{A}$ makes the key extraction queries, $\mathcal{B}'$ responds as in Game 4.

Challenge. When $\mathcal{A}$ makes the challenge query for the challenge identity id^* and a message m , $\mathcal{B}'$ sets the challenge ciphertext as

$$c_0^* = y_0 + \lceil \frac{q}{2} \rceil \cdot \mu, \quad \mathbf{c}_1^* = \left[\text{ReRand} \left(\begin{bmatrix} \mathbf{I}_k & \mathbf{R}_0 + \mathbf{R}_{\text{id}^*} \\ \mathbf{0}^\top & \mathbf{e}_{\text{id}^*}^\top \end{bmatrix}, \mathbf{y}, \sigma_0, \frac{\sigma_1}{2\sigma_0} \right) \right].$$

Guess. At the end of the game, if $\mathcal{A}$ successfully guesses the correct (simulated) challenge bit from $\mathcal{B}'$, then $\mathcal{B}'$ guesses that the challenge samples are RLWE samples and outputs 1; otherwise, $\mathcal{B}'$ guesses the uniform distribution and outputs 0.

When the RLWE challenge is from the real RLWE distribution, i.e., $y_i = u_i \cdot v + e_i$ for some $v \in \mathcal{R}_q$, the challenge ciphertext is distributed exactly as in Game 5.

When the RLWE challenge is truly random, i.e., $y_i = y'_i + e_i$ for some $y'_i \xleftarrow{\$} \mathcal{R}_q$, the challenge ciphertext is distributed exactly as in Game 6. We therefore conclude that Game 5 and Game 6 are indistinguishable by the $\text{RLWE}_{n,k+2,q,D_{\mathcal{R},\sigma_0}}$ assumption.

Proof of Lemma 18

Proof. In Game 6, we can re-write $\mathbf{y}' \xleftarrow{\$} \mathcal{R}_q^{k+1}$ as $\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ where $y_1 \xleftarrow{\$} \mathcal{R}_q^k$ and $y_2 \xleftarrow{\$} \mathcal{R}_q$, then we have that

$$\begin{aligned} \bar{\mathbf{c}}_1^* &= \text{ReRand} \left(\begin{bmatrix} \mathbf{I}_k & \mathbf{R}_0 + \mathbf{R}_{\text{id}^*} \\ \mathbf{0}^\top & \mathbf{e}_{\text{id}^*}^\top \end{bmatrix}, \mathbf{y}' + \mathbf{e}'_0, \sigma_0, \frac{\sigma_1}{2\sigma_0} \right) \\ &= \mathbf{y}'^\top \cdot \begin{bmatrix} \mathbf{I}_k & \mathbf{R}_0 + \mathbf{R}_{\text{id}^*} \\ \mathbf{0}^\top & \mathbf{e}_{\text{id}^*}^\top \end{bmatrix} + (\mathbf{e}'_2)^\top \\ &= [y_1^\top | y_2] \cdot \begin{bmatrix} \mathbf{I}_k & \mathbf{R}_0 + \mathbf{R}_{\text{id}^*} \\ \mathbf{0}^\top & \mathbf{e}_{\text{id}^*}^\top \end{bmatrix} + (\mathbf{e}'_2)^\top \\ &= [y_1^\top | y_1^\top (\mathbf{R}_0 + \mathbf{R}_{\text{id}^*}) + y_2 \cdot \mathbf{e}_{\text{id}^*}^\top] + (\mathbf{e}'_2)^\top. \end{aligned}$$

By Lemma 3, we have that the distribution of $\mathbf{e}'_2$ is statistically close the discrete Gaussian distribution $D_{\mathcal{R}^{2k}, \sigma_1}$. Thus, Game 6 and Game 7 are statistically indistinguishable.

Proof of Lemma 19

Proof. First, it is easy to see that c_0^* is uniformly random over $\mathcal{R}_q$ in both Game 7 and Game 8. We also have to show that the distribution of $\mathbf{c}_1^*$ is close to the uniform distribution over $\mathcal{R}_q^{2k+1}$. To see this, we first observe the following

$$(\mathbf{b}^\top, \mathbf{b}^\top \mathbf{R}_0, \mathbf{y}_1^\top, \mathbf{y}_1^\top \mathbf{R}_0) \approx (\mathbf{b}^\top, \mathbf{b}'^\top, \mathbf{y}_1^\top, (\mathbf{y}'_1)^\top) \approx (\mathbf{b}^\top, \mathbf{b}^\top \mathbf{R}_0, \mathbf{y}_1^\top, (\mathbf{y}'_1)^\top), \quad (18)$$

where $\mathbf{b}, \mathbf{b}' \xleftarrow{\$} \mathcal{R}_q^k$, $\mathbf{R}_0 \leftarrow [-\rho, \rho]^{k \times k}$, $\mathbf{y}_1, \mathbf{y}'_1 \xleftarrow{\$} \mathcal{R}_q^k$. It can be seen that the first and the second distributions are statistically close by apply Lemma 5 for $\begin{bmatrix} \mathbf{b}^\top \\ \mathbf{y}_1^\top \end{bmatrix} \in \mathcal{R}_q^{2 \times k}$ and $\mathbf{R}_0 \leftarrow [-\rho, \rho]_{\mathcal{R}}^{k \times k}$. It can also be seen that the second and the third distributions are statistically close by applying the same lemma for $\mathbf{b}$ and $\mathbf{R}_0$. From the above, we have that the following distributions are statistically close:

$$\begin{aligned} & (\mathbf{b}^\top, \mathbf{b}^\top \mathbf{R}_0, \mathbf{y}_1^\top + (\mathbf{e}'_{2,1})^\top, \mathbf{y}_1^\top \mathbf{R}_0 + \mathbf{y}_1^\top \mathbf{R}_{\text{id}^*} + y_2 \mathbf{e}_{\text{id}^*}^\top + (\mathbf{e}'_{2,2})^\top) \\ & \approx (\mathbf{b}^\top, \mathbf{b}^\top \mathbf{R}_0, \mathbf{y}_1^\top + (\mathbf{e}'_{2,1})^\top, (\mathbf{y}'_1)^\top + \mathbf{y}_1^\top \mathbf{R}_{\text{id}^*} + y_2 \mathbf{e}_{\text{id}^*}^\top + (\mathbf{e}'_{2,2})^\top) \\ & \approx (\mathbf{b}^\top, \mathbf{b}^\top \mathbf{R}_0, \mathbf{y}_1^\top, (\mathbf{y}'_1)^\top). \end{aligned}$$

The first and the second distributions above are statistically close by Eq. (18), whereas the second and the third distributions are statistically close by the fact that $\mathbf{e}'_{2,1}$, $\mathbf{e}'_{2,2}$ and $\mathbf{R}_i$, which are used to compute $\mathbf{R}_{\text{id}^*}$, are chosen independently random from other variables. Therefore, Game 7 and Game 8 are statistically indistinguishable.

E Parameter Constraints for our IBE construction in Sect. 5.3

Denote the security parameter as λ and the degree of ring $\mathcal{R}$ by n . To satisfy the correctness and security requirement, for any real number ω , we require:

- For the requirement of the partition function as in Def. 6, Lemma 22, we need

$$n \text{ is a power of } 2, \quad q \equiv 3 \pmod{8}, \quad tp \leq n, \quad m = 2n, \quad L + 1 \leq m^\eta.$$

- For the requirements of [TrapGen](#) algorithm and gadget vector $\mathbf{g}$ in Def. 3, we need

$$\rho < \frac{1}{2} \sqrt{q/n}, \quad k \geq 2 \log_\rho q, \quad k \geq \lceil \log_b q \rceil.$$

- For the requirement of [SampleLeft_{hybrid}](#) algorithm in Theorem 4, we need

$$\sigma \geq \eta_\epsilon(\mathbb{Z}^{nk}) \cdot \|\widetilde{\mathbf{T}}_{\mathbf{b}}\|,$$

where $\|\widetilde{\mathbf{T}}_{\mathbf{b}}\| \leq O(b\rho\sqrt{n\log_\rho q})$ by Lemma 7.

- For the requirement of [SamPre](#) in [SampleRight_{hybrid}](#) algorithm in Theorem 5, we need

$$s \geq \eta_\epsilon(\mathbb{Z}^{nk}) \cdot \|\widetilde{\mathbf{T}}_{\mathbf{g}}\|,$$

where $\|\widetilde{\mathbf{T}}_{\mathbf{g}}\| \leq \sqrt{b^2 + 1}$ by Def. 3.

- For the requirement of [SampleRight_{hybrid}](#) algorithm in Theorem 5, we need

$$\sigma^2 \geq (s^2 + \eta_\epsilon^2(\mathbb{Z}^{nk})) \cdot (\|\mathbf{R}_{\text{id}}\|^2 + 1),$$

where $\|\mathbf{R}_{\text{id}}\| \leq nkb \cdot \|\mathbf{R}\| \leq nkb \cdot C\rho\sqrt{n} \cdot (2\sqrt{k} + \omega)$ with probability $2e^{-\pi\omega^2}$ by Theorem 3 and Lemma 4.

- For the requirement of [SampleRight_{hybrid}](#) algorithm in Theorem 5, we need

$$r^2 \geq (\sigma^2 + \eta_\epsilon^2(\mathbb{Z}^{nk})) \cdot (\|\mathbf{e}_{\text{id}}\|^2 + 1).$$

Note that we consider the *plain model* and *CRS model* respectively, as in [1, Corollary 4.9].

- In *plain model*, $\|\mathbf{e}_{\text{id}}\| \leq \|\mathbf{e}\|4(L+1)tmn^2(kb)^3\eta\|\mathbf{R}'\|$ by Theorems 1 to 3.
- In *CRS model*, $\|\mathbf{e}_{\text{id}}\| \leq \|\mathbf{e}\|\tilde{O}(tLmnk^2b^3\eta)\|\mathbf{R}'\|$.

Additionally, by the definition of $\mathbf{e}$ (in Sect. 2.5) and Lemma 2, $\|\mathbf{e}\| \leq \sigma'_0\sqrt{2nk}$. By Lemma 4, $\|\mathbf{R}'\| \leq C\rho\sqrt{n} \cdot (2\sqrt{2k} + \omega)$ with probability $2e^{-\pi\omega^2}$.

- For the requirement of [ReRand](#) algorithm, we need

$$\sigma_1 \geq 2\sigma_0 \cdot \|\mathbf{V}\| \text{ for } \mathbf{V} = \begin{bmatrix} \mathbf{I}_k & \mathbf{R}_{\text{id}}^* \\ \mathbf{0}^\top & \mathbf{e}_{\text{id}}^* \end{bmatrix}.$$

- For the correctness of our IBE scheme (Lemma 11), we need modulus q satisfy

$$q \geq 5\omega(\sigma_0 + r\sigma_0\sqrt{n} + \sigma_1\sigma\sqrt{2nk})$$

to achieve correctness with concrete probability at least $1 - (6e^{-\pi\omega^2} + 2^{-2nk} + 2^{-n})$.

F Application for LWE-based IBE

We note that our novel cross-multiplication design and several techniques are not only restricted to RLWE-based IBE [1, 25], but also can be applied to LWE-based IBE [2, 38, 39].

Our several novel tools/techniques include: a) homomorphic computation outputting BGG+-style encoding with two distinct-norm trapdoors; b) sampling algorithm with hybrid Gaussian outputs; c) partial rerandomization. In this section, we explain how to apply these techniques to LWE-based IBE, respectively.

F.1 Apply Technique a) to LWE-based IBE

Usually in LWE-based IBE [2, 38, 39], there are a trapdoor pair $(\mathbf{B}, \mathbf{T}_\mathbf{B}) \in \mathbb{Z}_q^{n \times nk} \times \mathbb{Z}^{nk \times nk}$, and some public uniform matrices $\mathbf{C}_i$, which are simulated as $\mathbf{C}_i = \mathbf{B}\mathbf{R}_i + x_i\mathbf{G}$ in the security proof (where $\mathbf{G} \in \mathbb{Z}_q^{n \times nk}$ and $k \geq \log q$). As described in Sect. 1.2, our technique a) follows the steps below:

1. First, we simulate $\mathbf{C}_i$ in GSW-style encryption form.
2. Then, we homomorphically compute the partition function $H(\cdot)$ and get a GSW-style encryption $\mathbf{C}_H$ of the partition function.
3. Also, we provide a public random matrix $\mathbf{M}$ in the construction, and simulate it in a BGG+-style encoding of the secret key of GSW.
4. Finally, using the BGG+-style encoding $\mathbf{M}$, we homomorphically transform the GSW-style encryption $\mathbf{C}_H$ into an (approximate) BGG+-style encoding of the partition function, i.e.,

$$\mathbf{C}_{\text{id}} = \mathbf{B}\mathbf{R}_{\text{id}} + H(\text{id})\mathbf{G} + \mathbf{E}_{\text{id}}. \quad (19)$$

A straightforward but failing approach to adapting to LWE

1. First, we simulate

$$\mathbf{C}_i = \begin{bmatrix} \mathbf{A} \\ \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \end{bmatrix} \mathbf{R}_i + x_i \begin{bmatrix} \mathbf{G} \\ \mathbf{g}^\top \end{bmatrix},$$

where $\left(\begin{bmatrix} \mathbf{A} \\ \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \end{bmatrix}, \mathbf{s} \right)$ are the public key and secret key of GSW scheme.

2. Then, we homomorphically compute the partition function $H(\cdot)$ and get $\mathbf{C}_H = \begin{bmatrix} \mathbf{A} \\ \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \end{bmatrix} \mathbf{R}_H + H(\text{id}) \begin{bmatrix} \mathbf{G} \\ \mathbf{g}^\top \end{bmatrix}$. For the last k columns of $\mathbf{C}_H$, i.e., $\begin{bmatrix} \hat{\mathbf{C}}_0 \\ \hat{\mathbf{c}}_1^\top \end{bmatrix} = \begin{bmatrix} \mathbf{A} \hat{\mathbf{R}}_H \\ (\mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top) \hat{\mathbf{R}}_H + H(\text{id}) \mathbf{g}^\top \end{bmatrix}$, then we have

$$\hat{\mathbf{c}}_1^\top - \mathbf{s}^\top \hat{\mathbf{C}}_0 = H(\text{id}) \mathbf{g}^\top + \mathbf{e}^\top \hat{\mathbf{R}}_H.$$

3. Also, we add a public random matrix $\mathbf{M}$ in the construction, and simulate each row of $\mathbf{M}$ in the security proof. Suppose $\mathbf{B} = \begin{bmatrix} \mathbf{b}_1^\top \\ \vdots \\ \mathbf{b}_n^\top \end{bmatrix}$, we sample a small-norm matrix $\mathbf{R}_s \in \mathbb{Z}^{nk \times nk}$ and set

$$\mathbf{M} = \begin{bmatrix} \mathbf{m}_1^\top \\ \vdots \\ \mathbf{m}_n^\top \end{bmatrix} = \begin{bmatrix} \mathbf{b}_1^\top \mathbf{R}_s + \mathbf{s}^\top \mathbf{G} \\ \vdots \\ \mathbf{b}_n^\top \mathbf{R}_s + \mathbf{s}^\top \mathbf{G} \end{bmatrix} = \mathbf{B} \mathbf{R}_s + \begin{bmatrix} \mathbf{s}^\top \\ \vdots \\ \mathbf{s}^\top \end{bmatrix} \mathbf{G}.$$

4. Finally, we compute

$$\hat{\mathbf{c}}_1^\top - \mathbf{m}_i^\top \cdot \mathbf{G}^{-1}(\hat{\mathbf{C}}_0) = \mathbf{b}_i^\top \cdot (-\mathbf{R}_s \mathbf{G}^{-1}(\hat{\mathbf{C}}_0)) + H(\text{id}) \mathbf{g}^\top + \mathbf{e}^\top \hat{\mathbf{R}}_H$$

for $i \in [n]$. Since we want to get a result as the form of Eq. (19), we concatenate these n vectors on the diagonal (we try to use the vectors $\mathbf{g}^\top$ to make up the matrix $\mathbf{G}$) as follows:

$$\begin{bmatrix} \mathbf{b}_1^\top \cdot (-\mathbf{R}_s \mathbf{G}^{-1}(\hat{\mathbf{C}}_0)) + H(\text{id})\mathbf{g}^\top + \mathbf{e}^\top \hat{\mathbf{R}}_H & & \\ & \ddots & \\ & & \mathbf{b}_n^\top \cdot (-\mathbf{R}_s \mathbf{G}^{-1}(\hat{\mathbf{C}}_0)) + H(\text{id})\mathbf{g}^\top + \mathbf{e}^\top \hat{\mathbf{R}}_H \end{bmatrix}.$$

Obviously, such a matrix is not what we expect (i.e., Eq. (19)), since it's only non-zero on the diagonal.

A novel and successful approach to adapting to LWE

From the above “incorrect” idea, we know that we cannot use the vectors $\mathbf{g}^\top$ to make up the matrix $\mathbf{G}$, the only way to generate the final matrix we want (i.e., Eq. (19)), is to generate the complete matrix $\mathbf{G}$ directly in the process of homomorphic computation. Therefore, we propose another idea:

1. First, we simulate

$$\mathbf{C}_i = \begin{bmatrix} \mathbf{A} \\ \mathbf{s}_1^\top \mathbf{A} + \mathbf{e}_1^\top \\ \vdots \\ \mathbf{s}_n^\top \mathbf{A} + \mathbf{e}_n^\top \end{bmatrix} \mathbf{R}_i + x_i [\mathbf{G} \quad \mathbf{G}].$$

2. Then, we homomorphically compute the partition function $H(\cdot)$ and get

$$\begin{aligned} \mathbf{C}_H &= \begin{bmatrix} \mathbf{A} \\ \mathbf{s}_1^\top \mathbf{A} + \mathbf{e}_1^\top \\ \vdots \\ \mathbf{s}_n^\top \mathbf{A} + \mathbf{e}_n^\top \end{bmatrix} \mathbf{R}_H + H(\text{id})[\mathbf{G} \quad \mathbf{G}]. \text{ For the last } nk \text{ columns of } \mathbf{C}_H, \text{ i.e.,} \\ \begin{bmatrix} \hat{\mathbf{C}}_0 \\ \hat{\mathbf{C}}_1 \end{bmatrix} &= \begin{bmatrix} \mathbf{A} \hat{\mathbf{R}}_H \\ \begin{bmatrix} \mathbf{s}_1^\top \mathbf{A} + \mathbf{e}_1^\top \\ \vdots \\ \mathbf{s}_n^\top \mathbf{A} + \mathbf{e}_n^\top \end{bmatrix} \hat{\mathbf{R}}_H + H(\text{id})\mathbf{G} \end{bmatrix}, \text{ we have} \\ \hat{\mathbf{C}}_1 - \begin{bmatrix} \mathbf{s}_1^\top \\ \vdots \\ \mathbf{s}_n^\top \end{bmatrix} \hat{\mathbf{C}}_0 &= H(\text{id})\mathbf{G} + \begin{bmatrix} \mathbf{e}_1^\top \\ \vdots \\ \mathbf{e}_n^\top \end{bmatrix} \hat{\mathbf{R}}_H. \end{aligned}$$

3. Also, we add a public random matrix $\mathbf{M}$ in the construction, and simulate

$$\mathbf{M} = \mathbf{B} \mathbf{R}_s + \begin{bmatrix} \mathbf{s}_1^\top \\ \vdots \\ \mathbf{s}_n^\top \end{bmatrix} \mathbf{G}.$$

4. Finally, we compute

$$\hat{\mathbf{C}}_1 - \mathbf{M} \cdot \mathbf{G}^{-1}(\hat{\mathbf{C}}_0) = \mathbf{B} \underbrace{(-\mathbf{R}_s \cdot \mathbf{G}^{-1}(\hat{\mathbf{C}}_0))}_{\mathbf{R}_{\text{id}}} + H(\text{id})\mathbf{G} + \underbrace{\begin{bmatrix} \mathbf{e}_1^\top \\ \vdots \\ \mathbf{e}_n^\top \end{bmatrix} \hat{\mathbf{R}}_H}_{\mathbf{E}_{\text{id}}}.$$

Obviously, $\mathbf{R}_{\text{id}}$ is small-norm and $\mathbf{E}_{\text{id}}$ is large-norm. So we construct a matrix version of the sampling vector as in Eq. (19), and then we can get

$$\mathbf{PK}_{\text{id}} = [\mathbf{B}|\mathbf{B}\mathbf{R}_{\text{id}} + H(\text{id})\mathbf{G} + \mathbf{E}_{\text{id}}] \in \mathbb{Z}^{n \times 2nk}$$

to sample the hybrid secret.

F.2 Apply Technique b) to LWE-based IBE

Similar to Sect. 1.2, we sample $\mathbf{h} \in \mathbb{Z}^n$, $\bar{\mathbf{p}} \in \mathbb{Z}^{nk}$, $\mathbf{p} \in \mathbb{Z}^{2nk}$ as perturbations, which follow some elliptical Gaussian distributions. Let $\tilde{\mathbf{x}} \in \mathbb{Z}^{nk}$ be a short vector such that $\mathbf{G}\tilde{\mathbf{x}} = H(\text{id})^{-1} \cdot (\mathbf{u} + \mathbf{h} + \mathbf{E}_{\text{id}} \cdot \bar{\mathbf{p}} - \mathbf{PK}_{\text{id}} \cdot \mathbf{p})$. We then construct the hybrid secret $(\mathbf{w} \in \mathbb{Z}^n, \mathbf{x} \in \mathbb{Z}^{2nk})$ as follows.

$$\underbrace{[\mathbf{B}|\mathbf{B}\mathbf{R}_{\text{id}} + H(\text{id})\mathbf{G} + \mathbf{E}_{\text{id}}]}_{\mathbf{PK}_{\text{id}}} \cdot \underbrace{(\mathbf{p} + [\begin{smallmatrix} -\mathbf{R}_{\text{id}} \\ \mathbf{I} \end{smallmatrix}] \tilde{\mathbf{x}})}_{\mathbf{x}, \text{small}} = \mathbf{u} + \underbrace{(\mathbf{h} + \mathbf{E}_{\text{id}} (\bar{\mathbf{p}} + \tilde{\mathbf{x}}))}_{\mathbf{w}, \text{large}}$$

It's easy to verify that the above equation holds:

$$\begin{aligned} \mathbf{PK}_{\text{id}} \cdot \mathbf{x} &= [\mathbf{B}|\mathbf{B}\mathbf{R}_{\text{id}} + H(\text{id})\mathbf{G} + \mathbf{E}_{\text{id}}] \cdot (\mathbf{p} + [\begin{smallmatrix} -\mathbf{R}_{\text{id}} \\ \mathbf{I} \end{smallmatrix}] \tilde{\mathbf{x}}) \\ &= \mathbf{PK}_{\text{id}} \cdot \mathbf{p} + H(\text{id})\mathbf{G}\tilde{\mathbf{x}} + \mathbf{E}_{\text{id}} \cdot \tilde{\mathbf{x}} \\ &= \mathbf{PK}_{\text{id}} \cdot \mathbf{p} + (\mathbf{u} + \mathbf{h} + \mathbf{E}_{\text{id}} \cdot \bar{\mathbf{p}} - \mathbf{PK}_{\text{id}} \cdot \mathbf{p}) + \mathbf{E}_{\text{id}} \cdot \tilde{\mathbf{x}} \\ &= \mathbf{u} + (\mathbf{h} + \mathbf{E}_{\text{id}} \cdot \bar{\mathbf{p}} + \mathbf{E}_{\text{id}} \cdot \tilde{\mathbf{x}}) \\ &= \mathbf{u} + \mathbf{w} \end{aligned}$$

F.3 Apply Technique c) to LWE-based IBE

Similar to Sect. 1.2, we construct the hybrid error as

$$\mathbf{c}_1^\top = \mathbf{v}^\top \cdot \mathbf{D} \cdot [\mathbf{I}_n, \mathbf{PK}_{\text{id}}] + [\mathbf{e}_1^\top, \mathbf{e}_2^\top] \in \mathbb{Z}^{2nk+n},$$

where $\mathbf{v} \in \mathbb{Z}^n$ is a random vector, $\mathbf{D} \in \mathbb{Z}^{n \times n}$ is an invertible matrix, $\mathbf{e}_1 \leftarrow D_{\mathbb{Z}^n, \sigma_0}$ is a small error and $\mathbf{e}_2 \leftarrow D_{\mathbb{Z}^{2nk}, \sigma_1}$ is a large error.

Similar to Sect. 1.2, we first use the re-randomization algorithm to generate a partial ciphertext

$$\begin{aligned} (\tilde{\mathbf{c}}_1^*)^\top &= \text{ReRand}([\begin{smallmatrix} \mathbf{I}_{nk} & \mathbf{R}^* \\ \mathbf{0} & \mathbf{E}^* \end{smallmatrix}], \mathbf{v}^\top \cdot \mathbf{D} \cdot [\mathbf{B}, \mathbf{I}_n] + (\mathbf{e}'_0)^\top, \sigma'_0, \frac{\sigma_1}{2\sigma_0}) \\ &= \mathbf{v}^\top \cdot \mathbf{D} \cdot [\mathbf{B}, \mathbf{I}_n] \cdot [\begin{smallmatrix} \mathbf{I}_{nk} & \mathbf{R}^* \\ \mathbf{0} & \mathbf{E}^* \end{smallmatrix}] + (\mathbf{e}'_2)^\top \\ &= \mathbf{v}^\top \cdot \mathbf{D} \cdot [\mathbf{B}, \mathbf{B}\mathbf{R}^* + \mathbf{E}^*] + (\mathbf{e}'_2)^\top, \end{aligned}$$

where the distribution of $\mathbf{e}'_2$ is close to $\mathbf{e}_2$. We concatenate the last part of $\mathbf{v}^\top \cdot \mathbf{D} \cdot [\mathbf{B}, \mathbf{I}_n] + (\mathbf{e}'_0)^\top$ and $\tilde{\mathbf{c}}_1^*$, and get the ciphertext $\mathbf{c}_1^*$ as follows

$$(\mathbf{c}_1^*)^\top = \mathbf{v}^\top \cdot \mathbf{D} \cdot [\mathbf{I}_n, \mathbf{B}, \mathbf{B}\mathbf{R}^* + \mathbf{E}^*] + [\mathbf{e}'_0^\top, \mathbf{e}'_2^\top].$$